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Language models generalize beyond natural proteins

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Abstract

Learning the design patterns of proteins from sequences across evolution may have promise toward generative protein design. However it is unknown whether language models, trained on sequences of natural proteins, will be capable of more than memorization of existing protein families. Here we show that language models generalize beyond natural proteins to generate *de novo* proteins. We focus on two protein design tasks: fixed backbone design where the structure is specified, and unconstrained generation where the structure is sampled from the model. Remarkably although the models are trained only on sequences, we find that they are capable of designing structure. A total of 228 generated proteins are evaluated experimentally with high overall success rates (152/228 or 67%) in producing a soluble and monomeric species by size exclusion chromatography. Out of 152 experimentally successful designs, 35 have no significant sequence match to known natural proteins. Of the remaining 117, sequence identity to the nearest sequence match is at median 27%, below 20% for 6 designs, and as low as 18% for 3 designs. For fixed backbone design, the language model generates successful designs for each of eight experimentally evaluated artificially created fixed backbone targets. For unconstrained generation, sampled proteins cover diverse topologies and secondary structure compositions, and have high experimental success rate (71/129 or 55%). The designs reflect deep patterns linking sequence and structure, including motifs that occur in related natural structures, and motifs that are not observed in similar structural contexts in known protein families. The results show that language models, though only trained on sequences, learn a deep grammar that enables the design of protein structure, extending beyond natural proteins.

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Introduction

Generative artificial intelligence for biology has potential to open up a space of protein design beyond natural proteins. Since amino acid sequences are the fundamental codes of proteins, learning to read and write these codes with a language model may have promise. Language models have played a central role in recent advances in artificial intelligence (1), including developments in complex reasoning, mathematical problem solving, image generation, and natural language generation (2–4). Scaling laws link performance with the compute, data, and number of parameters used to train the models (5), and emergence of higher level capabilities is observed with increasing scale (6). In biology, recent work on evolutionary scale language models of proteins has shown that a deep knowledge of intrinsic biological properties emerges from training on protein sequences (7). Information about the folded three dimensional structure of proteins develops within the models, extending to atomic resolution structure (8). This information emerges through training on sequences alone. At the same time the structural information that emerges as a result of training on sequences has been shown to depend on the available evolutionary information, varying as a function of the number of related proteins in the training data (8, 9). It is an open question across domains to what extent language models are capable of generalizing outside their training data. In biology, it is unknown whether language models can be used to explore a design space beyond that of natural proteins.

Here we demonstrate that language models generalize beyond natural proteins to generate *de novo* proteins, different in sequence and structure from natural proteins. We experimentally validate a large number of designs spanning diverse topologies and sequences. We find that although language models are trained only on the sequences of proteins, they are capable of designing protein structure, including structures of artificially engineered *de novo* proteins that are distinct from those of natural proteins. Given the backbone of a *de novo* protein structure as a target, the language model generates sequences that are predicted to fold to the specified structure. When the sequence and structure are both free, language models produce designs that span a wide range of fold topologies and secondary structure compositions, creating proteins which overlap the natural sequence distribution as well as extend beyond it. Designs succeed experimentally across the space of sampled proteins, including many designs that are distant in sequence from natural proteins. The model generates motifs that link sequence to the design of structure and can apply them in new sequence and structural contexts, including motifs such as complex hydrogen bond

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语言模型超越天然蛋白质进行概括

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摘要

学习跨越进化的序列中蛋白质的设计模式，可能对生成性蛋白质设计有前景。然而，目前尚不清楚训练于天然蛋白质序列的语言模型，是否能够超越对现有蛋白质家族的记忆。我们在此展示，语言模型超越天然蛋白质，能够生成新设计蛋白质。我们专注于两项蛋白质设计任务：固定主链设计（即结构已指定）和无约束生成（即结构从模型中采样）。值得注意的是，尽管模型仅在序列上训练，我们发现它们能够设计结构。共评估了228种生成的蛋白质，在尺寸排除色谱法中，高达67%的成功率（152/228），能够产生可溶且为单体的物种。在152种实验成功设计中，有35种与已知天然蛋白质没有显著的序列匹配。在剩下的117种中，最近序列匹配的序列相似性中位数为27%，6种设计低于20%，3种设计低至18%。对于固定主链设计，语言模型为每个八个实验评估的人工创建的固定主链目标生成成功设计。在无约束生成中，采样的蛋白质涵盖了多样的拓扑结构和二级结构组成，并且具有高实验成功率（71/129或55%）。这些设计反映出序列与结构之间的深层模式，包括在相关天然结构中出现的基序，以及在已知蛋白质家族的相似结构背景中未观察到的基序。结果表明，尽管语言模型仅在序列上训练，但它们学习了一种深层语法，使得蛋白质结构的设计超越了天然蛋白质。

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介绍

生成式人工智能在生物学领域有潜力开启超越天然蛋白质的蛋白质设计空间。由于氨基酸序列是蛋白质的基本编码，因此用语言模型学习读取和编写这些编码可能具有前景。语言模型在最近的人工智能进展中发挥了核心作用（1），包括在复杂推理、数学问题解决、图像生成和自然语言生成中的发展（2–4）。缩放法则将模型的性能与训练时使用的计算、数据和参数数量联系起来（5），并且随着规模的增加，观察到更高水平能力的出现（6）。在生物学中，最近关于蛋白质的进化规模语言模型的工作表明，从蛋白质序列训练中获得的深刻生物学内在特性的知识显现出来（7）。关于蛋白质折叠的三维结构的信息在模型中发展，并延伸到原子分辨率结构（8）。这一信息通过仅在序列上进行训练而显现。同时，作为训练序列的结果所出现的结构信息已被证明依赖于可用的进化信息，随训练数据中相关蛋白质数量的变化而变化（8，9）。在各个领域，关于语言模型在多大程度上能够概括超出其训练数据仍然是一个开放的问题。在生物学中，目前不清楚语言模型是否可以用来探索超越天然蛋白质的设计空间。

在这里，我们展示了语言模型在生成新设计蛋白质方面的泛化能力，这些蛋白质在序列和结构上与天然蛋白质不同。我们通过实验验证了大量不同拓扑和序列的设计。我们发现，尽管语言模型仅在蛋白质的序列上进行训练，但它们能够设计蛋白质结构，包括与天然蛋白质明显不同的人工设计新设计蛋白质的结构。以新设计蛋白质结构的骨架作为目标，语言模型生成预测将折叠为指定结构的序列。当序列和结构都是自由时，语言模型产生的设计跨越广泛的折叠拓扑和二级结构组成，创造出既重叠天然序列分布又超越它的蛋白质。在样本蛋白质空间中，设计在实验中成功，包括许多在序列上与天然蛋白质相距甚远的设计。该模型生成了将序列与结构设计联系起来的模体，并能在新的序列和结构上下文中应用这些模体，包括复杂氢键等模体。

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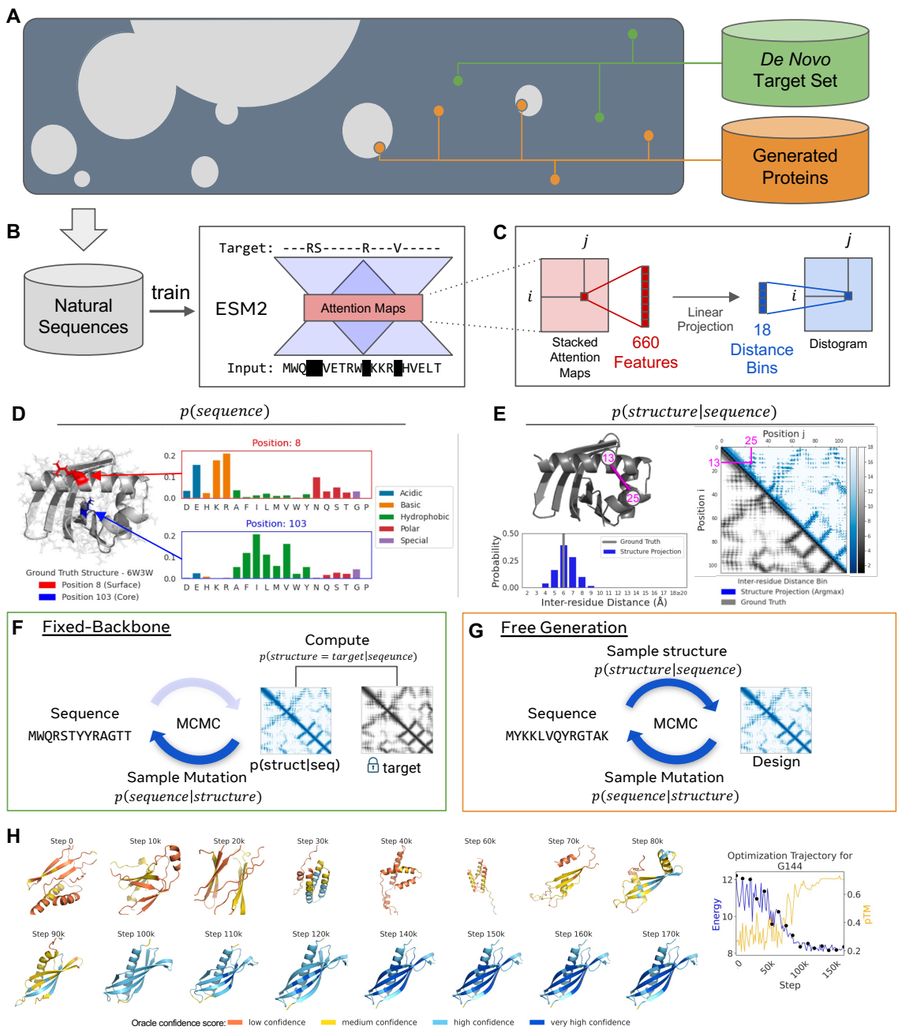

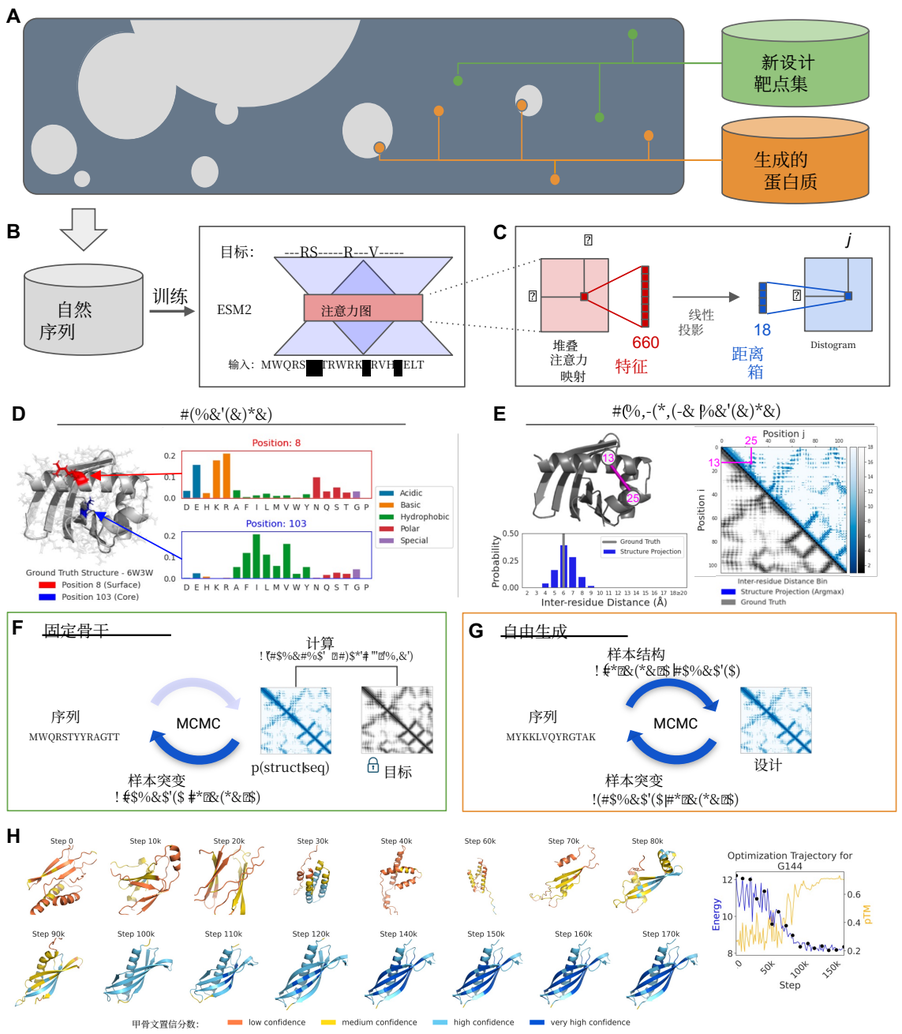

Figure 1. Overview. (**A**) Illustration of protein sequence space. Natural sequences (gray) cover a fraction of possible protein sequences. To generalize beyond natural sequences language models will need to access underlying design patterns. We evaluate language models on (i) a fixed backbone sequence design task with a set of *de novo* designed proteins (green), and (ii) via an unconstrained *de novo* protein generation task (orange). (**B**) The language model ESM2 is trained using masked language modeling over millions of diverse natural proteins across evolution. (**C**) After training, information about tertiary structure can be identified in the internal attention states of the model. A linear projection translates the attention at a pair of positions in the sequence to a distribution over inter-residue distances. (**D**) Probability of a sequence. The model outputs a probability for each amino acid at every position in the protein, here shown for the designed protein 6W3W. The model gives a higher probability to hydrophilic amino acids at a surface residue and hydrophobic ones at a residue in the core. (**E**) Probability of a structure given a sequence. For a given sequence the projection measures the compatibility of the internal representations of the language model with a structure. Tertiary structure is identified by probability mass on inter-residue distances less than 8Å. For 6W3W there is a good match between the projected structure (above diagonal) and ground truth structure (below diagonal). (**F**) The two terms giving the probability of sequences and structures are used to generate sequences. For fixed target design we use MCMC to generate sequences given a specified backbone structure, by sampling from the conditional distribution of sequences given a structure. (**G**) For unconstrained generation we allow both the sequence and structure to vary. (**H**) Predicted structures (using AlphaFold) are shown at even intervals across a single free generation trajectory. The model samples a range of possible topologies before narrowing to the refinement of one topology.

图 *1.* 概述. (**A**) 蛋白质序列空间的示意图。天然序列（灰色）覆盖了可能的蛋白质序列的一小部分。为了超越天然序列，语言模型需要访问潜在的设计模式。我们在以下两个任务中评估语言模型：(i) 使用一组 新设计 蛋白质（绿色）的固定骨架序列设计任务，以及 (ii) 通过一个不受限制的 新设计 蛋白质生成任务（橙色）。(**B**) 语言模型 ESM2 使用在进化过程中对数百万种不同天然蛋白质的掩码语言建模进行训练。 (**C**) 训练后，模型的内部注意力状态可以识别关于三级结构的信息。线性投影将序列中一对位置的注意力转换为氨基酸之间距离的分布。(**D**) 序列的概率。模型为蛋白质中每个位置的每种氨基酸输出一个概率，这里展示的是设计的蛋白质 6W3W。模型对表面残基的亲水性氨基酸给予更高的概率，而对核心中的疏水性氨基酸给予较低的概率。(**E**) 给定序列的结构概率。对于给定的序列，投影测量语言模型的内部表示与结构的兼容性。通过在氨基酸之间距离小于 8 ˚A 的概率质量来识别三级结构。对于 6W3W，投影结构（对角线以上）与真实结构（对角线以下）之间有很好的匹配。(**F**) 生成序列的概率和结构的两个项用于生成序列。对于固定目标设计，我们使用 MCMC 生成给定特定骨架结构的序列，方法是从给定结构的序列条件分布中抽样。(**G**) 对于不受限制的生成，我们允许序列和结构同时变化。(**H**) 预测的结构（使用 AlphaFold）在单一自由生成轨迹中以均等间隔显示。模型在缩小到一个拓扑的细化之前，先对一系列可能的拓扑进行抽样。

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networks that are not found in sequence- or structurally-similar known proteins. Overall experimental success rates are high with 152 out of a total of 228 (67%) experimentally evaluated proteins producing a soluble and monomeric species by size exclusion chromatography (SEC). The high success rate extends to proteins that are distant from natural proteins where 31 out of a total of 49 (63%) experimentally evaluated proteins succeed.

A deep grammar of protein sequences

We hypothesize that there exists a deep underlying grammar in protein sequences that makes it possible for the language model to generalize. To generalize beyond natural proteins, language models will need to access design patterns that extend outside the space of natural proteins. Classically this form generalization has been enabled by an energy function grounded in physics that captures the native folded state (10). Recently deep learning based methods grounded in structure have been proposed as a new approach to this problem by inverting structure prediction (11, 12), or conditioning on backbone structures (13–15). By modeling the structure explicitly during training, new deep learning approaches may capture something similar to the physical energy (16). The success of language models on this problem suggests that deep patterns in sequences may offer an alternative path to generalization, independent of an explicit model of the underlying physics.

The classical perspective of evolutionary inference from sequences is that information about the properties of proteins is encoded into the sequence patterns of evolutionarily related proteins through conservation and coevolution. This view develops from the observation that the statistics of protein families reflect the constraints acting on the evolution of the sequences including biological structure and function (17, 18). This insight has formed the basis for the inference of structure and function from sequences in a protein family (19), and has also recently been applied with success by generative models to generate new examples from existing protein families (20–22). To date experimental validation of sequence based models for protein design has been limited to natural protein families.

Accessing a *de novo* design space distant from naturally occurring protein families is a fundamentally more challenging problem. This problem by definition cannot be solved by generating new samples from naturally occurring protein families. To solve this problem with a model grounded in sequences, it will be necessary to learn sequence patterns that generalize beyond individual protein families. Evolutionary scale language models go beyond classic protein family models by training on diverse sequences across evolution which means that they have the potential to learn deep patterns across all proteins, including where there is no experimental structure. There is evidence for local patterns in sequences that generalize beyond individual protein families, in the form of motifs that are local in the sequence (23) as well as motifs that are local in 3d space (24). However, the mapping between sequence and structure is not one-to-one (25), and designing sequences to reach a well-folded native state requires solving an exponentially large combinatorial problem to select a set of local sequence patterns which interact non-locally to specify a coherent structure (26). To design protein structure, the language model will have to develop an implicit understanding of how sequence determines structure, including local rules that link the design of structure with sequence, as well as global rules that determine whether a sequence is coherent and will fold into a native state.

Generative protein design with language models

We evaluate language models generatively, focusing on generalization beyond natural proteins. The known protein sequences sampled by evolution represent only a small fraction of the vast number of possible proteins (Fig. 1A). To generalize outside the space of proteins that has been explored by evolution it will be necessary to access deep patterns of protein design that apply outside this space. We focus on two generative protein design tasks. The first is fixed backbone design where the objective is to generate a sequence that folds to the target structure. This task assesses the ability of the language model, which has been trained only on sequences, to design protein structures. The second task is free generation, where the structure is unconstrained and allowed to vary along with the sequence. This enables characterization of the full generative capability of the model across diverse sequences and structural patterns to understand the space of proteins accessible to the model.

A test set of *de novo* designed artificial proteins is used to assess generalization beyond natural protein structures. The test set includes a diverse selection (N = 39) of structurally validated artificial protein structures from the Protein Data Bank (PDB) (27), which span a range of lengths ($67 \leq L \leq 184$), and topologies (Fig. S1 and Appendix A.1). Importantly, these *de novo* proteins have meaningful structural differences from proteins belonging to natural folds, including with respect to ideality, exact repetition, and symmetry of elements. Since the language model has not been trained on protein structures, generating designs for these backbones tests for the ability of the model to generalize to structures unlike those of the natural proteins whose sequences it has been trained on.

The language model, ESM2, is an evolutionary scale model of protein sequences that has been trained across the full extent of *natural* protein sequences (28). The training dataset excludes artificial sequences, as well as any sequences having similarity to the test set of *de novo* proteins used in the evaluations (Appendix A.1). ESM2 is trained with the masked language modeling objective (29) to recover the identity of amino acids from their context in the rest of the sequence (Fig. 1B). This training objective has been shown to materialize information about the folded structure of proteins in the internal representations of the model (7–9, 30). Since the training of the language model is only on sequences, information about structure that emerges must be the result of the unsupervised learning of patterns in sequences.

A linear projection from the attention maps of the language model identifies internal states that reflect protein structure. Previous work has shown that specific attention maps in transformer protein language models such as ESM2 encode the proximity of residue

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这些网络在序列上或结构上与已知蛋白质相似的情况下并不存在。总体实验成功率较高，在 228 个被评估的蛋白质中，有 152 个（67%）通过尺寸排斥色谱法（SEC）产生了可溶且单体的物种。这一高成功率适用于与天然蛋白质相远离的蛋白质，其中 49 个实验评估的蛋白质中有 31 个（63%）成功。

蛋白质序列的深层语法

我们假设蛋白质序列中存在深层的基础语法，使得语言模型能够进行泛化。为了超越天然蛋白质，语言模型需要访问扩展到天然蛋白质空间之外的设计模式。传统上，这种形式的泛化是通过一个基于物理的能量函数实现的，该函数捕捉到本征折叠状态（10）。最近，基于结构的深度学习方法被提出作为解决这一问题的新方法，通过反转结构预测（11，12），或依赖主链结构（13–15）。通过在训练期间显式建模结构，新型深度学习方法可能会捕捉到类似于物理能量的东西（16）。语言模型在这个问题上的成功表明，序列中的深层模式可能提供一种独立于所基于物理学的明确模型的泛化替代路径。

从序列进行进化推理的经典观点是，关于蛋白质特性的信息通过保护和共同进化在进化相关的蛋白质的序列模式中进行编码。这一观点源自于对蛋白质家族统计的观察，这些统计反映了作用于序列进化的约束，包括生物学的结构与功能（17, 18）。这一洞察为从蛋白质家族的序列推测结构与功能奠定了基础（19），并且最近也成功地被生成模型应用于从现有的蛋白质家族中生成新的示例（20–22）。迄今为止，基于序列的蛋白质设计模型的实验验证仅限于自然蛋白质家族。

访问一个 新设计 的设计空间，这与自然产生的蛋白质家族相距甚远，是一个根本上更具挑战性的问题。根据定义，这个问题无法通过生成来自自然发生的蛋白质家族的新样本来解决。要用一个基于序列的模型解决这个问题，就必须学习超越单个蛋白质家族的序列模式。进化规模语言模型通过在进化过程中对多样序列进行训练，超越经典的蛋白质家族模型，这意味着它们有潜力学习所有蛋白质的深层模式，包括那里没有实验结构的地方。已有证据表明，在序列中存在超越单个蛋白质家族的局部模式，这些模式以模体的形式存在，模体在序列中是局部的 (23)，以及模体在三维空间中是局部的 (24)。然而，序列与结构之间的映射并不是一一对应的 (25)，而设计序列以达到良好折叠的本征状态需要解决一个指数级大的组合问题，以选择一组局部序列模式。

这些模式进行非局部交互，以指定一个连贯的结构 (26)。为了设计蛋白质结构，语言模型必须发展出对序列如何决定结构的隐含理解，包括将结构设计与序列链接的局部规则，以及确定序列是否连贯并将折叠成本征状态的全局规则。

使用语言模型进行生成性蛋白质设计

我们评估语言模型的生成能力，重点关注超越天然蛋白质的泛化。通过进化采样的已知蛋白质序列仅代表了可能的蛋白质数量的一个小部分（图。1A）。要在进化探索过的蛋白质空间之外进行泛化，将有必要获取适用于该空间之外的蛋白质设计的深层模式。我们专注于两个生成蛋白质设计任务。第一个是固定主链设计，目标是生成一个可以折叠成目标结构的序列。该任务评估了仅仅在序列上进行训练的语言模型设计蛋白质结构的能力。第二个任务是自由生成，结构不受限制，可以与序列一起变化。这使得能够表征模型在多样的序列和结构模式中全面的生成能力，以理解模型可访问的蛋白质空间。

一个测试集新设计的人工作蛋白质用于评估超越天然蛋白质结构的泛化能力。测试集包括一系列多样化的选择（N = 39）来自蛋白质数据银行（PDB）中经过结构验证的人工蛋白质结构（27），这些结构跨越不同的长度（$67 \leq L \leq 184$）和拓扑（图 S1 及附录 A.1）。重要的是，这些 新设计的蛋白质在理想性、完全重复性和元素对称性等方面与属于天然折叠的蛋白质存在显著的结构差异。由于语言模型没有在蛋白质结构上进行训练，生成这些骨架的设计考验了模型对与其训练序列的天然蛋白质不同的结构进行泛化的能力。

语言模型ESM2是一个蛋白质序列的进化规模模型，已在天然蛋白质序列的全部范围内进行训练（28）。训练数据集排除了人工序列，以及任何与用于评估的新设计蛋白质测试集相似的序列（附录A.1）。ESM2使用掩码语言建模目标进行训练（29），以从序列的其余部分的上下文中恢复氨基酸的身份（图 1B）。该训练目标已被证明能够在模型的内部表示中体现蛋白质的折叠结构信息（7–9，30）。由于语言模型的训练仅基于序列，因此出现的结构信息必须是对序列中模式的无监督学习结果。

来自语言模型注意力图的线性投影识别出反映蛋白质结构的内部状态。之前的研究表明，变压器蛋白质语言模型中特定的注意力图，如ESM2，编码了残基的相近性

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Language models generalize beyond natural proteins

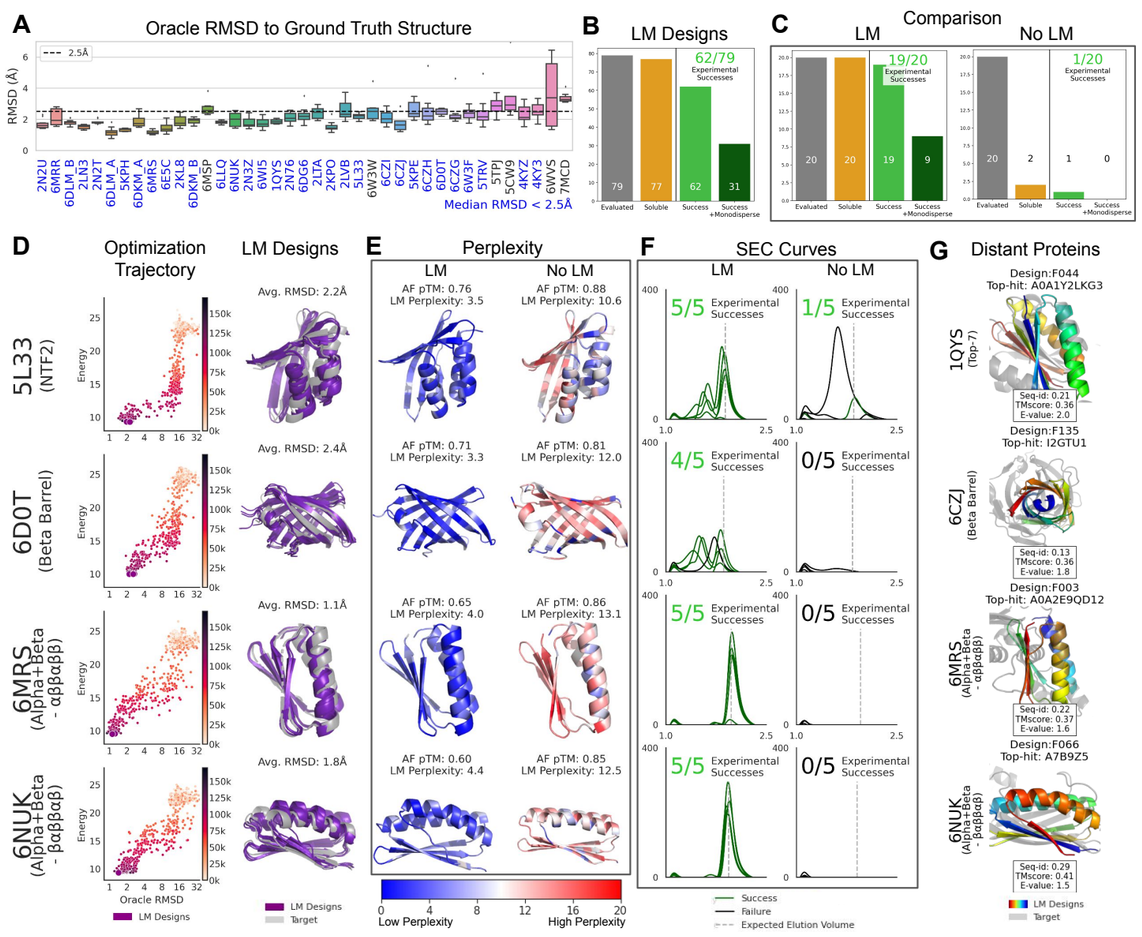

Figure 2. Design of sequences for de novo *structures.* (**A**) Overall evaluation of designs for the *de novo* target set using an *in silico* oracle. Root-mean-square deviation (RMSD) between C-alpha atoms designed structure (oracle prediction) and target structure is plotted for the top 10 designs by optimization objective for each target. Targets are ordered by increasing length. The language model generates sequences that are predicted to fold to the target structure for a large majority of *de novo* backbones in the test set. (33/39 achieve median RMSD < 2.5Å). (**B**) Experimental outcomes for ESM designs. A total of 79 designs across 6 *de novo* backbone targets were selected by a variety of criteria including sequence novelty and manual inspection for interesting motifs. Designs are considered a success if they are soluble and there is a peak at the expected elution volume by size-exclusion chromatography (SEC). Designs are categorized as monodisperse when the only peak is at the expected elution volume. Overall, 78% succeed, and 39% are monodisperse. (**C**) Experimental outcomes for comparison of designs with and without the language model. For each of the four targets, the top 5 out of 200 designs by optimization objective were selected for experimental evaluation. Overall 95% of designs with a language model succeed, while most designs without a language model fail due to insolubility. (**D**) (Left) Optimization trajectory showing energy specified by the language model vs RMSD to target, over the course of MCMC optimization. Energy decreases and funnels to low RMSD. (Right) Visualization of the top 5 designs selected by energy at the end of each trajectory. (**E**) Language modeling perplexity of designs. Language model designs are seen as probable by the language model, while high perplexity for the baseline designs indicates their sequences are seen as improbable. This coincides with experimental success. (**F**) Comparison of SEC traces between designs with and without a language model. The vast majority of language model designs are soluble and have a peak at the expected elution volume; in comparison few designs without a language model are soluble. (**G**) A subset of additional, successful language model designs are novel with respect to known natural proteins. Examples for four different backbones are shown with the design superimposed on the predicted structure of the top-significance hit from a sequence search against natural proteins. In each case the closest natural sequence has low sequence identity (<0.3) and predicted structure with different topology.

语言模型在天然蛋白质之外的推广

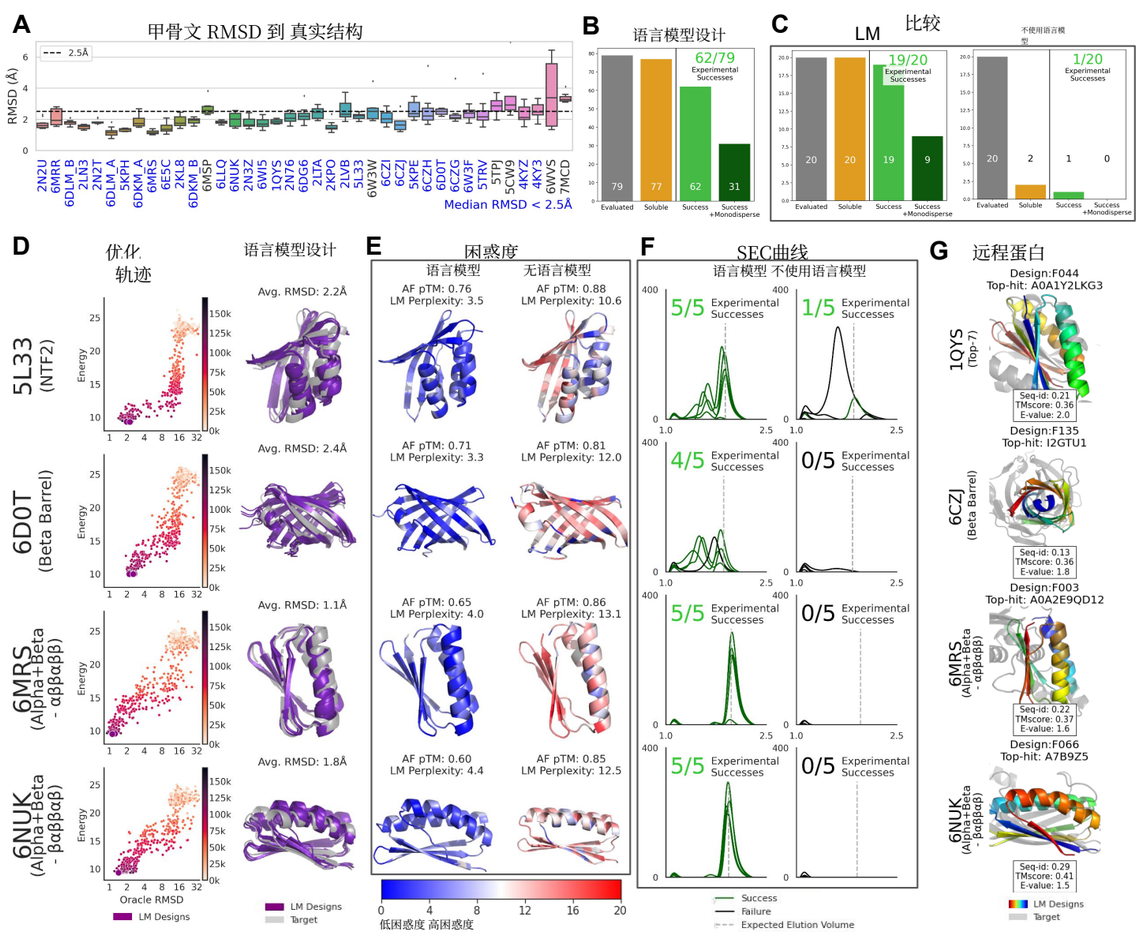

图 *2.*序列设计用于新设计结构。(**A**)使用计算机模拟甲骨文对新设计目标集的总体评估。设计结构（甲骨文预测）和目标结构之间的C-α原子的均方根偏差（RMSD）图示为每个目标的优化目标排名前10的设计。目标按长度递增排序。语言模型生成的序列在测试集中对于大多数新设计骨架预计可以折叠成目标结构。（33/39的中位RMSD为 < 2.5 ˚A）。(**B**) ESM设计的实验结果。总共选择了79个设计，涵盖6个新设计骨架目标，依据多个标准，包括序列新颖性和对有趣基序的人工检查。设计在可溶且在大小排阻色谱法（SEC）中预期洗脱体积出现峰值时被视为成功。当唯一峰位于预期洗脱体积时，设计被分类为单分散。整体上，78%成功，39%为单分散。(style id='21'>C) 比较使用和不使用语言模型的设计的实验结果。对于四个目标，按优化目标选择了200个设计中排名前5的进行实验评估。整体上，有语言模型的设计成功率为95%，而没有语言模型的大多数设计由于不可溶而失败。(style id='23'>D)（左）优化轨迹显示由语言模型指定的能量与目标的RMSD随MCMC优化过程的变化。能量减少并向低RMSD汇聚。（右）每个轨迹结束时按照能量选择的前5个设计的可视化。(style id='25'>E) 设计的语言建模困惑度。语言模型设计被语言模型视为可能，而基线设计的高困惑度表明它们的序列被视为不太可能。这与实验成功相吻合。(style id='27'>F) 使用和不使用语言模型的设计之间SEC痕迹的比较。绝大多数语言模型设计是可溶的，并在预期洗脱体积处有峰值；相比之下，少数没有语言模型的设计是可溶的。(style id='29'>G) 一组额外的成功语言模型设计在已知天然蛋白质的背景下是新颖的。展示了四种不同骨架的示例，设计与针对天然蛋白质的序列搜索的最高重要性命中的预测结构重叠。在每种情况下，最接近的天然序列具有低序列相似性（<0.3）和预测的不同拓扑结构。

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pairs in the structure (9, 30). We fit a linear projection that takes the attention between two positions in the protein sequence and outputs a distribution over pairwise distances (Fig. 1C). This maps an internal attention state of 660 dimensions into 18 inter-residue distance bins. Because of the limited number of parameters (660 per distance bin for a total of 11,898 including a bias for each distance bin), far too few to represent the immense complexity of possible protein structures, the output can be interpreted as a projection of the structure captured by the internal states of the model. The projection defines an energy landscape (a function of the representation states of the language model rather than a physical energy) that can be used to evaluate the compatibility of any given structure with the representation of the sequence produced by the language model. Application to the *de novo* target set shows understanding of existing *de novo* proteins (Table S1 and Figs. S2 and S3).

Together, the models of sequence, and structure given sequence, specify a generative model of proteins defined by the language model. The sequence model assigns a probability to any sequence, by giving a probability for each amino acid at every position in the protein (Fig. 1D). For natural proteins these probabilities reflect the functional effects of mutations, structural preferences of amino acids, and aspects of biochemical function (31). The projection of structure gives a compatibility between the language model's representation of a sequence with a three dimensional structure (Fig. 1E). In this work, we consider these models to specify a generative model for protein design:

$$p(sequence, structure) = p(structure|sequence)p(sequence)$$

For fixed backbone design, protein sequences are generated by taking low temperature samples from the conditional distribution specified by the language model via Markov chain Monte Carlo (MCMC) with simulated annealing (Fig. 1F, Appendix A.3.1). Free generation removes the constraint on structure entirely and generates new proteins by sampling from the joint distribution of sequence and structure specified by the language model. A blocked Gibbs sampling approach is introduced which alternates between sampling a new structure conditioned on the current sequence, and sampling a new sequence conditioned on the current structure (Fig. 1G, Appendix A.3.3). An example free generation trajectory is shown in Fig. 1H. As the temperature is lowered, the trajectory proceeds from a phase where it samples a range of possible topologies before narrowing into a single topology that is refined into a confidently predicted structure in the final stage of optimization.

We perform extensive experimental testing of a total of 228 designs from the language model. Designs are considered a success that are well expressed, soluble, and pass a size exclusion chromatography (SEC) test for molecular (hydrodynamic) radius indicative of a properly-folded monomeric species (Appendix A.7). Experimental success of a significant fraction of the generated proteins, along with independent computational evaluation of the structures, demonstrates that language models are able to access a design space beyond that of natural proteins.

Language models design sequences that fold to *de novo* structures

Fixed backbone design evaluates generation of sequences to realize a specified target structure. Use of *de novo* designed structures as targets requires the model to generalize beyond natural proteins, necessitating the use of more general patterns for the design of structure. Success at this task would indicate that the model has an understanding of the underlying design principles of protein structure generalizing to structures not encoded by natural sequences.

Across the test set of 39 artificially designed *de novo* protein structures, fixed backbone designs generated by the language model are predicted to closely match the target structures by the AlphaFold high-resolution structure prediction oracle. We generate 200 different designs for each of the *de novo* target structures (Appendix A.4). The generative model succeeds in producing low-RMSD designs for the vast majority of targets in the *de novo* test set (Fig. 2A). Subsetting to the best 10 of 200 designs by the language model's optimization objective, median RMSD is < 2.5Å for 84% (33/39) of targets and minimum RMSD is < 2Å for 90% (35/39) of targets. Structures are also confidently predicted, with median pTM $>$ 0.7 for 56% (22/39) and maximum pTM > 0.7 for 90% (35/39). Average sequence identity with the targets is low (22%), indicating that the language model is finding solutions to the design problem that differ from the original sequence.

Generated proteins have high overall experimental success rates in the laboratory. We ran an additional set of fixed backbone design trajectories to explore the diversity of design motifs generated from the model. A total of 79 fixed backbone designs spanning 6 *de novo* targets were selected from a pool including the additional trajectories for evaluation by a variety of criteria including the presence of interesting structural motifs (Appendix A.6). Out of this set of experimentally tested proteins, 97% (77/79) were soluble, 78% (62/79) were successful, passing a SEC test for the presence of a peak at the expected elution volume indicating a folded monomeric species, and 39% (31/79) were monodisperse, exhibiting only a single SEC peak at the expected elution volume (Fig. 2B). Successes span a range of topologies, including a success for length 182 *de novo* TIM-barrel 6WVS which has a highly idealized symmetric structure (Fig. S4). Across the set of experimental successes, sequence identity to the original sequence of the target structure is low (mean = 24%), which suggests that the language model is exploring a new design space for the target structures.

We perform a controlled experiment to understand the role of the language model in experimental success of designs. For comparison we use AlphaFold as a model of the probability of structure given sequence. For a set of four fixed backbone *de novo* targets with distinct folds, we generate 200 designs using each method, with the top 5 by optimization objective for each method selected for experimental evaluation (Appendix A.3). Experimentally, 95% (19/20) of language model sequence designs and 5% (1/20) designs without a language model were successful (Fig. 2C). Augmenting AlphaFold with an n-gram prior, fails to rescue the designs (0% success rate, 0/20) (Tables S3 and S4).

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结构中的对 (9，30)。我们拟合一个线性投影，它将蛋白质序列中两个位置之间的注意力进行处理，并输出成对距离的分布 (图。1C)。这将660维的内部注意力状态映射到18个残基间距箱中。由于参数数量有限（每个距离箱660个，总计为11,898个，包括每个距离箱的偏差），远远不足以表示可能蛋白质结构的巨大复杂性，因此输出可以被解释为模型内部状态捕获的结构投影。该投影定义了一个能量景观（一个表示语言模型表征状态的函数，而不是物理能量），可以用来评估任意给定结构与语言模型生成的序列表示之间的兼容性。对新设计 目标集的应用显示出对现有新设计 蛋白质的理解（表 S1和图。S2 和S3）。

模型的序列和给定序列的结构一起指定了由语言模型定义的蛋白质生成模型。序列模型为任何序列分配一个概率，通过给出蛋白质每个位置上每个氨基酸的概率（图。1D）。对于天然蛋白质，这些概率反映了突变的功能效应、氨基酸的结构偏好以及生化功能的一些方面（31）。结构的投影给出了语言模型的序列表示与三维结构之间的兼容性（图。1E）。在这项工作中，我们考虑这些模型来指定蛋白质设计的生成模型：

$$p(sequence, structure) = p(structure|sequence)p(sequence)$$

对于固定主链设计，蛋白质序列是通过在马尔可夫链蒙特卡洛（MCMC）和模拟退火的条件下，从语言模型指定的条件分布中提取低温样本生成的（图。1F，附录 A.3.1）。自由生成完全移除了对结构的约束，并通过从语言模型指定的序列和结构的联合分布中抽样来生成新蛋白质。引入了一种阻塞吉布斯采样方法，该方法交替进行：在当前序列的基础上抽样新结构，以及在当前结构的基础上抽样新序列（图。1G，附录 A.3.3）。图中显示了一个自由生成轨迹的示例。1H。随着温度的降低，轨迹从一个抽样多种可能拓扑结构的阶段开始，随后收敛到一个拓扑结构，并在优化的最后阶段精炼为一个自信预测的结构。

我们对来自语言模型的228个设计进行了广泛的实验测试。被认为成功的设计具备良好的表达、可溶性，并通过大小排斥色谱（SEC）测试，其分子（流体动力学）半径能指示正确折叠的单体物种（附录A.7）。生成蛋白质中相当一部分的实验成功，以及对其结构的独立计算评估，表明语言模型能够进入超越天然蛋白质的设计空间。

语言模型设计的序列折叠成新设计结构

固定主链设计评估序列的生成，以实现特定的目标结构。使用新设计结构作为目标要求模型超越天然蛋白质进行概括，因此需要采用更一般的模式来设计结构。该任务的成功表明模型理解蛋白质结构的基础设计原则，并能够概括出未被天然序列编码的结构。

在39个人工设计的新设计蛋白质结构的测试集中，由语言模型生成的固定骨架设计被预测与目标结构紧密匹配，使用的是AlphaFold高分辨率结构预测甲骨文。我们为每个新设计目标结构生成200个不同的设计（附录A.4）。生成模型在绝大多数目标中成功地产生了低均方根偏差设计，在新设计测试集中（图。2A）。通过语言模型的优化目标对200个设计中最好的10个进行子集筛选，中位均方根偏差为 < 2。5 ˚A，对于84%（33/39）的目标，最小均方根偏差为 < 2 ˚A，对于90%（35/39）的目标。结构的预测也具有很高的信心，中位pTM >为0.7，适用于56%（22/39），最大pTM > 0 为0.7，适用于90%（35/39）。与目标的平均序列相似性较低（22%），这表明语言模型正在寻找与原始序列不同的设计问题解决方案。

生成的蛋白质在实验室中的整体实验成功率很高。我们进行了额外的一组固定骨架设计轨迹，以探索从模型生成的设计基序的多样性。总共选择了79个固定骨架设计，涵盖6新的设计 目标，从一个池中选择，包括附加的轨迹，以多种标准进行评估，包括有趣的结构基序的存在 （附录A.6）。在这组经过实验测试的蛋白质中，97%（77/79）是可溶的，78%（62/79）是成功的，经过SEC测试，显示出在预期洗脱体积处存在峰，指示折叠的单体物种，39%（31/79）是单分散的，只在预期洗脱体积处展示出一个SEC峰（图。2B）。成功的设计横跨多种拓扑结构，包括一个长度为182的成功新设计 TIM桶 6WVS，其具有高度理想化的对称结构 (图。S4)。在这组实验成功中，目标结构原始序列的序列同一性较低（平均 = 24%），这表明语言模型正在为目标结构探索新的设计空间。

我们进行了一项受控实验，以了解语言模型在设计实验成功中的作用。作为比较，我们使用AlphaFold作为给定序列的结构概率模型。对于一组四个固定主链新设计目标，这些目标具有不同的折叠，我们使用每种方法生成200个设计，并为每种方法选择优化目标前5名进行实验评估（附录A.3）。实验结果显示，95%（19/20）的语言模型序列设计成功，而没有语言模型的设计成功率为5%（1/20）（图.2C）。通过n-gram先验增强AlphaFold，未能挽救这些设计（成功率为0%，0/20）（表格S3 和 S4）。

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Language models generalize beyond natural proteins

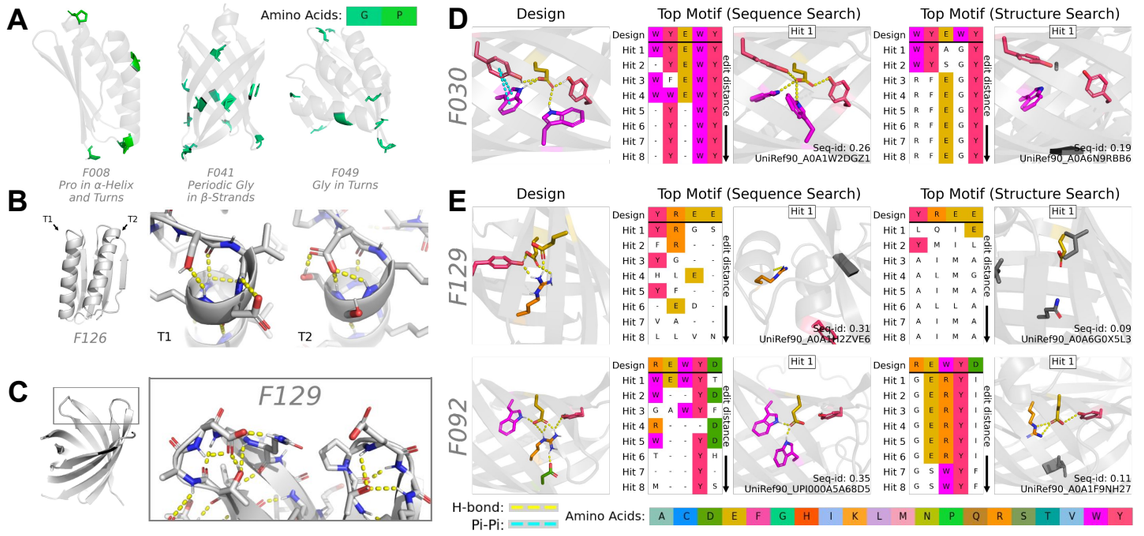

Figure 3. Language models materialize deep patterns of protein design, generating native-like and de novo *motifs.* (**A**) Placement of proline or glycine within three different designed proteins induce curvature on alpha-helices, beta-sheets, and turns. (**B,C**) Hydrogen bond networks in turns. (**B**) Helix dipole capping forms hydrogen bonds to obscure polar backbone atoms in the final helix turn. (**C**) Hydrogen bond networks formed in turns involving beta-sheets. (**D,E**) Comparison of motifs in designed and natural proteins. Designs (left) are compared against the nearest motif in natural proteins found by sequence search (center), and structure search (right). Hits are sorted by matching amino acids only at motif positions. (**D**) Example of a hydrogen bond motif used in one of the designs. Sequence matches are found that have the same motif in aligned positions. However the surrounding sequence context is significantly different, having 26% sequence identity. (**E**) Examples of possible *de novo* hydrogen-bond networks. Not only is the sequence context different, the motif itself is not present in the aligned positions of any matching natural sequences or structures.

Language model perplexity separates success from failure across both design methods. MCMC trajectories for the language model funnel to low RMSD with decreasing energy, with average RMSD values ranging from 1.1Å to 2.4Å (Fig. 2D). Notably, while AlphaFold confidently predicts the structures of language model designs, the language model does not assign high sequence likelihoods to AlphaFold designs. Language model perplexities of select AlphaFold-designed sequences range from 10.6 to 13.1 (Fig. 2E), significantly higher than the average *de novo* target sequence perplexity of 6.7. Other metrics have limited ability to identify experimental success (Fig. S5 and Table S4): the Rosetta all-atom energy function for modeling and design (32, 33) judges both sets to be good designs, packing metrics are similar but slightly favor the (unsuccessful) AlphaFold designs, while hydrophobicity and SAP score favor the language model designs. Recently autoregressive inverse folding models directly conditioned on the target structure have demonstrated high experimental success rates in the laboratory (15). We generated sequences with ProteinMPNN and ESM-IF1 (14). Both models achieve high local confidence pLDDT ($>$ 90 mean). Their ESM pseudo-perplexity is 5.76 and 5.79 respectively, higher than ESM designs and significantly lower than AlphaFold designs (Table S2), in line with high experimental success rates reported for those methods.

Experimental evaluation of both design sets (with and without the language model) indicates that 19/20 of language model designs are successful and 9/20 are monomeric (Fig. 2F). Target 6D0T has no monomeric designs from the language model, though the ground truth *de novo* sequence was also found to not be monomeric, when tested as a positive control (Appendix A.7). Designs without a language model largely fail due to insolubility.

Including the results of the controlled comparison, and the larger set of designs evaluated, the language model produced experimentally successful designs for all of a total of 8 *de novo* backbones. One possibility is that language model designs succeed because the model retrieves a protein similar to the target from its training set. To rule this out, we analyze the overall set of 81 experimental successes. Each design is searched against UniRef90 (which fully includes the sequences used to train the language model) to identify similar sequences (Appendix A.5). For 17 successful designs spanning 4 backbones, there are no significant (E-value $<$ 1) sequence matches whatsoever in the training set. Four of these are shown in Fig. 2G. Of the remaining 64, sequence identity to the nearest sequence match is only 27% on average, and is $<$ 30% for 41 of the 64, spanning each of the 8 tested backbones. This suggests that the model is not solving the design problem by retrieving similar sequences it has memorized.

语言模型的泛化超越了天然蛋白质

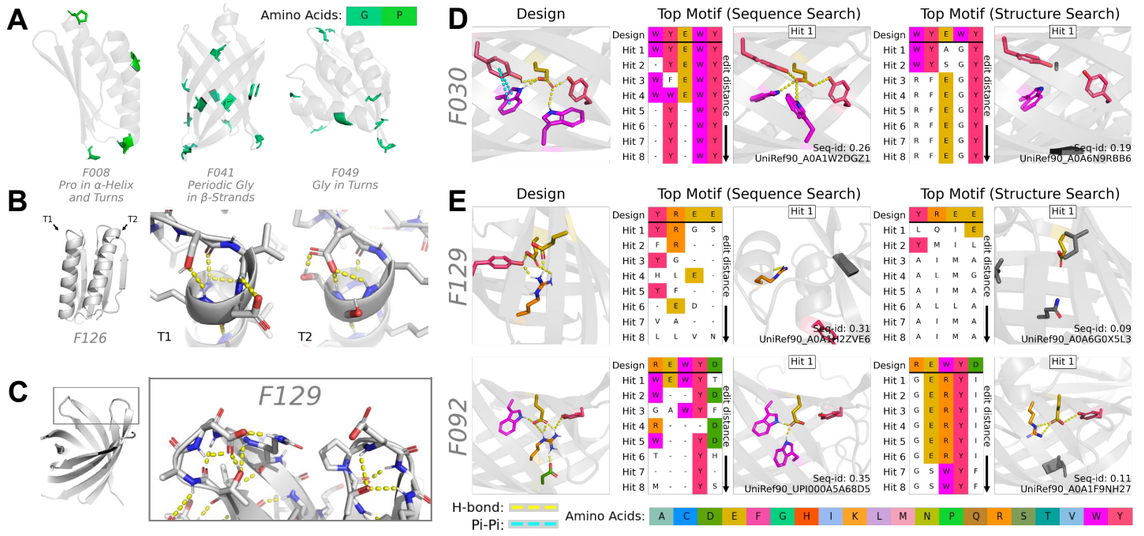

图 *3.*语言模型具体化了蛋白质设计的深层模式，生成类似天然的和新设计模体。(**A**) 脯氨酸或甘氨酸在三种不同设计蛋白质中的位置引发了α-螺旋、β-折叠和转角的弯曲。(**B,C**) 转角中的氢键网络。(**B**) 螺旋偶极末端形成氢键以遮蔽最终螺旋转角中的极性主链原子。(**C**) 涉及β-折叠的转角中形成的氢键网络。(**D,E**) 设计与天然蛋白质中模体的比较。设计（左）与通过序列搜索（中）和结构搜索（右）找到的天然蛋白质中最接近的模体进行了比较。命中按模体位置匹配的氨基酸仅进行排序。(**D**) 用于其中一个设计的氢键模体示例。在对齐位置中发现的序列匹配具有相同的模体。然而，周围的序列上下文有显著不同，序列相似性为26%。（**E**）可能的新设计氢键网络的示例。不仅序列上下文不同，模体本身在任何匹配的天然序列或结构的对齐位置中都不存在。

语言模型困惑度将成功与失败区分开来，适用于两种设计方法。语言模型的MCMC轨迹随着能量降低而汇聚到低均方根偏差，平均均方根偏差值范围从1.1 ˚A到2.4 ˚A（图2D）。值得注意的是，尽管AlphaFold自信地预测语言模型设计的结构，但语言模型对AlphaFold设计的序列赋予的高可能性不高。选定的AlphaFold设计序列的语言模型困惑度范围从10.6到13.1（图。2E），显著高于平均的新设计目标序列困惑度6.7。其他指标识别实验成功的能力有限（图。S5 和表S4）：用于建模和设计的Rosetta全原子能量函数（32, 33）认为这两组都是良好的设计，包装指标相似，但稍微偏向（不成功的）AlphaFold设计，而疏水性和SAP得分则偏向语言模型设计。最近，自回归逆折叠模型直接基于目标结构进行了条件处理，展示了在实验室中的高成功率（15）。我们使用ProteinMPNN和ESM-IF1生成了序列（14）。这两个模型都实现了高局部置信度pLDDT（$>$ 90 均值）。它们的ESM伪困惑度分别为5.76和5.79，高于ESM设计且显著低于AlphaFold设计（表S2），与这些方法报告的高实验成功率一致。

对两个设计集（有语言模型和没有语言模型）进行的实验评估表明，19/20 的语言模型设计成功，9/20 为单体 (图 2F)。目标 6D0T 从语言模型中没有单体设计，尽管真实值 新设计序列在作为阳性对照时也被发现为非单体（附录A.7）。没有语言模型的设计大多由于不溶解性而失败。

包括受控比较的结果以及评估的更大设计集，语言模型实验性地为总共8个新设计骨架生成了成功的设计。一种可能性是，语言模型设计之所以成功，是因为模型从训练集中检索到与目标相似的蛋白质。为了排除此可能性，我们分析了81个实验成功案例的整体集。每个设计都与UniRef90（该数据库完全包括用于训练语言模型的序列）进行了比对，以识别相似序列（附录A.5）。在涵盖4个骨架的17个成功设计中，训练集中根本没有显著的（E值 $<$ 1）序列匹配。这其中有四个展示在图2G中。在剩余的64个设计中，平均与最近序列匹配的序列相似性仅为27%，并且对于64个设计中的41个，相似性仅为$<$ 30%。这表明模型并不是通过检索它所记住的相似序列来解决设计问题的。

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To further understand whether the model is using homology at the threshold of detection by sequence similarity, we obtained AlphaFold predicted structures of hits, including those that do not reach the significance cutoff (Appendix A.5; Fig. S6). For 19/81 experimental successes top Jackhmmer hits are not structural matches to the design. For 19 designs spanning 4 backbones, the top-10 jackhmmer hits (including those that do not reach the significant threshold) all have TM-score < 0.6. For 8 of those designs spanning the same 4 backbones, top-10 hits are all likely to be a different fold (TMscore < 0.5). This suggests that while in some cases the model is able to use sequence homology at the threshold of detection, there are also cases where it appears to have generalized beyond that, further evidence that in many cases the language model is generating novel solutions to the design problem which differ both from the ground-truth sequence, and natural proteins.

Language models materialize deep patterns of protein design

Generated proteins show evidence of using deep patterns of the design of protein structure. These patterns occur in the form of structural motifs used in the design of natural proteins applied in significantly differing sequence contexts, as well as the formation of motifs which cannot be found in related structures. Two well-studied ways that sequence determines structure are through amino acids that constrain backbone-geometry, and through the role of chemically diverse side chains in determining the intermolecular forces that stabilize a protein's particular folded conformation. Two amino acids which influence backbone geometry are proline and glycine. These two amino acids add flexibility to and bend protein backbones, respectively. In three example designs, the language model places these residues to induce curvature in various secondary structure elements: a proline bends an alpha-helix, regular placement of glycines in beta-sheets promote the flexibility to form a beta-barrel, and all but one glycine are placed in loops in an NTF2 design (Fig. 3A). A side chain based motif present through fixed backbone designs is helix dipole capping, where side chains of amino acids at the ends of alpha-helices obscure otherwise exposed polar backbone atoms in the final alpha-helix turn (Fig. 3B). A second side chain based motif is hydrogen-bond networks in bulge-containing beta-turns, which are present in fixed backbone designs for beta-barrels, such as 6D0T and 6CZJ (Fig. 3C). This and to a larger extent the periodic glycines in beta-strands in Fig. 3A were identified as natural motifs that enabled successful *de novo* design of the target beta-barrel in (34).

Designs also exhibit complex hydrogen bonding networks. Some design successes include hydrogen bonding networks between four or more polar and even charged residues in the interior of structures. Design of buried polar and charged interactions is difficult due to the geometric constraints of energetically satisfying such interactions (35). Notably, the bond networks shown span a range of intermolecular force categories: among predicted structures, F129, a beta-barrel, contains a salt-bridge, F025 contains a pi-cation bond, and F030 contains a T-shaped pi-pi interaction (Fig. S7). The original designs for the examples shown have purely hydrophobic interiors. While these hydrogen bonding networks can only be fully confirmed by high-resolution structural studies, the biophysical properties observed (high yield of monodisperse protein with the expected retention volume) is consistent with their accuracy, since inaccurate placement of these residues is likely to lead to mis-folding and aggregation.

The hydrogen-bonding networks with polar residues are realized in new sequence contexts, indicating a strong form of generalization beyond the sequences used for training the model. We retrieve the most similar aligned sequences via Jackhmmer search of UniRef90 and similar, aligned structures via Foldseek (36) search of AlphaFold DB (37). Returned sequences are all sorted by minimum edit distance at aligned motif positions, and the closest matching motif is shown. (Appendix A.5.4). For generated protein F030 (Fig. 3D, Fig. S7), sequence search does recover a natural protein with this motif in aligned positions. However the surrounding sequence context in the design is dissimilar, having a full-sequence identity of 26%. For F129 and F092 (Fig. 3E, Fig. S7), not only does the surrounding sequence context have low sequence identity, the motif itself is not present in the aligned positions of any matching natural sequences or structures. Use of these motifs in fixed backbone designs is a remarkable form of generalization, since the model is applying them in new sequence contexts, and structures that are distinct from natural proteins.

Language models generate novel structures and sequences

Language models generate new protein sequences that differ significantly from natural sequences. We sample a large set (N = 25,000) of proteins of fixed length (L = 100) without constraint on the structure. The blocked Gibbs sampling method which traverses the joint landscape of sequence and structure provides a more diverse set of proteins than previous unconstrained generation methods (Table S5).

Generations cover a variety of topologies with sequences overall dissimilar from natural proteins. Structures are predicted for all generated sequences using Alphafold, and generations are projected into two dimensions using t-SNE based on their pairwise structural distance measured by TM-score (Fig. 4A). In a hierarchical clustering of the structures, 7,663 distinct clusters were identified at a TM-score threshold of 0.75. The distribution of the generated secondary structures reveals a range of patterns with 52% of generations containing mostly alpha helices, 22% containing mostly beta sheets, and 28% a mix of alpha helices and beta sheets (Fig. 4B). A large fraction of the generations are well predicted by the oracle (median pLDDT = 84.49, 70% pLDDT > 70; Fig. 4C).

Many of the generations are distant in sequence from natural proteins. We measure the distance of generated sequences from natural proteins by searching each generation against the 200M natural sequences in AlphaFold DB (37). This also enables comparison of the structure of the nearest sequence match to that of the generated protein. Overall the language model generates proteins which

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为了进一步了解模型是否在序列相似性的检测阈值上使用同源性，我们获得了AlphaFold预测的命中结构，包括那些没有达到显著性截止值的结构（附录A.5；图S6）。在19/81个实验成功案例中，排名最高的Jackhmmer命中并未与设计的结构匹配。在覆盖4个骨架的19个设计中，排名前10的Jackhmmer命中的TM-score均为 < 0.6。对于这些设计中涵盖相同4个骨架的8个设计，排名前10的命中都可能是不同的折叠（TM-score < 0.5）。这表明，尽管在某些情况下模型能够在检测阈值上使用序列同源性，但也存在一些案例显示它似乎已经超越了这一点，进一步证据表明在许多情况下，语言模型生成了与真实序列和天然蛋白质有所不同的新颖设计解决方案。

语言模型实现蛋白质设计的深层模式

生成的蛋白质显示出使用蛋白质结构设计的深层模式的证据。这些模式以结构模体的形式出现，应用于在显著不同的序列背景下设计的天然蛋白质，以及形成在相关结构中无法找到的模体。序列决定结构的两个广泛研究的方法是通过氨基酸限制主链几何形状，以及化学多样化侧链在确定稳定蛋白质特定折叠构象中的作用。影响主链几何的两种氨基酸是脯氨酸和甘氨酸。这两种氨基酸分别增加了蛋白质主链的柔韧性和弯曲性。在三个示例设计中，语言模型放置这些残基以诱导各种次级结构元素的曲率：脯氨酸弯曲α-螺旋，甘氨酸在β-折叠中的规律性放置促进了形成β-桶的灵活性，而在NTF2设计中除了一个甘氨酸外，其余的都放置在环中（图3A）。一个基于侧链的模体存在于固定骨架设计中，称为螺旋偶极子封闭，其中α-螺旋末端氨基酸的侧链遮蔽了最终α-螺旋转弯中暴露的极性主链原子（图3B）。第二个基于侧链的模体是在含有肿块的β-回转中的氢键网络，这些网络存在于β-桶的固定骨架设计中，如6D0T和6CZJ（图。3C）。这个以及在图中的β-链中的周期性甘氨酸在更大程度上被识别为促进成功新设计目标β-桶的天然模体（34）。

设计还体现出复杂的氢键网络。一些设计成功的案例包括在结构内部四个或更多极性甚至带电残基之间的氢键网络。由于能量上满足这些相互作用的几何约束，设计埋藏的极性和带电相互作用是困难的（35）。值得注意的是，所示的键网络跨越了一系列分子间力类别：在预测的结构中，F129，一个β-桶，包含盐桥，F025包含π-阳离子键，F030包含T型π-π相互作用

带有极性残基的氢键网络在新的序列上下文中被实现，显示出一种超越用于训练模型的序列的强烈泛化形式。我们通过Jackhmmer查询UniRef90检索最相似的对齐序列，并通过Foldseek查询相似的对齐结构（36）在AlphaFold DB（37）中。返回的序列全部按对齐的基序位置的最小编辑距离排序，并显示出最接近的匹配基序。（附录A.5.4）。对于生成的蛋白质F030（图.3D，图.S7），序列搜索确实在对齐位置中找到了一个带有此基序的天然蛋白质。然而，设计中的周围序列上下文是不相似的，具有26%的全序列相似性。对于F129和F092（图.3E，图.S7），周围的序列上下文不仅具有低序列相似性，而且该基序本身在任何匹配的天然序列或结构的对齐位置中都不存在。在固定骨架设计中使用这些基序是一种显著的泛化形式，因为模型在新的序列上下文以及与天然蛋白质不同的结构中应用它们。

语言模型生成新的结构和序列

语言模型生成的新蛋白质序列与天然序列有显著不同。我们没有对结构施加约束，采样了一大组 (N = 25,000) 固定长度 (L = 100) 的蛋白质。阻塞吉布斯采样方法遍历了序列和结构的联合景观，提供了比以前的无约束生成方法更加多样化的蛋白质集合 (表 S5)。

生成的序列涵盖了多种拓扑，整体上与天然蛋白质明显不同。使用AlphaFold预测所有生成序列的结构，并基于TM-score测量的成对结构距离，使用t-SNE将生成结果投影到二维 (图.4A)。在结构的层次聚类中，识别出了7,663个独特的簇，TM-score阈值为0.75。生成的次级结构分布显示出多种模式，其中52%的生成结果主要包含α螺旋，22%主要包含β-片，28%则是α螺旋和β-片的混合 (图.4B)。大量的生成结果得到了甲骨文的良好预测（中位数pLDDT = 84.49，70% pLDDT > 70；图. 4C）。

许多生成的序列与天然蛋白质的距离较远。我们通过在AlphaFold数据库中搜索与200M天然序列的每个生成序列来测量生成序列与天然蛋白质的距离（37）。这也使得可以比较最近序列匹配的结构与生成的蛋白质的结构。总体而言，语言模型生成的蛋白质

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Language models generalize beyond natural proteins

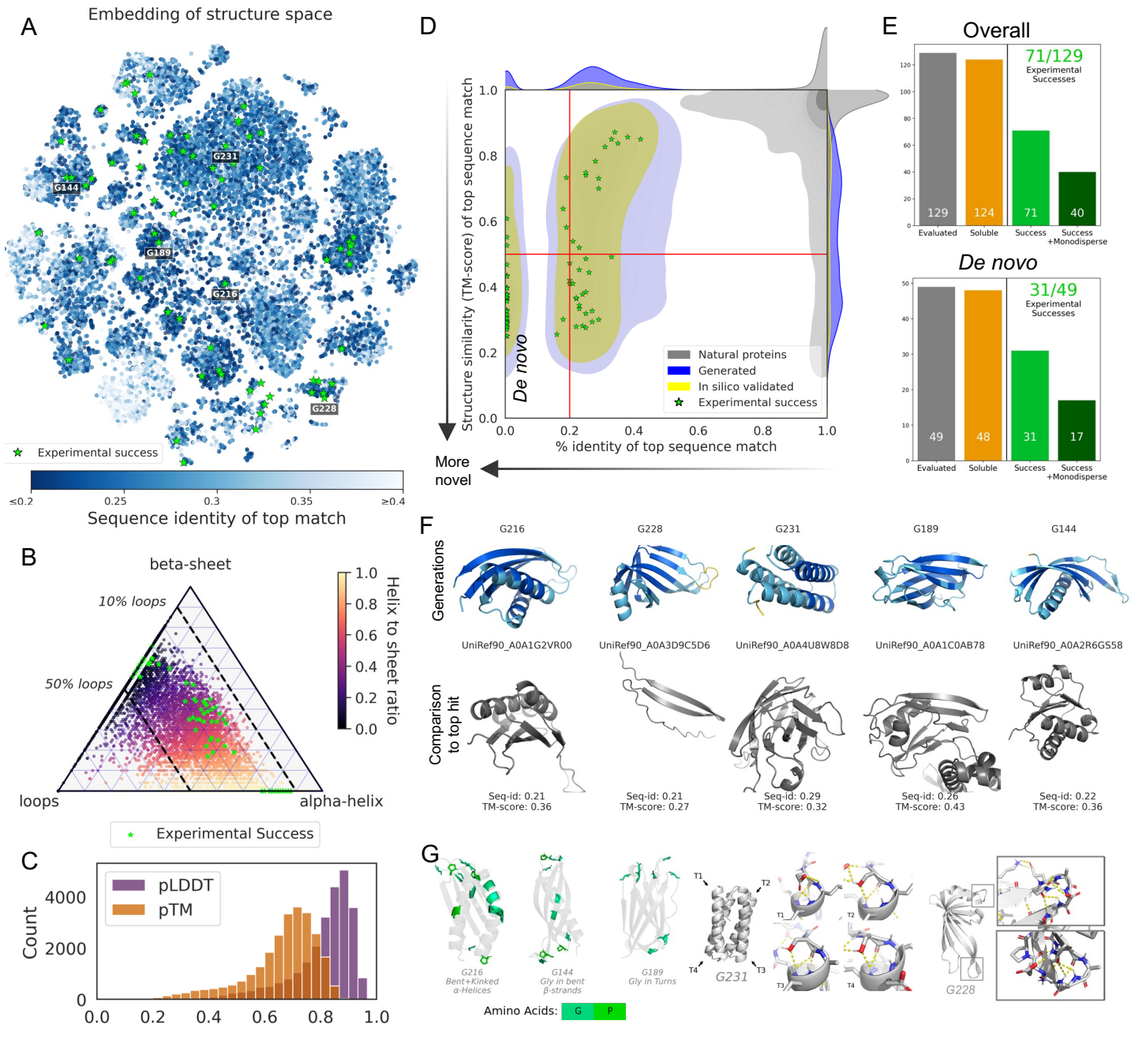

Figure 4. Language models generate novel structures and sequences. (**A**) Embedding of the structural space spanned by the generated proteins using t-SNE. Color indicates sequence identity to the best matching native sequence. A large fraction of the space has low sequence identity to natural proteins with 16% of generations having no significant sequence match to a natural protein. Designs that succeeded in experimental evaluation are indicated with green stars. (**B**) Distribution of secondary structure for generations. Experimental successes (green) are observed over differing compositions of secondary structure. (**C**) Distributions of pLDDT and pTM indicate designs are well predicted (median pLDDT of 84.5) by the *in silico* oracle. (**D**) Density plot of sequence and structural similarity to natural proteins. For each generated protein the best matching native sequence is retrieved from AlphaFoldDB. Each generated protein is plotted by its sequence similarity (x-axis) and structure similarity (y-axis) to the match, with hits that do not pass the significance threshold marked at zero on the x-axis. Generated proteins occupy a part of the space distinct from natural proteins, with a fraction having minimal sequence similarity to natural proteins (lower left quadrant). Designs passing *in silico* filters and experimental successes are coextensive with the overall distribution of generations. (**E**) Overall outcome of experimental evaluations. The majority of tested designs (55%) passed the solubility test and had an elution volume peak in the correct confidence interval (top). Additionally a high fraction (63%) of the evaluated proteins distant from natural sequences are successful (bottom). (**F**) Predicted structures of six experimental successes (top). Structures are aligned against the oracle predicted structure of their top significant hit from a sequence search of natural proteins (bottom); in all examples the predicted topology is different. (**G**) For generations in panel F, the same motifs as in Fig. 3A - 3C are shown: Proline and Glycine inducing curvature, helix capping, and hydrogen-bond networks in turns. Even on proteins with minimal similarity to naturals, the language model produces known motifs.

语言模型在天然蛋白质之外进行泛化

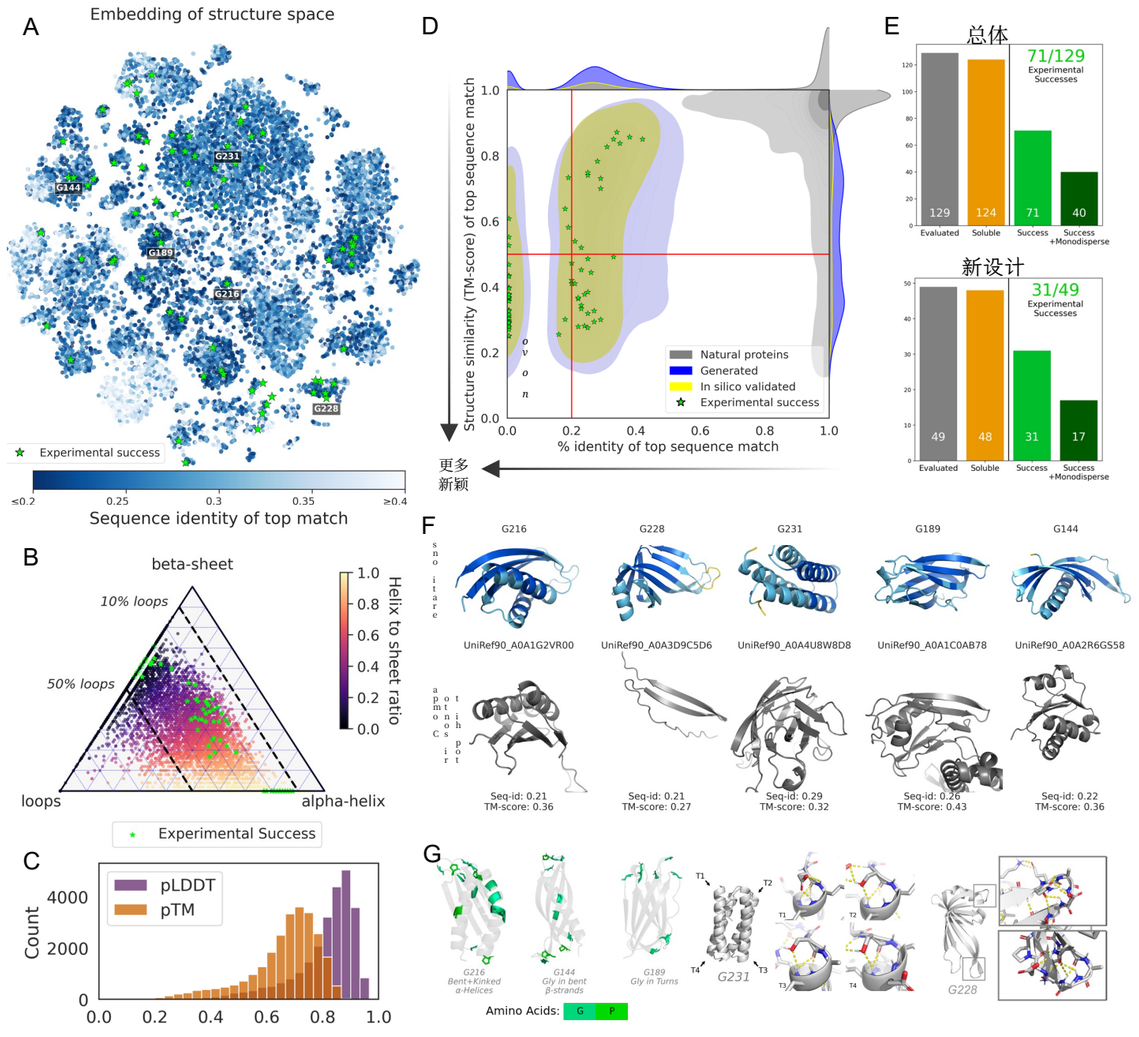

图 *4.* 语言模型生成新颖的结构和序列。（**A**）使用 t-SNE 嵌入生成蛋白质所跨越的结构空间。颜色指示与最佳匹配的天然序列的序列相似性。很大一部分空间与天然蛋白质的序列相似性较低，16% 的生成物与天然蛋白质没有显著的序列匹配。通过绿色星星标示实验评估中成功的设计。（**B**）生成物的二级结构分布。在不同的二级结构组成中观察到实验成功的设计（绿色）。（**C**）pLDDT 和 pTM 的分布表明设计是良好预测的（中位数 pLDDT 为 84.5）通过 计算机模拟 甲骨文。（**D**）序列和与天然蛋白质结构相似性的密度图。每个生成的蛋白质都从 AlphaFoldDB 检索出最佳匹配的天然序列。每个生成的蛋白质通过其序列相似性（x轴）和与匹配的结构相似性（y轴）进行绘制，未通过显著性阈值的命中在 x 轴标记为零。生成的蛋白质占据与天然蛋白质不同的空间的一部分，其中一些在与天然蛋白质的序列相似性较小（左下象限）。与通过 计算机模拟 筛选和实验成功的设计与生成的整体分布重合。（**E**）实验评估的整体结果。大多数测试设计（55%）通过了溶解度测试，并在正确的置信区间内具有洗脱体积峰（顶部）。此外，高百分比（63%）的与天然序列相距较远的评估蛋白质是成功的（底部）。（**F**）六个实验成功预测的结构（顶部）。这些结构与天然蛋白质的序列搜索中其最显著命中的甲骨文预测结构进行了对齐（底部）；在所有示例中，预测的拓扑结构都是不同的。（**G**）对于面板 F 中的生成物，显示与图 3A - 3C 中相同的基序：脯氨酸和甘氨酸诱导的曲率、螺旋封顶和转弯中的氢键网络。即使在与天然蛋白质相似性较小的蛋白质上，语言模型也能产生已知的基序。

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show a clear separation from the distribution of natural proteins, including a fraction that are distant from known proteins. Fig. 4D shows the distribution of similarity to known proteins, where each generation is plotted according to its sequence (x-axis) and structural (y-axis) similarity to its top sequence hit, with insignificant (E-value $>$ 1) hits placed at x=0 (16.6% of generations, in total). A large part of the distribution of generated proteins have structures different from those predicted for their nearest sequence match, further evidence that the model is not simply memorizing known proteins. A set of 15k natural proteins are also shown. Natural proteins cluster in the upper right corner, while generated proteins occupy a distinct part of the space. A significant fraction of the language model generated proteins (15.5%) have minimal similarity to natural proteins (lower left quadrant), with minimal sequence similarity (Seq-id $<$ 0.2) of the nearest match, and a predicted structure likely to be a different fold (TM-score $<$ 0.5).

A large fraction of the designs, including those that are distant from natural proteins, succeed experimentally. We selected a number of designs that passed our *in silico* quality filters for experimental evaluation. Out of the total set of generations, 20% (N = 5,198) passed the quality filters (Appendix A.4). A total of 129 of that set were expressed and evaluated, and 55% (71/129) were found to be experimentally successful. The 71 structures and their metrics are shown in Fig. S8, marked with a green star in Figs. 4A, 4B and 4D. Overall, 96% of the free generations that were evaluated were soluble, 55% had an elution volume peak in the correct confidence interval, and 30% were monodisperse (Fig. 4E top, Appendix A.7).

A high success rate is also observed for generations that are distant from natural proteins. For a set of 49 distant generations (Fig. 4D, bottom-left quadrant), 31 of 49 (63%) are successful in experimental evaluation. For these 31 experimental successes we perform a deeper analysis of similarity to natural proteins. We further search each against UniRef90 which provides comprehensive coverage of natural proteins and fully contains the language model's training set. Out of 31 distant designs, 16 have no significant (E-value $<$ 1) sequence matches whatsoever (Fig. S9). We obtain predicted structures for the top-10 sequence matches regardless of their significance. For 12 out of the 31 distant designs (5 of which are shown in Fig. 4F), none of the sequence matches are likely to have the same fold (TM-score $<$ 0.5) (Fig. S9). Predicted structures are generally confident (78% of predictions with pLDDT $>$ 70, average pLDDT = 81.24). Structural motifs observed in fixed backbone designs such as proline and glycine placement, helix capping, and hydrogen-bond networks, also appear within *de novo* generations (Fig. 4G). As a whole these results show that the language model generalizes outside the space of natural proteins to generate *de novo* proteins.

Evolutionary scale language models

Transformer protein language models were introduced by (7), which found evidence for the emergence of information about function and tertiary structure from the unsupervised training. Concurrent work at a smaller scale examined LSTM-based models (38–40). Large scale protein language models with billions of parameters have now been open sourced (8, 41–43). Generative use of language models has recently been explored by *in silico* studies (44, 45), and experimentally with confirmation of function for new sequences generated for existing protein families (22). To the best of our knowledge, experimentally validated work (20, 22, 46) with sequence based models has not crossed the threshold of $<$ 30% identity to natural proteins.

Conclusions

The classical picture of sequence space as being constituted by independent local evolutionary landscapes around each protein family would suggest that language models will be limited to a memorization of the space of natural proteins. Consistent with this, the information about structure that emerges in language models of proteins has been shown to depend on the evolutionary information available to the model during training, which would appear to be unencouraging for the potential to use language models generatively beyond natural proteins. Here we have presented evidence counter to this: language models generalize beyond natural protein families to generate proteins in a sequence space distant from natural proteins. Our results are the first time purely sequence based approaches have been shown to generalize beyond natural proteins, and are promising for sequence based generative artificial intelligence for *de novo* protein design, where we have demonstrated that there exists a space of *de novo* proteins, distant from those in nature, that are designable by generative language models.

This generalization points to a deeper structure underlying natural sequences, and to the existence of a deep grammar that is learnable by a language model. Our results suggest that the vast extent of protein sequences created through evolution contains an image of biological structure and function that reveals design patterns that apply across proteins, that can be learned and recombined by a fully sequence based model. The generalization beyond natural proteins does not necessarily indicate that language models are learning a physical energy. Language models may still be learning patterns, rather than the physical energy, but speculatively, in the limit of infinite sequence data, these patterns might approximate the physical energy. At a minimum the language model must have developed an understanding of the global coherence of a protein connecting the sequence and folded structure.

The existence of a deep grammar across proteins would explain the two observations which prima facie seem to contradict each other: that the understanding of natural proteins depends on evolutionary support in the training data, and also that the language models generalize outside of known natural protein families. If there is a power law distribution of learnable patterns, then it is expected that many protein structures will be designable with the common patterns that have the most support in the training data. At the same time, the frequency that patterns are observed in the training data will correspond with the learnability of the patterns. It will take greater amounts of training data, and model capacity, to learn rare patterns. This is consistent with the observation of both generalization to a new design space (that is accessible via the patterns that have been learned), and dependence on support in training

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与天然蛋白质的分布清晰分隔，包括一部分远离已知蛋白质的样本。图。4D显示了与已知蛋白质相似性的分布，其中每一代根据其序列（x轴）和结构（y轴）与其最优序列命中相似性进行绘制， insignificant (E值 $>$ 1) 命中被置于 x=0(总共 16.6% 的世代)。生成的蛋白质的结构与其最近序列匹配的预测结构不同，这进一步证明该模型并非仅仅记忆已知的蛋白质。还显示了一组 15k 的天然蛋白质。天然蛋白质聚集在右上角，而生成的蛋白质则占据了这一区域的不同部分。语言模型生成的蛋白质中有相当一部分（15.5%）与天然蛋白质的相似性极低（左下象限），与最近匹配的序列相似性最小（Seq-id $<$ 0.2），且预测的结构可能属于不同的折叠（TM-score $<$ 0.5）。

大部分设计，包括那些与天然蛋白质相距较远的设计，在实验上都成功。我们选择了一些通过我们计算机模拟质量过滤器进行实验评估的设计。在全部生成中，20% (N = 5,198) 通过了质量过滤器（附录A.4）。该组中总共有 129 个设计被表达和评估，其中 55% (71/129) 被发现是实验成功的。这 71 个结构及其指标显示在图中。S8，在图中以绿色星标记。4A， 4B 和 4D。总的来说，96% 的被评估的自由生成是可溶的，55% 的洗脱体积在正确的置信区间内出现峰值，30% 是单分散的（图。4E 顶部，附录 A.7）。

对于远离天然蛋白质的生成，观察到较高的成功率。对于一组49个远离的生成（图。4D，左下象限），49个中有31个（63%）在实验评估中成功。对于这31个实验成功案例，我们对其与天然蛋白质的相似性进行更深入的分析。我们进一步将每个案例与UniRef90进行对比，UniRef90提供了天然蛋白质的全面覆盖，并完全包含了语言模型的训练集。在31个远离的设计中，16个没有显著的（E值 $<$

1) 序列匹配 (图。S9)。无论其重要性如何，我们获得了前10个序列匹配的预测结构。在31个远端设计中，12个的序列匹配中（其中5个显示在图。4F），没有一个序列匹配很可能具有相同的折叠 (TM-score $<$ 0.5) (图。S9)。预测的结构通常是可信的 (78%的预测具有 pLDDT $>$ 70，平均 pLDDT = 81.24)。在固定骨架设计中观察到的结构基序，如脯氨酸和甘氨酸的位置、螺旋封头以及氢键网络，也出现在新设计 的生成中 (图。4G)。总体来看，这些结果表明，语言模型在天然蛋白质之外进行泛化，以生成新设计 蛋白质。

进化规模语言模型

变压器蛋白质语言模型由（7）引入，研究发现无监督训练产生了关于功能和三级结构的信息。在较小规模的同时工作中，检查了基于LSTM的模型（38–40）。大规模蛋白质语言模型参数达到数十亿-

参数现在已开源（8， 41–43）。生成语言模型的使用最近已通过计算机模拟 研究（44， 45），并通过确认生成的新序列对已有蛋白质家族的功能进行了实验证实（22）。截至目前，我们所知，基于序列的模型经过实验证实的工作（20， 22， 46）尚未达到与天然蛋白质的%身份阈值的 $<$ 30。

结论

经典的序列空间图景将其视为围绕每个蛋白质家族的独立局部进化景观，会暗示语言模型仅限于对天然蛋白质空间的记忆。与此一致的是，在蛋白质的语言模型中出现的关于结构的信息被证明依赖于在训练期间模型可用的进化信息，这似乎对生成性地使用语言模型超越天然蛋白质的潜力并不乐观。我们在这里提供了相反的证据：语言模型超越天然蛋白质家族对序列空间进行概括，以生成远离天然蛋白质的蛋白质。我们的结果是首次证明纯基于序列的方法能够超越天然蛋白质进行概括，并且对基于序列的生成性人工智能在新设计蛋白质设计方面充满希望，在那里我们证明存在一个远离自然界的新设计蛋白质空间，可以被生成语言模型设计。

这种概括指向了潜在于自然序列的更深层结构，及一个可由语言模型学习的深层语法的存在。我们的结果表明，通过进化生成的广泛的蛋白质序列包含生物结构和功能的图像，揭示了可跨蛋白质应用的设计模式，这些模式可以被完全基于序列的模型学习和重新组合。超越天然蛋白质的概括并不一定表明语言模型正在学习物理能量。语言模型可能仍在学习模式，而不是物理能量，但可以推测，在无限序列数据的极限下，这些模式可能接近物理能量。至少，语言模型必须发展出蛋白质的全球一致性的理解，连接序列和折叠结构。

在蛋白质之间存在深层语法将解释表面上似乎相互矛盾的两个观察：一是对天然蛋白质的理解依赖于训练数据中的进化支持，二是语言模型在已知的天然蛋白质家族之外进行概括。如果可学习模式存在幂律分布，那么预期许多蛋白质结构将能够使用在训练数据中获得最多支持的常见模式进行设计。同时，模式在训练数据中出现的频率将与这些模式的可学习性相关。学习稀有模式需要更多的训练数据和模型容量。这与观察到的新设计空间的概括（通过已学习的模式可访问）及对训练中支持的依赖是一致的。

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data (the proteins composed of rare patterns are harder to learn). If scaling laws continue to hold for protein language models we can expect their generative ability will continue to improve. As models and data scale, the existence of a learnable underlying grammar would predict that the rare patterns will be learned, expanding both the predictive ability of the model, and the design space that is accessible.

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数据（由稀有模式组成的蛋白质更难以学习）。 如果缩放法则继续适用于蛋白质语言模型，我们可以预期它们的生成能力将继续提高。 随着模型和数据的扩大，存在可学习的基础语法将预测稀有模式将被学习，从而扩展模型的预测能力和可访问的设计空间。

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A. Methods

A.1. Data

A.1.1. *De Novo* Target Set

A held-out set of *de novo* proteins is used for the task of design with a fixed target backbone. A diverse set ($N = 39$) of *de novo* structures from the Protein Data Bank (27) was selected, which span a range of lengths ($67 \leq L \leq 184$), folds (e.g. alpha-bundle, beta-barrel, NTF2, Rossman) and *de novo* design methods (26, 34, 47–56). See Fig. S1 for a visual display of all x-ray crystal structures comprising the *de novo* target set. These proteins were designed by humans, rather than by natural evolutionary processes. Importantly, these *de novo* proteins have meaningful structural differences from proteins belonging to natural folds. For example, NTF2 targets have unnatural binding pockets (55), beta-barrels are narrower and have short beta-turns (34), and some designs were entirely new folds at the time of their creation (47). Although these proteins are by definition distinct in both sequence and structure from natural proteins, each protein in the target set is queried against UniRef100 (28), which subsumes the training set of ESM2, and all sequences returned as matches by Jackhmmer search are excluded from the language model's training, see next section.

The Protein Data Bank Identification Codes (PDB IDs) of the *de novo* target set are:
`1QYS,2KL8,2KPO,2LN3,2LTA,2LVB,2N2T,2N2U,2N3Z,2N76,4KY3,4KYZ,5CW9,5KPE,5KPH,5L33,5TPJ,5TRV,6CZG,6CZH,6CZI,6CZJ,6D0T,6DG6,6DKM_A,6DKM_B,6DLM_A,6DLM_B,6E5C,6LLQ,6MRR,6MRS,6MSP,6NUK,6W3F,6W3W,6WI5,6WVS,7MCD`

A.1.2. Sequence dataset used to train ESM2

The language model used throughout this work is `ESM2_650M` (8). Therefore, all pretraining settings described in that work apply for the language model used here.

To test whether the language model's understanding of proteins generalizes from natural to *de novo* space, it is critical that the model did not see *de novo* proteins at train time. To this end, we first remove all sequences from ESM2's train set labeled as "artificial construct" on the UniProt (57) website, when `2021_04` was the most recent release (1,027 total proteins). To guard against mislabeled proteins, and to further remove sequences in the train set which may bear similarity to the target set, we additionally perform Jackhmmer (58) searches of each *de novo* sequence against UniRef100 `2021_04` with flags `--num-iter 1 {max}`, and remove all hits returned by the tool from ESM2's training set (58,462 proteins).

A.1.3. Structure projection dataset

The structure projection network was trained on a nonredundant dataset from PDB consisting of 15,051 proteins (structure release dates prior to 1 May 2018) used in Yang, et. al. (59).

A.1.4. Heldout set of natural proteins

A small ($N = 214$) set of natural proteins with structures in the PDB was selected to serve as a baseline comparison when evaluating language model *de novo* protein understanding in Figs. S1 and S2 and Table S1. The set is composed of PDBs available on July 2020 that have sequence identity < 0.3 to the dataset used to train the structure projection, according to mmseqs2 (60). A length filter of $50 \leq L < 250$ was applied to roughly match the length distribution of the *de novo* target set ($67 \leq L \leq 184$).

A.2. Models

A.2.1. ESM2

We use `ESM2_650M` (8) as our choice of large-scale protein language model throughout this work. ESM2 is a Transformer model trained via masked language modeling over the universe of known, natural protein sequences. At training time, protein sequences are shown to the model with a fraction of their residues masked, randomly permuted to a different amino acid, or left unmodified, according to standard BERT noise probabilities (7). The model's task is to predict those masked residues given bi-directional context of all unmasked residues in the input. ESM2 is trained only on natural protein sequences; sequences annotated as artificially constructed and sequences matched by sequence search with *de novo* target set queries were removed from the language model's training set Appendix A.1.2.

The language model is used to approximate $p(sequence)$ via the pseudo-likelihood (61). Let us first define the probability $p_\theta(x_i|x_{-i})$ over the possible amino acids at position i in sequence x, conditioned on the remainder of sequence. This conditional probability is obtained by constructing x_{-i} where amino acid i is replaced with `<mask>`, and computing the language model probabilities at position i. The pseudo-likelihood is then defined as $\prod_i p(x_i|x_{-i})$.

A.2.2. Structure Projection

The structure projection is a single learned affine projection (linear projection with bias term) from ESM2 internal representations to inter-residue distance, applied identically to each position-pair $[i, j]$ of the protein.

In its implementation, the ($N = 660$) attention maps computed during ESM2 inference for a given sequence are used as input to a linear projection. At position $[i, j]$ we compute $z_{ij} = W_{projection}\text{attention_maps}_{ij} + b_{projection}$. The vectors z_{ij} are the softmax logits which define the categorical distribution $p(d_{ij}|sequence)$ over binned inter-residue distance between the carbon-beta atoms, known as distogram (62). Under a conditional pairwise independence assumption we use $\prod_{ij,j\neq i} p(d_{ij}|sequence)$ to approximate $p(structure|sequence)$. There are $660 * 18 + 18 = 11,898$ total learned parameters in the structure projection. The binning resolution of the model is ≈1Å, with 16 bins spanning the range [2.5Å, 20Å). The very

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A. 方法

A.1. 数据

A.1.1. 新设计目标集合

一个保留的集合新设计的蛋白质用于固定目标骨架的设计任务。一组多样的（$N = 39$）新设计的结构来自蛋白质数据银行（27），这些结构涵盖了一系列的长度（$67 \leq L \leq 184$）、折叠（例如，α-束，β-桶，NTF2，Rossman）以及新设计的方法（26，34，47–56）。请参见图 S1，以查看所有构成新设计目标集合的X射线晶体结构的可视化展示。这些蛋白质是由人类设计的，而非自然进化过程产生的。重要的是，这些新设计的蛋白质与属于自然折叠的蛋白质在结构上具有实质性的差异。例如，NTF2目标具有不自然的结合口袋（55），β-桶更狭窄，并且具有短的β-回转（34），某些设计在创建时完全是新的折叠（47）。尽管这些蛋白质在序列和结构上被定义为与天然蛋白质不同，但目标集合中的每个蛋白质都与UniRef100（28）进行查询，该数据库包含ESM2的训练集，所有通过Jackhmmer搜索返回的匹配序列都被排除在语言模型的训练之外，详见下一节。

蛋白质数据银行识别码（PDB 编号）为新设计目标集提供了：
`1QYS,2KL8,2KPO,2LN3,2LTA,2LVB,2N2T,2N2U,2N3Z,2N76,4KY3,4KYZ,5CW9,5KPE,5KPH,5L33,5TPJ,5TRV,6CZG,6CZH,6CZI,6CZJ,6D0T,6DG6,6DKM_A,6DKM_B,6DLM_A,6DLM_B,6E5C,6LLQ,6MRR,6MRS,6MSP,6NUK,6W3F,6W3W,6WI5,6WVS,7MCD`

A.1.2. 用于训练 ESM2 的序列数据集

本研究中使用的语言模型是`ESM2 650M`（8）。因此，_该文中描述的所有预训练设置适用于这里使用的语言模型。

为了测试语言模型对蛋白质的理解是否能从自然蛋白质推广到新设计 领域，关键是模型在训练时没有看到 新设计 蛋白质。为此，我们首先从ESM2的训练集中移除所有在UniProt (57) 网站上标记为“人造构造”的序列，当时`2021 04`是最近的发布版本（总共有1,027个蛋白质）。为了防止标签错误的蛋白质，并进一步移除训练集中可能与目标集相似的序列，我们还对每个 新设计 序列执行Jackhmmer (58) 搜索，针对UniRef100 `202104` 使用标记`--num-iter 1 {max`，并从 ESM2 的训练集中移除工具返回的所有命中（58,462个蛋白质）。

A.1. 3. 结构投影数据集该结构投影网络是在来自PDB的非冗余数据集上训练的，包含15,051个蛋白质（结构发布日期在2018年5月1日之前），用于Yang等人（59）。

A.1. 4. 自然蛋白质的留出集一个小（$N = 214$）的天然蛋白质集，其在PDB中具有结构，被选中作为基线比较，在评估语言模型新设计蛋白质理解时，如图S1和S2 及表S1中。该数据集由2020年7月可用的PDB组成，具有序列相似性 < 0.3 与用于训练结构投影的数据集相比，依据mmseqs2（60）。应用了一个长度过滤器 $50 \leq L < 250$，大致匹配新设计目标集的长度分布（$67 \leq L \leq 184$）。

A.2. 模型

A.2.1. ESM2

我们使用 `ESM2650M` (8) 作为我们在本研究中选择的大规模蛋白质语言模型。ESM2 是一个通过掩码语言建模训练的 Transformer 模型，涉及已知的自然蛋白质序列。在训练时，蛋白质序列以部分残基被掩码、随机重排为不同的氨基酸或保持不变的方式呈现给模型，遵循标准 BERT 噪声概率。

(7). 模型的任务是根据输入中所有未遮蔽氨基酸的双向上下文预测那些遮蔽的残基。ESM2 仅在自然蛋白质序列上进行训练；被标注为人工构建的序列以及通过序列搜索与新设计目标集查询匹配的序列从语言模型的训练集中移除，附录A.1.2。该语言模型用于近似 $p(sequence)$ 通过伪似然（61）。首先定义在位置$p_\theta(x_i|x_{-i})$ 上可能氨基酸的概率 i 在序列中x，条件是序列的其余部分。这个条件概率是通过构造 x_{-i} 用 i 替换的氨基酸 <掩码>，并计算在位置 i的语言模型概率获得的。然后定义伪似然为$\prod_i p(x_i|x_{-i})$。

A.2.2. 结构投影

结构投影是一个从ESM2内部表示到残基间距离的单一学习的仿射投影（带有偏置项的线性投影），它被应用于蛋白质的每个位置对$[i, j]$。

在其实现中，ESM2推理期间为给定序列计算的($N = 660$)注意力图作为线性投影的输入。在位置 $[i, j]$ 处，我们计算 z_{ij}，其维度为($N = 18$)：$z_{ij} = W_{projection}$注意力图$_{ij} + b_{projection}$。向量 z_{ij} 是softmax logits，定义了在碳β原子之间的分箱残基间距离的类别分布 $p(d_{ij}|sequence)$，称为distogram (62)。在条件对称独立假设下，我们使用$\prod_{ij,j\neq i} p(d_{ij}|sequence)$来近似 $p(structure|sequence)$。总共有660个

$* 18 + 18 = 11$,在结构投影中总共有898个学习参数。模型的分箱分辨率为≈1 ˚A，16个箱覆盖范围 [2.5 ˚A， 20 ˚A）。非常

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first bin represents <2.5Å, and the very last bin represents >20Å. Symmetry was applied to prediction logits, since distograms are by definition symmetric ($d_{ij} = d_{ji}$). Weights of the ESM2 model were frozen during training of the structure projection.

The structure projection was trained on a random subset of 80% of the sequence and structure pairs published in Yang, et. al (59) Appendix A.1.3. As in that work, distograms are constructed from inferred Carbon-beta coordinates of protein backbones. We trained the model for 10 epochs with a batch size of 4 and learning rate of 1e-2 using categorical cross-entropy loss between all $[i, j]$ pairs in the predicted distogram and ground truth distogram. There are no common structures between the dataset used to learn the structure projection and the *de novo* target set.

A.2.3. N-GRAM PRIOR

Background distributions of uni-, bi-, and tri- gram (n-gram) amino acid frequencies were determined via the amino acid frequencies in UniRef50, release `2018_03`. During design, the Kullback–Leibler divergence (D_{KL}) is calculated between the n-gram frequencies of the background distribution and of the designed sequence. D_{KL} terms are added with equal weight to produce a single n-gram energy term. Conceptually this can also be seen as using the n-grams as a language model $p(sequence)$ (63) which can be combined with the ESM transformer language model. Concretely, the energy function is defined between n-gram frequencies of our design sequence and background:

$$E_{ngram} = \sum_{i \in \{1,2,3\}} D_{KL}(ngram_i(x), ngram_{i,bg})$$

A.3. Tasks

A.3.1. FIXED BACKBONE DESIGN

The goal of fixed backbone design is to generate a protein sequence x for a target backbone y. As in (59), the backbone is derived from the set of 3D coordinates of the protein's Carbon-beta atoms (inferred for glycines), with length equal to the number of residues in the protein. These 3D coordinates are converted to a distogram of binned pairwise distances Appendix A.2.2.

We'd like to sample sequences x with high likelihood, conditioned on the target backbone, Y:

$$x \sim p(x|y = Y)$$

To sample from this distribution, we first note from Bayes rule that this is equivalent to sampling from the unnormalized product of unconditional sequence prior $p(x)$ and a conditional structure distribution $p(y|x)$:

$$p(x|y = Y) = \frac{p(x)p(y = Y|x)}{p(y = Y)} \propto p(x)p(y = Y|x)$$

- $p(x)$: approximated by the language model's pseudo-likelihood computed by multiplying marginal likelihoods when masking out each individual token, and the n-gram prior.
- $p(y = Y|x)$: approximated by the distogram distribution from the language model with structure projection head, evaluated for the target Y.
- $p(y = Y)$: constant, in the case of a fixed target.

To sample $p(x|y = Y)$, we utilize an energy-based sampling procedure, via Markov-Chain Monte-Carlo (MCMC). Formally, our full energy function for sampling from $p(x)p(y = Y|x)$ is the following expression for a randomly selected sequence index i:

$$E(x) = \lambda_p E_{projection}(y = Y|x) + \lambda_{LM} E_{LM}(x) + \lambda_n E_{ngram}(x)$$

where:

$$E_{projection}(y = Y|x) = - \sum_{ij, d_{ij} < 10\mathring{A}} logp(y_{ij} = Y_{ij}|x)/L^2$$

$$E_{LM}(x) = -\sum_i \log p(x'_i|x_1)\mathbb{1}(x'_1 = x_1)/L$$

$$E_{ngram}(x) = \sum_{i \in \{1,2,3\}} D_{KL}(ngram_i(x), ngram_{i,bg})$$

In the above energy function $E(x)$, the first term $E_{projection}$ specifies sequence-structure consistency, taking only pair-positions into account which are in contact in the target, i.e. have inter-residue distance $d_{ij} < 10\mathring{A}$. The term E_{LM} specifies sequence negative log likelihood, and the term E_{ngram} is based on the n-gram model of sequence likelihood. The terms are composed together with separate weights $\lambda_p = 3$, $\lambda_{LM} = 2$, $\lambda_n = 1$ enforcing different prioritizations of each objective, which were determined by hyperparameter sweeps. The overall energy function $E(x)$ defines a Boltzmann distribution:

$$p(x) = \frac{1}{Z} e^{-E(x)/T} \propto e^{-E(x)/T}$$

A uniform random amino acid mutation x'_i at a randomly selected sequence index i is proposed at each step with Metropolis acceptance rate α:

$$\alpha = \min(1, e^{-E(x')/T} / e^{-E(x)/T})$$

Mutations to cysteine were disallowed, as their presence would interfere with experimental evaluation. Note that by defining acceptance as the ratio of $E(x')$ and $E(x)$, the relative ratio between $E_{LM}(x')$ and $E_{LM}(x)$ can be efficiently approximated in 1 forward pass through the language model (by computing marginal substitution likelihoods at the substituted position), as opposed to the L forward passes required to explicitly compute $E_{LM}(x)$.

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第一个区间表示 <2.5 ˚A，而最后一个区间表示 >20 ˚A。对预测对数进行了对称处理，因为根据定义，Distogram 是对称的 ($d_{ij} = d_{ji}$)。在结构投影的训练过程中，ESM2 模型的权重被冻结。

结构投影是在Yang等（59）附录A.1.3中发布的80%序列和结构对的随机子集上训练的。与该工作中一样，利用推断的蛋白质主链的碳β坐标构建distogram。我们对模型进行了10个epoch的训练，批大小为4，学习率为1e-2，使用分类交叉熵损失函数，计算预测的distogram与真实值distogram之间的所有 $[i, j]$ 对。用于学习结构投影的数据集与新设计目标集之间没有共同的结构。

A.2.3. N-gram 先验

一元、二元和三元(n-gram)氨基酸频率的背景分布是通过在UniRef50，发布于2018年03中氨基酸频率确定的。在设计过程中，计算背景分布的n-gram频率与设计序列的n-gram频率之间的Kullback–Leibler散度（D_{KL}）。D_{KL} 这些项以相等的权重添加，以生成一个单一的n-gram能量项。从概念上讲，这也可以看作是将n-gram用作语言模型 $p(sequence)$ (63)，可以与ESM变换器语言模型结合。具体而言，能量函数是定义于我们设计序列和背景的n-gram频率之间：

$$E_{ngram} = \sum_{i \in \{1,2,3\}} D_{KL}(ngram_i(x), ngram_{i,bg})$$

A.3. 任务

A.3.1. 固定主链设计

固定主链设计的目标是生成一个蛋白质序列 x，针对目标主链 y。如同 (59) 中所述，主链是由蛋白质的碳-β原子的三维坐标（针对甘氨酸进行推断）所派生，其长度等于蛋白质中残基的数量。这些三维坐标被转换为一个距离的distogram，按照附录A.2.2.

我们希望根据目标主链 Y，采样出高可能性的序列 x：

$$x \sim p(x|y = Y)$$

为了从这个分布中采样，我们首先注意到根据贝叶斯法则，这相当于从无条件序列先验 $p(x)$和条件结构分布 $p(y|x)$ 的未归一化乘积中进行采样：

$$p(x|y = Y) = \frac{p(x)p(y = Y|x)}{p(y = Y)} \propto p(x)p(y = Y|x)$$

- $p(x)$: 由通过乘法边际似然计算的语言模型伪似然近似在屏蔽每个单独的标记时，以及 n-gram 先验。
- $p(y = Y|x)$：通过具有结构投影头的语言模型中的distogram分布进行近似，针对目标 Y 进行已评估。
- $p(y = Y)$: 常量，在固定目标的情况下。

为了从 $p(x|y = Y)$ 进行采样，我们利用基于能量的采样过程，通过马尔可夫链蒙特卡洛 (MCMC)。形式上，我们用于从 $p(x)p(y = Y|x)$ 进行采样的完整能量函数是针对随机选择的序列索引 i 的以下表达式：

$$E(x) = \lambda_p E_{projection}(y = Y|x) + \lambda_{LM} E_{LM}(x) + \lambda_n E_{ngram}(x)$$

其中：

$$E_{projection}(y = Y|x) = - \sum_{ij, d_{ij} < 10\mathring{A}} logp(y_{ij} = Y_{ij}|x)/L^2$$

$$E_{LM}(x) = -\sum_i \log p(x'_i|x_1)\mathbb{1}(x'_1 = x_1)/L$$

$$E_{ngram}(x) = \sum_{i \in \{1,2,3\}} D_{KL}(ngram_i(x), ngram_{i,bg})$$

在上述能量函数 $E(x)$ 中，第一个项 $E_{projection}$ 指定了序列-结构一致性，仅考虑在目标中接触的成对位置，即具有残基间距离 $d_{ij} < 10\mathring{A}$。项 E_{LM} 指定序列的负对数似然，而项 E_{ngram} 基于序列似然的 n-gram 模型。各项通过单独的权重 $\lambda_p = 3$、$\lambda_{LM} = 2$、$\lambda_n = 1$ 组合在一起，以强制每个目标的不同优先级，这些优先级是通过超参数遍历确定的。整体能量函数 $E(x)$ 定义了 玻尔兹曼分布：

$$p(x) = \frac{1}{Z} e^{-E(x)/T} \propto e^{-E(x)/T}$$

在每一步中，提议以均匀随机的氨基酸突变 x'_i 在随机选择的序列索引 i 处进行，接受率为Metropolis α：

$$\alpha = \min(1, e^{-E(x')/T} / e^{-E(x)/T})$$

由于半胱氨酸的存在会干扰实验评估，因此不允许进行半胱氨酸突变。请注意，通过将接受定义为 E（x'）与 E（x）的比率，可以在语言模型中通过一次正向传递有效地近似 E_{LM}（x'）与 E_{LM}（x）之间的相对比率（通过计算替换位置的边际替换似然），而不是需要L次正向传递来明确计算 E_{LM}（x）。

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We perform 170,000 steps of MCMC sampling. We use a geometrically decaying temperature schedule for simulated annealing. Every 10,000 steps, we decay the temperature T by 2, from an initial value of 8 to a terminal value of approximately 6e-5. Full design trajectories take ≈ 10 hours for a fixed backbone design with sequence length ≈ 100 on a single 32GB Volta gpu. We could achieve successful designs on most targets (low target RMSD according to the oracle) with fewer steps, but the step count was increased to achieve best performance, especially for longer fixed backbone designs, e.g. 6WVS ($L = 182$).

A.3.2. Fixed backbone design without a language model

Designs from the language model ("LM") were compared against designs from a baseline with a powerful structure predictor, but no language model ("no-LM"). For this baseline, AlphaFold was used as the structure model.

To keep the comparison with LM designs matched, AlphaFold's pairwise distance (distogram) output was used as $p(structure = Target|sequence)$. Since no transformer language model is used, there is no $p(sequence)$ term and fixed backbone design without a language model optimizes for sequences that have high likelihood $p(structure|sequence)$. Additionally, to ensure a fully matched comparison against the LM designs, a second set of no-LM designs were generated, which include the same E_{ngram} term used for LM designs Appendix A.2.3. The additional n-gram term can be interpreted as adding a weak n-gram language model. The coefficient of this n-gram term was selected via a line sweep Table S3. In the main comparison, we only feature results from the set without the n-gram term, since that set was more successful experimentally (1/20 successes vs. 0/20 successes with n-gram term).

A gradient-based public algorithm for producing AlphaFold-based designs was used. Baseline designs were produced by ColabDesign (12, 16) (commit hash e7bb3def), using the `design_3stage()` AfDesign recipe, which alternatingly and then simultaneously optimizes across all 5 AlphaFold pTM model replicas. It was found that more steps improved the convergence to low target RMSD over the course of optimization, so the default number of steps used was scaled up by a factor of 5, for a total of 1500 soft_iters, 500 temp_iters, and 50 hard_iters. This design protocol requires less steps of optimization due to employing gradient-based optimization; the algorithm can update each position in the sequence at each step, whereas the MCMC protocol we employ only makes a single mutation at each step. Although AlphaFold's distogram output was optimized rather than its atomic structure prediction output, all designs were verified to have $< 1\mathring{A}$ RMSD to target and > 0.8 pTM according to the AlphaFold Oracle Appendix A.4.1.

LM and no-LM protocols were used to produce 200 designs per target each. Simple selection of the top 5/200 seeds (per target) according to each protocol's optimization objective was used to select designs for experimental evaluation Appendix A.6.2.

A.3.3. Free Generation

The goal of free (unconstrained) generation is to design a new protein sequence x which is sampled from the universe of possible sequences x and their associated backbones y. As in the previous section, backbones of designs are represented by the distogram distribution over pairwise distances. In particular, we wish to sample sequences x and associated structures y with high joint probability:

$$x, y \sim p(x, y)$$

We utilize an energy-based sampling procedure to sample both sequence x and structure backbone y from this joint distribution. In particular, we utilize a blocked Gibbs MCMC sampling procedure where, starting from an initially random sequence x, we sample a definite backbone y for the current sequence

$$y_{sampled} \sim p(y|x)$$

and then sample an updated sequence x' given the current backbone y.

$$x' \sim p(x|y = y_{sampled})$$

During the $p(y|x)$ sampling phase, inter-residue distances are sampled independently at all pair-positions in the distogram. During the $p(x|y)$ sampling phase, 3 steps of the MCMC protocol for fixed backbone design are performed (see prior section), where the sampled backbone $y_{sampled}$ is used as a target.

In total, 170,000 steps of MCMC are performed, where a step is comprised of a $p(y|x)$ sampling phase and a $p(x|y)$ sampling phase. For $p(x|y)$ sampling, the same temperature schedule is used as in fixed backbone design: temperature is decayed by a factor of 2 every 10,000 steps, from 8 to $\approx 6e-5$. For the structure sampling step, a fixed temperature of 1 is used. Annealing both temperatures led to low diversity (alpha-bundle) solutions which indeed have very high $p(y|x)$ and $p(x|y)$, which did not happen with fixed $p(y|x)$ temperature. With this protocol to sample from the joint distribution, a diverse set of topologies was generated with varied secondary structure content (Fig. 4A; Fig. 4B), respectively. Finally, as in fixed backbone design, mutations to cysteine were disallowed, as their presence would interfere with experimental testing.

A.4. *In silico* Quality Metrics

A.4.1. Structure Oracle

Designed sequences were given as input to AlphaFold for an *in silico* assessment of their structure. AlphaFold serves as a powerful orthogonal predictor of protein structure, as the AlphaFold model differs from ESM2 in its architecture, objective, and training data. When evaluation designs, sequences are input without generating a multiple sequence alignment (MSA) nor using any

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我们执行了 170,000 步 MCMC 采样。我们使用几何衰减的温度计划进行模拟退火。每 10,000 步，我们将温度 T 从初始值 8 衰减 2，最终值约为 6e-5。对于序列长度 ≈ 100 的固定骨架设计，完整的设计轨迹需要 ≈ 10 小时，在一台 32GB Volta GPU 上。我们能够在大多数目标上实现成功的设计（根据甲骨文的低目标均方根偏差），而所需步骤较少，但步数增加以达到最佳性能，尤其是对于较长的固定骨架设计，例如 6WVS ($L = 182$)。

A.3.2. 无语言模型的固定骨架设计

与无语言模型（“no-LM”）的强大结构预测基线相比，语言模型（“LM”）的设计被进行比较。对于该基线，使用 AlphaFold 作为结构模型。

为了保持与语言模型设计的比较一致，使用了AlphaFold的成对距离（距离图）输出作为 $p(structure = Target|sequence)$。由于未使用变换器语言模型，因此没有 $p(sequence)$ 项，固定主链设计在没有语言模型的情况下优化具有高可能性的序列$p(structure|sequence)$。此外，为了确保与语言模型设计之间的完全匹配比较，生成了第二组无LM设计，这些设计包括用于LM设计的相同 E_{ngram} 项，附录A.2.3。额外的n-gram项可以解释为添加一个弱n-gram语言模型。该n-gram项的系数是通过一个线性扫查表S3选择的。在主要比较中，我们仅展示来自不带n-gram项的那组的结果，因为该组实验上更成功（1/20成功 vs. 0/20成功有n-gram项）。

使用了一种基于梯度的公共算法来生成基于AlphaFold的设计。通过ColabDesign生成了基线设计（12，16）（提交哈希e7bb3def），使用了设计`3stage()`AfDesign配方，该配方交替并同时在所有5个AlphaFold pTM模型副本上进行优化。发现更多的步骤有助于优化过程中收敛到低目标RMSD，因此使用的默认步骤数量增加了5倍，共计1500个软迭代、500个温度迭代和50个硬迭代。由于采用基于梯度的优化，这一设计协议需要的优化步骤较少；该算法可以在每一步更新序列中的每个位置，而我们采用的MCMC协议每一步只能进行一次突变。尽管优化的是AlphaFold的距离图输出，而不是其原子结构预测输出，但所有设计都已验证与目标的 $< 1\mathring{A}$ RMSD和> 0.8 根据AlphaFold甲骨文的pTM附录A.4.1。

使用LM和无LM协议为每个目标生成200个设计。根据每个协议的优化目标，简单地选择前5/200种子（每个目标）用于实验评估设计的选择附录A.6.2.

A.3.3. 自由生成

自由（无约束）生成的目标是设计一个新的蛋白质序列x，该序列从所有可能序列x及其相关骨架y的宇宙中抽样。与前一部分一样，设计的骨架由成对距离的distogram分布表示。特别地，我们希望以高联合概率抽样序列x和相关结构y：

$$x, y \sim p(x, y)$$

我们利用一种基于能量的采样程序，从这个联合分布中对序列 x 和结构主链 y 进行采样。具体来说，我们采用阻塞吉布斯 MCMC 采样程序，从一个最初随机的序列 x 开始，对当前序列采样一个确定的主链 y。

$$y_{sampled} \sim p(y|x)$$

然后，根据当前的主链 y 采样更新的序列 x’ 。

$$x' \sim p(x|y = y_{sampled})$$

在 $p(y|x)$采样阶段，所有配对位置的 distogram 中的残基间距离是独立采样的。在 $p(x|y)$ 采样阶段，执行固定主链设计的 MCMC 协议的 3 步（见前一节），其中采样的主链 $y_{sampled}$ 被用作目标。

总共执行 170,000 步 MCMC，其中一步包括一个$p(y|x)$ 采样阶段和一个 $p(x|y)$ 采样阶段。对于 $p(x|y)$ 采样，使用的温度计划与固定主链设计中的相同：温度每 10,000 步减小一个因子 2，从 8 降到 $\approx 6e-5$。在结构采样步骤中，固定温度为 1。退火这两个温度导致了低多样性（α-束）解决方案，实际上具有非常高的 $p(y|x)$ 和 $p(x|y)$，而在固定 $p(y|x)$ 温度下并未发生这种情况。通过使用这个联合分布的采样协议，生成了一组多样的拓扑结构，具有不同的二级结构内容（如图4A；图4B）。最后，与固定主链设计一样，不允许对半胱氨酸进行突变，因为它们的存在会干扰实验测试。

A.4. 计算机模拟**质量指标**

A.4.1. 甲骨文

设计的序列作为输入被提供给AlphaFold，以对其结构进行计算机模拟评估。AlphaFold作为一种强大的正交预测器，能够预测蛋白质结构，因为AlphaFold模型在其架构、目标和训练数据上与ESM2不同。在评估设计时，序列作为输入，但不生成多序列比对（MSA）或使用任何

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templates. We follow the standard protocol of predicting structure and confidence scores (pTM captures global confidence, pLDDT captures local confidence) with all 5 publicly released models, then select the most confident output by pLDDT. Amber relaxation was performed on the selected, predicted structure. All predicted structures of designs in this study come from this pipeline, and structural metrics described in the following sections are calculated using these predictions. Confidence metrics pTM and pLDDT, as well as RMSD to target structure where available, can be used for selecting designs as well Appendix A.6.

A.4.2. Solubility and aggregation metrics

Three *in silico* metrics are used for the purpose of filtering out candidates with strong evidence that they will not be soluble or monomeric:

1. Hydrophobic Solvent Accessible Surface Area (SASA) computes the SASA for each hydrophobic residue. It quantifies how much of the protein's surface, which is accessible to the solvent, is hydrophobic. High hydrophobic SASA is problematic as for monomeric proteins we expect the SASA to be mostly polar in order to stay in solution rather than aggregating via exposed hydrophobic surface area.
2. Net Charge: a simplified sequence-based net charge by counting positively and negatively charged amino acids in the sequence, to try to avoid proteins with zero net charge, as this may lead to aggregation in polar solvents.
3. The spatial aggregation propensity (SAP) is a metric introduced to quantify the aggregation propensity, i.e. whether the protein will aggregate into non-functional and typically insoluble assemblies (64). SAP score identifies surface exposed patches of hydrophobic residues and computes a weighted sum of atoms with positive hydrophobicity score. The SAP metric is useful to study and filter both fixed backbone designs and free generations, as in both cases we aim to design monomeric proteins. This requires the monomer to be soluble and have mostly hydrophilic amino acids on the surface, corresponding to low SAP score (65). This metric is correlated but can be complementary to the hydrophobic SASA. For comparison across sequence lengths, we use average SAP score, i.e. averaged over residues.

For fixed backbone designs tested experimentally, we use SAP score during selection except for those in the "LM vs. no-LM" comparison where no additional in-silico metrics are used for filtering (Methods; Selection of designs for experimental evaluation). When filtering free generations, we use all three hydrophobicity metrics with relatively loose thresholds, and combine them with logical "and", i.e. the candidate has to pass all filters. Firstly, hydrophobic SASA < 1.7 times the "ideal surface" computed using the ideal sphere for the same length protein. Secondly, we require sequence-based net charge ≥ 2 or ≤ -2. Finally, we filter for averaged SAP scores ≤ 0.4, and relax this threshold to 0.5 when the predicted structure contains at least 25% beta strands.

A.4.3. Packing Metrics

Two metrics are used to filter candidates which are likely not well-packed:

1. Protein Packing is quantified with the Rosetta PackStat filter, and is an approximate implementation of RosettaHoles (66). This is a stochastic algorithm, so it is averaged across 100 repeats. It returns a score between 0 and 1, where 1 means perfect packing. We keep free generation candidates only if packing > 0.55.
2. Shape Complementarity of secondary structure elements in the structure (67) is implemented in the Rosetta SSShapeComplementarity filter with loops="true" helices="true". This metric aims to quantify whether the surface normals from different interacting secondary structures are well-aligned, indicating that secondary structure elements fit well together. We keep free generation candidates only if shape complementarity > 0.6.

The Packing and Shape Complementary metrics were computed twice: once on the structure from the AlphaFold pipeline after Amber relaxation, and once after an additional step of Rosetta minimization with the `beta_nov16` (32). Logical "or" between structure filters is used: if either of the structures passes the filter, the filter is satisfied.

A.4.4. Globularity metrics

A final set of metrics are used to screen out proteins which are not globular and have oblong shapes such as extended helix bundles. We follow Dill et al. (68) and define the idealized radius of a protein based on its number of residues as 2.24 * (num_residues ** 0.392) (68), and its corresponding ideal surface area based on this radius. Using these as reference values, we define relative SASA and relative radius of gyration. The following metrics and thresholds are used:

1. The Radius of Gyration is the root mean square distance from the center of mass (not taking residue weights into account). We keep candidates if the relative radius of gyration is < 1.5.
2. Total solvent accessible surface area (SASA) computed by Rosetta TotalSasa. We keep candidates if the relative SASA is < 3.
3. Contact Order was computed but we did not filter on this metric. The sample of free generations spanned a range of Contact Order values.

A.5. Comparison to natural proteins

A.5.1. Settings used for sequence search

Designed sequences are tested for their distance from natural protein sequences via querying them against large-scale sequence databases. We emphasize that comparisons described throughout

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模板。我们遵循标准协议，利用所有5个公开发布的模型来预测结构和信心评分（pTM捕捉全球信心，pLDDT捕捉局部信心），然后通过pLDDT选择最自信的输出。对所选预测结构进行了Amber松弛。本研究中设计的所有预测结构均来自此流程，接下来的章节中描述的结构指标是基于这些预测计算的。信心指标pTM和pLDDT，以及可用时与目标结构的RMSD，可以用于选择设计，另见附录A.6。

A.4.2. 可溶性和聚集指标

为了过滤出那些有强烈证据表明它们不会是可溶或单体的候选者，使用了三个计算机模拟指标：

1. 疏水溶剂可接触表面积（SASA）为每个疏水残基计算SASA。它量化了蛋白质表面中，能够接触溶剂的部分有多少是疏水性的。高疏水性SASA是有问题的，因为对于单体蛋白质，我们期望SASA主要是极性的，以便保持在溶液中，而不是通过暴露的疏水表面积聚集。
2. 净电荷：通过计算序列中带正电和带负电的氨基酸数量来简化的基于序列的净电荷，以此尽量避免零净电荷的蛋白质，因为这可能导致在极性溶剂中聚集。
3. 空间聚集倾向 (SAP) 是一个引入的指标，用于量化聚集倾向，即蛋白质是否会聚集成非功能性且通常不溶解的组装（64）。SAP得分识别表面暴露的疏水残基斑块，并计算具有正疏水性得分的原子的加权总和。SAP指标对于研究和筛选固定骨架设计和自由生成都很有用，因为在这两种情况下，我们的目标是设计单体蛋白质。这要求单体能够溶解，并且表面大多数为亲水性氨基酸，对应于低SAP得分（65）。该指标与疏水性SASA相关，但可以作为补充。为了比较序列长度，我们使用平均SAP得分，即对残基进行平均。对于经过实验测试的固定骨架设计，我们在选择过程中使用SAP得分，除了“LM与无LM”比较的情况，在这些情况下不使用额外的计算指标进行筛选（方法；实验评估设计的选择）。在筛选自由生成时，我们使用所有三种疏水性指标，并且阈值相对宽松，并将它们通过逻辑“与”结合，即候选者必须通过所有筛选。首先，疏水性SASA < 1.7 乘以使用相同长度蛋白质的理想球体计算的“理想表面”。其次，我们要求基于序列的净电荷 ≥ 2 或 ≤ -2。最后，我们筛选平均SAP得分 ≤ 0.4，并将该阈值放宽至0.5 当预测结构包含至少25%的β链时。

A.4. 3. 打包度量：使用两个度量来过滤可能打包不良的候选者：

1. 蛋白质打包通过 Rosetta PackStat 过滤器进行量化，并且是 RosettaHoles 的一种近似实现 (66)。这是一种随机算法，因此它在100次重复中取平均。它返回一个介于 0 和 1之间的分数，其中1 意味着完美的打包。我们仅在打包 > 0.55时保留自由生成候选。

2. 结构中次级结构元素的形状互补性（67）在Rosetta SSShapeComplementarity过滤器中实现，带有环节=“真实”螺旋=“真实”。这个度量旨在量化来自不同相互作用的次级结构的表面法线是否对齐良好，表明次级结构元素之间配合良好。我们只保留形状互补性 > 0.6。打包和形状互补度量计算了两次：一次是在AlphaFold管道经过Amber松弛后的结构上，另一次是在Rosetta最小化的附加步骤后，`beta nov16` (32)。结构过滤器之间使用逻辑“或”：如果任一结构通过过滤器，则过滤器满足条件。

A.4.4. 球形度指标

一组最终的指标用于筛选出不是球形且具有长形状的蛋白质，例如延伸的螺旋束。我们遵循 Dill 等人 (68) 的定义，根据残基数量定义蛋白质的理想回转半径为 2.24 * (num residues ** 0.392) (68)，并基于该半径计算其理想表面积。使用这些作为参考值，我们定义相对 SASA 和相对回转半径。所使用的指标和阈值如下：

1. 回转半径是从质心的均方根距离（未考虑残基重量）。如果相对回转半径是 < 1.5，我们将保留候选者。
2. 由 Rosetta TotalSasa 计算的总溶剂可接触表面积 (SASA)。如果相对 SASA 是 < 3，我们将保留候选者。
3. 计算了接触顺序，但我们没有根据该指标进行筛选。自由生成的样本涵盖了一系列接触顺序值。

A.5. 与天然蛋白质的比较

A.5.1. 用于序列搜索的设置

设计的序列通过与大规模序列数据库查询，测试其与天然蛋白质序列的距离。我们强调，在整个过程中描述的比较

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this section (except in the case of motifs) are made for hits returned by sequence search; comparison of predicted structure for designs to known structure databases always returns hits likely to possess a similar fold (Fig. S10).

For sequence search, we use jackhmmer 3.3.2, a sequence search tool from the HMMER suite (58). Two jackhmmer settings were modified from their defaults, based on failure modes observed during during our analysis, which queries distant / de novo sequences against large-scale ($>$ 100M sequences) search databases:

1. **One Iteration.** Jackhmmer was run with only 1 iteration, instead of multiple (the default). This change was made because it was observed that additional iterations resulted in a growing amount of returned spurious hits, when given distant sequences as input, such as de novo ground truth sequences. Specifically, for query sequences with few natural sequence homologs, false positives increasingly dominated the query profile used on subsequent jackhmmer search iterations.

2. **Sorting by E-value of the best-scoring domain.** Throughout this paper, Jackhmmer results are always sorted according to *best-domain* (rather than full-sequence) E-value. It was found that ranking hits by full-sequence E-value frequently led to more spurious top hits. Specifically, designs—which comprise a single domain in fixed backbone design, and have a single-domain-like globular structure in free generation—tended to match long, repetitive hit sequences containing a repeated domain. In these cases, multiple weak per-domain matches resulted in a high cumulative full-sequence significance, even though the design had no strong match to any single domain in the hit sequence. This is a known potential failure mode stated in the user's guide (69). Though significance is determined by E-value for the best domain match, when top jackhmmer hits are subsequently analyzed (e.g. for calculating sequence identity and predicting structure) the *full* hit sequence is used.

Overall, Jackhmmer was run with non-default settings (-n 1 –seed 0).

Designs are compared against their sequence hits on three major axes: **E-value**, **Sequence Identity**, and **TM-score**:

1. **E-value.** Jackhmmer returns an E-value for each hit, which quantifies the significance of each hit's sequence match to the query. Specifically, E-value is the number of false positives that are expected to score as or more strongly than the given hit due to random chance. Hits with a (best-domain) E-value < 1 are considered significant. At this significance level, we expect one hit on average to be falsely considered significant, when querying each design against some large sequence database.

2. **Sequence Identity.** Sequence identity of the design to each of its hits was calculated via local alignment with Biotite's (70) `biotite.sequence.align.align_optimal()` given the BLOSUM62 substitution matrix applied to the full sequence of the design and the full sequence of the hit. Sequence identity was calculated as the number of matching characters in the two aligned sequences divided by the full length of the original query sequence (rather than just the length of the aligned region).

3. **TM-score.** Designs were also compared to some fraction of their top hits via TM-score of their predicted structures from the TM-align tool (71). Predicted structures of designs are obtained using the (AlphaFold, single-sequence) structure oracle (Appendix A.4.1). Predicted structures for (the full sequences of) top jackhmmer hits are obtained from AlphaFold DB, or the structure oracle given an MSA (instead of a single sequence) as input.

A.5.2. Comparison of all free generations to natural proteins in AlphaFold DB

In the case of Fig. 4D, each of the 25k free generations and the ≈15k natural proteins from (59) was queried against the sequences in AlphaFold DB (37), which comprise UniProt `2021_04` (57). Because all sequences in this database have a structure predicted by AlphaFold, searching against this database enables comparison of predicted structure at scale. We compare designs to only their single most significant (by best-domain E-value) hit, on the bases of sequence-identity and TM-score of predicted structures, fetched from the url: `https://alphafold.ebi.ac.uk/files/AF-<UniProtID>-F1-model_v3.pdb`. The bottom-left quadrant of Fig. 4D, where sequence-identity < 0.2 and TM-score of predicted structure < 0.5 was used to define a set of 49 distant free generations, of which 31 (67%) succeed experimentally. Generations that have no significant (best-domain E-value < 1) hits are displayed at 0 sequence identity in that plot, to distinguish them from generations possessing significant hits, visually. Results from this comparison are used in Fig. 4D, the definition of "49 distant free generations" in the Introduction, and the analysis of free generations.

A.5.3. Comparison of experimentally evaluated designs to natural proteins in UniRef90

In all other cases, when we compare experimentally evaluated designs to known natural proteins, we query against UniRef90 2021_04 (28), which fully contains the set of sequences seen by the language model during training.

Unlike in the comparison of designs natural proteins AlphaFold DB, where we consider top-hit statistics, only, we perform a more comprehensive analysis:

1. **E-value**. Same as the comparison to AlphaFold DB. (Best-domain) E-value of the top hit = *minimum* over *all* hits due to sorting.

2. **Sequence Identity.** Is calculated as a *maximum* over sequence identities for *all* significant (best domain E-value < 1) hits.

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本节（除非是基序的情况）是为序列搜索返回的命中而制作的；设计的预测结构与已知结构数据库的比较始终返回可能具有相似折叠的命中（图。S10）。

用于序列搜索，我们使用jackhmmer 3.3.2，这是HMMER套件中的一个序列搜索工具（58）。基于分析中观察到的失败模式，将两个jackhmmer设置从默认值进行了修改，旨在查询远序列/新设计序列对大规模（$>$ 100M序列）搜索数据库进行搜索：

1. **一次迭代**。Jackhmmer仅运行1次迭代，而不是多次（即默认设置）。之所以做出这一改变，是因为观察到在输入远序列时，额外的迭代导致返回的虚假命中数量逐渐增加，例如新设计的真实序列。具体来说，对于具有少量天然序列同源物的查询序列，假阳性越来越主导后续jackhmmer搜索迭代中使用的查询轮廓。

2. **按最佳得分域的E值排序**。在本文中，Jackhmmer结果始终根据最佳域(而不是全序列) E值进行排序。发现按全序列E值排名命中的方式常常导致更多的虚假顶级命中。具体来说，设计——由固定主链设计中的单一域组成，并在自由生成中具有类似单域的球状结构——往往会与包含重复域的长且重复的命中序列匹配。在这些情况下，多个较弱的每域匹配导致高的累计全序列显著性，即使该设计在命中序列中的任何单一域上没有强匹配。这是在用户指南中明确说明的潜在失败模式之一（69）。虽然显著性是通过最佳域匹配的E值来确定的，但当随后分析顶级jackhmmer命中时（例如，用于计算序列相似性和预测结构），将使用完整的命中序列。总体而言，Jackhmmer是使用非默认设置运行的（-n 1 –seed 0）。设计在三个主要轴上与其序列命中进行比较：**E值**，**序列相似性**，和**TM-score**：

1. **E值**。Jackhmmer 为每个命中返回一个 E 值，该值量化了每个命中序列与查询匹配的显著性。具体而言，E 值是指由于随机机会而预计会获得与给定命中同样或更高分数的假阳性数量。具有（最佳域）E 值 < 1 的命中被视为显著。在这一显著性水平下，我们预计每个设计在查询某个大型序列数据库时，平均会有一个命中被错误地认为是显著的。

2. **序列相似性**。 设计与其每个命中的序列相似性是通过与 Biotite 的本地比对计算得出的（70）`biotite.sequence.align.align_optimal()` 考虑到应用于设计的完整序列和命中的完整序列的 BLOSUM62 替代矩阵。序列相似性是通过两个对齐序列中匹配字符的数量除以原始查询序列的完整长度（而不仅仅是对齐区域的长度）来计算的。

3. **TM-score**。设计还通过 TM-align 工具的 TM-score 将其预测结构与一些顶级命中的部分进行比较（71）。设计的预测结构是使用 (AlphaFold, 单序列) 结构甲骨文获得的（附录A.4.1）。顶级 Jackhmmer 命中的（完整序列的）预测结构则是从 AlphaFold 数据库获得的，或在输入为 MSA（而不是单个序列）时从结构甲骨文获得。<style>

A.5.2. 将所有自由生成与AlphaFold数据库中的天然蛋白质进行比较

在图4D的案例中，查询了25k个自由生成和来自（≈15k个天然蛋白质）的序列，查询对象是AlphaFold数据库中的序列（37），这些序列包含UniProt `2021 04` (57）。由于该数据库中的所有序列都有AlphaFold预测的结构，因此针对该数据库的搜索可以使预测结构的比较规模化。我们仅将设计与其单个最显著的（按最佳域E值）命中进行比较，比较基础是预测结构的序列相似性和TM-score，数据源来自网址：`https://alphafold.ebi.ac.uk/files/AF-<UniProtID>-F1-model_v3.pdb`。图的左下象限4D中，序列相似性 < 0.2和预测结构的TM-score < 0.5 用于定义49个远程自由生成，其中31个（67%）在实验中成功。没有显著（最佳域E值 < 1）命中的生成在该图中的序列相似性为0，以便将它们与具有显著命中的生成在视觉上区分开。此比较的结果用于图4D、引言中“49个远程自由生成”的定义以及自由生成的分析。

A.5.3. 实验评估设计与自然蛋白质在UniRef90中的比较

在所有其他情况下，当我们将实验评估的设计与已知自然蛋白质进行比较时，我们会查询UniRef90 2021 04（28），该版本完全包含语言模型在训练期间看到的序列集合。

与在比较设计与天然蛋白质AlphaFold数据库时仅考虑顶部命中统计的情况不同，我们进行了更全面的分析：

1. **E值**。与对AlphaFold数据库的比较相同。（最佳域）顶部命中的E值 = 最低 由于排序而针对所有命中。

2. **序列相似性**。 计算为所有 显著（最佳域E值 < 1）命中的序列相似性的最大 值。

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3. **TM-score.** Is calculated as a maximum over the top-10 sequence hits. Predicted structures are acquired `https://alphafold.ebi.ac.uk/files/AF-<UniProtID>-F1-model_v3.pdb` where possible. However, a fraction of UniRef90 proteins ($\approx 20\%$) are not present in AlphaFold DB. For these proteins, predicted structures were obtained by folding their (full) sequences via the structure oracle, given an MSA produced by jackhmmer on UniRef90 (same settings as in Jumper et al. (72) as input rather than just the single hit sequence. Of the 228*10=2280 total jackhmmer hits considered throughout this paper, 4 (1 significant) had their TM-scores omitted from analysis, due to not being in AlphaFold DB and failing during oracle structure prediction because of GPU memory limitations (all are length > 1000). These errors do not affect the sequence statistics of jackhmmer hits (sequence identities, E-values).

UniRef90 is not the exact set of sequences seen by ESM2 during training. Two filters were applied to remove sequences labeled "ariticial" by UniProt (N = 1,027) and all sequences hit by Jackhmmer when querying with ground truth sequences of de novo targets (N = 58,462) Appendix A.1.2. It was discovered that for many fixed backbone designs, top-hits found in UniRef90 belonged to the sequences that had been removed. For this reason, we omit from consideration all hits that had been removed from ESM2's training set, when calculating the 3 (E-value, Sequence-identity, TM-score) metrics described above.

Results from this comparison are used in most statements of sequence novelty throughout this paper. Specifically: the statement of natural sequence dissimilarity in the Abstract and the detailed comparisons of experimentally evaluated fixed backbone and free generation designs to natural proteins (Fig. 2, Fig. 4F and 4G, Fig. S6, Fig S8).

A.5.4. Motif analysis

Hydrogen-bond network motifs were assessed for their similarity to (aligned) positions in natural proteins retrieved by both sequence- and structure- search. To test whether the language model is copying motifs from similar sequences in its training, designs were searched against UniRef90, again with `-n 1 --seed 0`. To test whether the language model is copying motifs from similar structures, designs were searched using Foldseek3 (36), an open-source tool for large-scale search of structures against structure databases. Version 7d0c07f89a was used, with non-default flags `--alignment-type 1` and the AlphaFold/UniProt (AlphaFold DB) Foldseek structure database. In both the sequence- and structure- search cases, the MSAs returned by each tool were sorted according to edit distance at motif positions only. MSAs were subject to minimal filtering, to focus edit distance calculations on significant or structurally similar hits. Specifically, jackhmmer results were filtered for best-domain E-value < 10, and Foldseek results were filtered for TM-score > 0.7. Predicted structures of hits are aligned to that of the design via TM-align.

Results of the comparison of designed motifs to minimum edit distance neighbors are shown in MSA-form (for motif positions, only) and graphically in Fig. 3D, 3E and Fig. S7.

A.6. Selection of designs for experimental evaluation

A.6.1. Overview

In total, 276 unique proteins were validated experimentally: 228 designs from the language model, 40 designs from the "no-LM" baseline, and 8 ground truth sequences corresponding to the *de novo* targets used in fixed backbone design. Designs are referred to by the scheme "FXXX" or "GXXX" for fixed backbone designs and generations, respectively, where XXX is an index in the range [0, 267], spanning all sequences tested, excluding the ground truths.

Experimental evaluation rounds:
Two total rounds of experimental evaluation were performed, using a consistent protocol:

1. Round 1 = 44 Fixed backbone designs, 48 free generations, 4 ground truths
2. Round 2 = 95 Fixed backbone designs, 81 free generations, 4 ground truths

A.6.2. Fixed backbone design

Design pools:
Two pools of candidate designs were considered for selection:

1. 200 designs using different random initializations and random seeds, for each *de novo* target.
2. An expanded set of designs - 9,060 additional designs were created roughly evenly split among the following targets: 1QYS (1990x), 6MRS (1500x), 6D0T (1604x), 6W3W (1968x).

Oracle Quality Filters:
The following oracle metrics were used for several (but not all) pools of experimentally tested designs:

1. Oracle (AlphaFold) RMSD < 2.5
2. Oracle (AlphaFold) pTM > 0.7
3. SAP score < 0.35

Round 1 (48x):
Goal: Select the most promising designs from the language model using information from the LM, the Oracle, Rosetta, and manual inspection.
Targets: 4 targets, selected for having x-ray crystal structures spanning a range of canonical topologies (especially those having high beta-sheet content, like 6CZJ) and sequence lengths.

- 1QYS (Top-7)
- 6W3W (NTF2)
- 6CZJ (Beta-barrel)
- 6WVS (TIM-barrel)

Source: 200 seeds run for each target (Design pool 1)
Filter (per-target):

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3.**TM-score**。计算为前10个序列命中中的最大值。预测的结构是通过折叠其（完整）序列获取的，使用jackhmmer 在 UniRef90 上生成的 MSA（设置与 Jumper 等人（72）相同）作为输入，而不仅仅是单个命中序列。228*10=2280个 jackhmmer 命中在整篇论文中考虑，有 4 个（1 个显著）因未在 AlphaFold 数据库中而被排除在分析之外，并由于 GPU 内存限制而在甲骨文结构预测过程中失败（所有都具有长度> 1000）。这些错误不会影响 jackhmmer 命中的序列统计（序列同一性、E值）。

UniRef90并不是ESM2在训练期间看到的确切序列集。应用了两个过滤器，以去除UniProt标记为“人工”的序列（N = 1,027）和在使用真实值序列查询时被Jackhmmer命中的所有序列（N = 58,462）附录A.1.2. 发现对于许多固定骨架设计，在UniRef90中找到的最佳匹配属于已被移除的序列。因此，在计算上述3个（E值、序列相似性、TM-score）指标时，我们将所有已从ESM2训练集中移除的匹配排除在外。

本比较的结果用于本文大部分关于序列新颖性的陈述中。具体来说：在摘要中关于天然序列差异性的陈述，以及对实验评估的固定主链和自由生成设计与天然蛋白质的详细比较（图。2，图。4F 和 4G，图。S6，图S8）。

A.5. 4. Motif分析氢键网络模体被评估其与通过序列和结构搜索检索到的天然蛋白质（对齐）位置的相似性。为了测试语言模型是否在其训练中从相似序列中复制模体，设计被搜索以对抗UniRef90，再次使用`-n 1 --seed 0`. 为了测试语言模型是否从相似结构中复制模体，设计通过使用Foldseek3（36）进行搜索，该工具是一个开源工具，用于对结构进行大规模搜索以匹配结构数据库。使用了版本7d0c07f89a，并且采用了非默认标志 `--alignment-type 1`以及AlphaFold/UniProt（AlphaFold数据库）Foldseek结构数据库。在序列和结构搜索的情况下，每个工具返回的MSA根据模体位置的编辑距离进行排序。MSA经历了最小过滤，以使编辑距离的计算集中在重要或结构相似的命中上。具体来说，jackhmmer的结果被过滤为最佳域E值 < 10，而Foldseek的结果被过滤为TM-score > 0.7. 预测的命中结构通过TM-align与设计结构对齐。设计的模体与最小编辑的比较结果

距离邻居以MSA形式显示（仅针对模体位置），并在图中呈现。3D，3E 和图。S7。

A.6. 实验评估设计的选择

A.6.1. 概述

总共验证了276种独特的蛋白质：来自语言模型的228种设计，来自“无LM”基线的40种设计，以及对应于新设计目标的8条真实序列，这些目标用于固定骨架设计。设计按“FXXX”或“GXXX”方案进行区分，分别表示固定骨架设计和生成，其中XXX是一个索引，范围为[0, 267]，，涵盖所有测试的序列，排除真实值。

实验评估轮次：
总共进行了两轮实验评估，使用了一致的协议：

1. 第 1 = 44 轮固定骨架设计，48个自由生成，4个真实值
2. 第 2 = 95 固定骨架设计，81个自由生成和4个真实值

A.6.2. 固定骨架设计

设计池：
考虑了两个候选设计池进行选择：

1. 200个设计使用不同的随机初始化和随机种子，针对每个新设计 目标。
2. 扩展设计集 - 创建了9,060个额外设计，目标大致平均分配：1QYS (1990x)、6MRS (1500x)、6D0T (1604x)、6W3W (1968x)。

甲骨文质量过滤器：
以下甲骨文指标用于几个（但不是所有）经过实验测试的设计池：

1. 甲骨文 (AlphaFold) 均方根偏差 < 2.5
2. 甲骨文 (AlphaFold) pTM > 0.7
3. SAP得分 < 0.35

第一轮 (48x)：
目标：从语言模型中选择最有前景的设计，使用来自语言模型、甲骨文、Rosetta 以及手动检查的信息。

目标：4个目标，因其 x 射线晶体结构涵盖了一系列典型拓扑（尤其是那些具有高 β-片段含量的，如 6CZJ）和序列长度而被选中。

- 1QYS (前7)
- 6W3W (NTF2)
- 6CZJ (β-桶)
- 6WVS (TIM桶)

来源：每个目标200种子运行（设计池1）过滤
（每个目标）：

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1. Optimization objective $\leq$ 75th percentile, across the 200 seeds
2. ⟨ Oracle Quality Filters ⟩
3. (6CZJ only) Manual filter for beta barrels that aren't fully closed according to the structure oracle.

Selection, post-filtration (per-target):

1. Top-5 by minimum whiten(Oracle RMSD) + whiten(SAP / len)
   - where the operation `whiten(x) = (x - np.mean(x))/np.std(x)` for an array of values `x`.
2. Top-1 by minimum Optimization Objective
3. Top-5 by minimum of (max sequence identity among Blastp hits with $E-value < 1$, per design)
   - BLAST was run against ESM2's exact train set, with all default settings.
4. 1 ground truth

Outcomes: (Evaluated / Soluble / Successful / +Monodisperse)

	Oracle RMSD and SAP	Optimization Objective	Blastp Seq-id	Ground Truth
1QYS	5 / 5 / 5 / 3	1 / 1 / 1 / 1	5 / 5 / 5 / 3	1 / 1 / 0 / 0
6CZJ	5 / 5 / 5 / 3	1 / 1 / 1 / 1	5 / 5 / 5 / 3	1 / 1 / 1 / 1
6W3W	5 / 5 / 2 / 0	1 / 1 / 1 / 0	5 / 5 / 5 / 3	1 / 0 / 0 / 0
6WVS	5 / 5 / 1 / 0	1 / 1 / 0 / 0	5 / 4 / 0 / 0	1 / 1 / 1 / 1

Round 2 (LM vs. no-LM) (64x):
Goal: Compare designs produced with an LM vs. a strong structure predictor (AlphaFold) without an LM, in a matched comparison. An n-gram prior was ablated for the no-LM method. Critically, no filtering was performed for this comparison, meaning that only the optimization objective used for design was considered for selecting top designs. It is interesting that this setting where the least filtering was performed is the one in which language model designs have the highest success rate (19/20). To achieve a clean, unbiased comparison, a new set of targets was chosen for this experiment, differing from those tested in Round 1.
Models: 4 targets * 5 backbones = 20 designs, each

- (20x) $p(y|x)$ = LM Structure Projection, $p(x)$ = LM + n-gram
- (20x) $p(y|x)$ = AlphaFold Distogram, $p(x)$ = Uniform
- (20x) $p(y|x)$ = AlphaFold Distogram, $p(x)$ = n-gram

Targets: 4 targets with crystal structure, different from those tested in Round 1, selected for having diverse structure and secondary structure content.

- 5L33 (NTF2)
- 6D0T (Beta Barrel)
- 6MRS (Foldit, Peak6)
- 6NUK (Foldit, Ferredog-Diesel)

Source: 200 seeds run for each target (Design pool 1)
Filter: None (in order to assess designs exclusively according to the preference of the models used to produce them)
Selection, after filtering (per-target):

- Top-5 by minimum Optimization objective
- 1 ground truth

Outcomes: (Evaluated / Soluble / Successful / +Monodisperse)

	LM Designs	AlphaFold Designs	AF + n-gram Designs	Ground Truth
5L33	5 / 5 / 5 / 0	5 / 2 / 1 / 0	5 / 0 / 0 / 0	1 / 1 / 1 / 1
6D0T	5 / 5 / 4 / 0	5 / 0 / 0 / 0	5 / 2 / 0 / 0	1 / 1 / 0 / 0
6MRS	5 / 5 / 5 / 4	5 / 0 / 0 / 0	5 / 1 / 0 / 0	1 / 1 / 0 / 0
6NUK	5 / 5 / 5 / 5	5 / 0 / 0 / 0	5 / 2 / 0 / 0	1 / 1 / 1 / 1

Round 2 (Distant sequences) (24x):
Goal: Test language model designs that are distant from natural proteins.
Source: Expanded set of designs (Design pool 2)
Targets:

- 1QYS (Top-7)
- 6CZJ (Beta-barrel)
- 6D0T (Beta Barrel)
- 6MRS (Foldit, Peak6)

Filter (per-target):

1. ⟨ Oracle Quality Filters ⟩
2. BlastP Non-redundant minimum $E_value > 1$
   - As a fast test of distance from natural proteins, designed sequences were searched against the BLAST (73) v5 non-redundant database downloaded Sept 12, 2022, with all default settings.
3. Jackhmmer top-hit (by best-domain E-value) TM-score < 0.5

Selection, after filtering: None
Outcomes: (Evaluated / Soluble / Successful / +Monodisperse)

	Outcomes
1QYS	8 / 8 / 7 / 4
6CZJ	6 / 6 / 6 / 4
6D0T	2 / 2 / 1 / 0
6MRS	8 / 7 / 7 / 4

Round 2 (Motifs) (11x):
Goal: Highlight interesting design motifs generated by the language model during fixed backbone design. Source: Expanded set of designs (Design pool 2)
Targets:

- 1QYS (Top-7)
- 6CZJ (Beta Barrel)
- 6D0T (Beta Barrel)

Filter: ⟨ Oracle Quality Filters ⟩
Selection, after filtering:

1. **Detection of buried polars residues.** A heuristic function was coded to roughly assess the number of polar amino acids not on the surface of the protein. Per-protein "depth" and solvent-accessible surface area (SASA) were calculated with the ShakeRupley and ResidueDepth classes from the BioPython (74) library respectively. The number of polar (IUPAC codes D,E,R,H,K) amino acids whose SASA percentile < 0.4 or depth percentile < 0.6 (across all amino acids in the designed sequence) were summed. All designs with a sum > 12 were selected for experimental evaluation.
2. **Detection of hydrogen-bond networks.** HBNetScore (Boyken et al. 2016) from

1. 优化目标 ≤75 百分位数，跨越200种子 2. ⟨ Oracle质量过滤器 ⟩3.（仅限6CZJ）手动过滤不完全闭合的β折叠结构，根据结构甲骨文。选择，过滤后（按目标）：

1. 前五名按最小白化（甲骨文 RMSD） +白化（SAP / 长度）•其中操作白化(x) = (x - np.mean(x))/np.std(x) 用于值 x 的数组。

2. 第一名按最小优化目标
3. 前五名按（与 $E-value < 1$ 的 Blastp 命中中最大序列相似性）最小值• BLAST 针对 ESM2 的精确训练集运行，使用所有默认设置。 4. 1 真实值

结果：（已评估 / 可溶 / 成功 / +单分散）

	甲骨文 RMSD 和 SAP	优化 优化目标	Blastp 序列标识	地面 真相
1QYS	5 / 5 / 5 / 3	1 / 1 / 1 / 1	5 / 5 / 5 / 3	1 / 1 / 0 / 0
6CZJ	5 / 5 / 5 / 3	1 / 1 / 1 / 1	5 / 5 / 5 / 3	1 / 1 / 1 / 1
6W3W	5 / 5 / 2 / 0	1 / 1 / 1 / 0	5 / 5 / 5 / 3	1 / 0 / 0 / 0
6WVS	5 / 5 / 1 / 0	1 / 1 / 0 / 0	5 / 4 / 0 / 0	1 / 1 / 1 / 1

第二轮 (语言模型 vs. 无LM) (64x):
目标：在匹配比较中，比较使用语言模型生成的设计与没有语言模型的强结构预测器（AlphaFold）生成的设计。无LM方法中去除了n-gram先验。关键是这一比较没有进行任何过滤，这意味着在选择顶级设计时，仅考虑用于设计的优化目标。有趣的是，在过滤最少的设置中，语言模型设计的成功率最高（19/20）。为了实现清晰、无偏的比较，本次实验选择了一组新的目标，与第一轮测试的目标不同。

模型：4个目标 * 5个框架 = 20设计，每个

- (20x) $p(y|x)$ = 语言模型结构投影， $p(x)$ = 语言模型 +n-gram
- (20x) $p(y|x)$ = AlphaFold Distogram， $p(x)$ = 均匀
- (20x) $p(y|x)$ = AlphaFold Distogram, $p(x)$ = n-gram

目标：4个具有晶体结构的目标，与第一轮中测试的结构不同，因具有多样的结构和二级结构内容而被选中。

- 5长度33 (NTF2)
- 6D0T (贝塔桶)
- 6 MRS (Foldit, Peak6)
- 6NUK (Foldit, Ferredog-Diesel)

来源：针对每个目标的200种子运行（设计池1）
过滤：无（以便根据用于生成设计的模型的偏好单独评估设计）

筛选后的选择（按目标）：

- 前五名按最小优化目标
- 1真实值

结果：（已评估 / 可溶 / 成功 / +单分散）

	LM 设计	AlphaFold 设计	AF +n-gram 设计	真实值 真实值
5L33	5 / 5 / 5 / 0	5 / 2 / 1 / 0	5 / 0 / 0 / 0	1 / 1 / 1 / 1
6D0T	5 / 5 / 4 / 0	5 / 0 / 0 / 0	5 / 2 / 0 / 0	1 / 1 / 0 / 0
6MRS	5 / 5 / 5 / 4	5 / 0 / 0 / 0	5 / 1 / 0 / 0	1 / 1 / 0 / 0
6NUK	5 / 5 / 5 / 5	5 / 0 / 0 / 0	5 / 2 / 0 / 0	1 / 1 / 1 / 1

第二轮 (远序列) (24x):目标：测试与天然蛋白质远离的语言模型设计。来源：扩展设计集 (设计池 2) 目标：

- 1QYS（前七名）
- 6CZJ（β-桶）
- 6D0T (贝塔桶)
- 6MRS (Foldit, Peak6)

过滤器（按目标）：

1. ⟨ 甲骨文质量过滤器 ⟩

2. BlastP 非冗余最小值 $E\ value > 1$• 作为对天然蛋白质的距离的快速测试，设计的序列在 BLAST (73) v5 非冗余数据库中被搜索，该数据库于2022年9月12日下载，使用所有默认设置。

3. Jackhmmer 最高命中（按最佳域 E 值）TM-score < 0.5

过滤后的选择：无
结果：（已评估 / 可溶 / 成功 / +单分散）

	结果
1QYS	8 / 8 / 7 / 4
6CZJ	6 / 6 / 6 / 4
6D0T	2 / 2 / 1 / 0
6MRS	8 / 7 / 7 / 4

第二轮（基序）（11x）：
目标：突出在固定主链设计过程中由语言模型生成的有趣设计基序。来源：扩展设计集（设计池 2）

目标：

- 1QYS (前7名)
- 6CZJ (贝塔桶)
- 6D0T (贝塔桶)

过滤器：⟨ 甲骨文质量过滤器 ⟩选择，过滤后：

1. **检测埋藏的极性残基**。编码了一个启发式函数，大致评估了不在蛋白质表面的极性氨基酸的数量。每个蛋白质的"深度"和溶剂可接触表面积（SASA）分别使用BioPython中的ShakeRupley和ResidueDepth类计算（74）。对所有氨基酸的SASA百分位 < 0.4或深度百分位 < 0.6 (在设计序列中所有氨基酸中)进行了求和。所有具有总和 > 12的设计被选中进行实验评估。

2.**氢键网络的检测**。HB- NetScore (Boyken等人 2016) 来源于

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`pyrosetta.rosetta.protocols.hbnet` was used to detect hydrogen bond networks in designs. An HB-NetScore score term was added to the `beta_nov16` Rosetta energy function (32) with weight 1, and that component score was calculated for each design. All designs with a score < -100 were selected for experimental evaluation.

Outcomes: (Evaluated / Soluble / Successful / +Monodisperse)

	Buried Polars	HBnets
1QYS	1 / 1 / 1 / 1	0 / 0 / 0 / 0
6CZJ	1 / 1 / 1 / 0	5 / 5 / 5 / 1
6D0T	2 / 2 / 2 / 0	1 / 1 / 1 / 0
6W3W	1 / 1 / 0 / 0	0 / 0 / 0 / 0

A.6.3. Free Generation

Design pools:
Two pools of free generations, using the same protocol, were considered for selection:

1. 10k generations, produced before Round 1 of experimental evaluation.
2. An additional set of 15k generations was produced, allowing selection of 25k total generations for Round 2 of experimental evaluation.

AlphaFold and Rosetta Quality Filters:
We use the structure oracle (AlphaFold) confidence predictions pLDDT and pTM, and select proteins where both pTM > 0.7 and pLDDT > 70. We then apply all three remaining categories of filters introduced in Appendix A.4 to select the promising free generation designs for experimental evaluation. From the 25k free generations, 20.1% ($N = 5,198$) pass these filters.

Round 1 (48x):
Source: 10k generations
Filtering:

- ⟨ AlphaFold and Rosetta Quality Filters ⟩

Selection:

1. 12 sequences with TM-score to nearest neighbor in PDB < 0.64
2. 36 sequences without significant BLAST hits; subsampled of the top 2 per structural cluster.
3. Structural cluster defined by agglomerative clustering based on pairwise structural similarity at a TM-score threshold of 0.75 across all *in silico* filtered generations.

Outcomes: (Evaluated / Soluble / Successful / +Monodisperse)

	Outcomes
Generations	48 / 46 / 22 / 11

Round 2 (Manual selection) (24x):
Source: 10k generations
Filter: ⟨ AlphaFold and Rosetta Quality Filters ⟩
Selection: Manual selection of a structurally interesting and diverse subset of *in silico* validated proteins.
Outcomes: (Evaluated / Soluble / Successful / +Monodisperse)

	Outcomes
Generations	24 / 22 / 14 / 8

Round 2 (Distant generations) (57x):
Source: 25k generations
Filter: ⟨ AlphaFold and Rosetta Quality Filters ⟩
Selection:

1. From the filtered set of proteins, we select a small subset of designs for experimental evaluation that are distant from natural proteins. For the sequence novelty of proteins, instead of following the approach stated at Appendix A.5 we used a separate tool (BLAST) to assess sequence novelty, so more diverse proteins are selected across the graph of Fig. 4D.
2. Sequences with no significant matches by BLAST (min E-value > 1) against UniRef90 are selected.
3. Out of the above, sequences with TM-score < 0.5 of top hit by Jackhmmer are selected.

Outcomes: (Evaluated / Soluble / Successful / +Monodisperse)

	Outcomes
Generations	57 / 56 / 35 / 21

A.7. Experimental evaluation

A.7.1. Plasmid construction

Plasmids for expressing proteins were constructed from synthetic DNA according to the following procedure, as in (75): Linear DNA fragments (Integrated DNA Technologies, IDT eblocks) encoding design sequences and including overhangs suitable for a BsaI restriction digest were cloned into custom target vectors using Golden Gate Assembly. All subcloning reactions resulted in C-terminally HIS-tagged constructs:MSG-design-GSGSHHWGSTHHHHHH (entry vector LM627), where the underlined sequence is the SNAC-tag (68) used for cleaving the HIS-tag (cleaving not used in this work), ot also contains a TRP residue to ensure proteins have measurable absorbance at 280 nm.

The entry vectors for Golden Gate cloning are modified pET29b+ vectors that contain a lethal ccdb gene between the BsaI restriction sites that is both under control of a constitutive promoter and in the T7 reading frame. The lethal gene reduces background by ensuring that plasmids that do not contain an insert (and therefore still carry the lethal gene) kill transformants. The vectors were propagated in ccdb resistant NEB Stable cells (New England biolabs C3040H, always grown from fresh transformants). LM627 is available via addgene (ID: 191551)

Golden Gate reactions (1 uL per well) were set up on a 96 well PCR plate using an ECHO acoustic liquid handler (Labcyte ECHO 525, Beckmann Coulter):

- 10x T4 Buffer 0.5 uL 10x T4 Buffer (New England Biolabs B0202S)
- Vector 3 fmol Vector (LM627)
- BsaI-HFv2 3U 0.0.06 uL BsaI-HFv2 (New England Biolabs

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`pyrosetta.rosetta.protocols.hbnet` 用于检测设计中的氢键网络。添加了一个HB-NetScore 分数项到 `beta nov16` 罗塞塔能量函数（32）权重为 1，并为每个设计计算该组件分数。所有得分< -100 的设计被选择用于实验评估。

结果：（已评估 / 可溶 / 成功 / +单分散）

	埋藏的极性	HBnets
1QYS	1 / 1 / 1 / 1	0 / 0 / 0 / 0
6CZJ	1 / 1 / 1 / 0	5 / 5 / 5 / 1
6D0T	2 / 2 / 2 / 0	1 / 1 / 1 / 0
6W3W	1 / 1 / 0 / 0	0 / 0 / 0 / 0

A.6. 3. 自由生成

设计池：
考虑了两个自由生成的池，使用相同的协议进行选择：

1. 10k个生成，产生于第一轮实验评估之前。

2. 额外产生了一组15k个生成，使得可以为第二轮实验评估选择总共25k个生成。

AlphaFold和Rosetta质量过滤器：
我们使用结构甲骨文（AlphaFold）信心预测pLDDT和pTM，并选择在两者pTM > 0.7 和pLDDT > 70均符合要求的蛋白质。然后，我们应用附录A.4中介绍的所有剩余三类过滤器，以选择有前景的自由生成设计进行实验评估。在这25k个自由生成中，20.1%（$N = 5,198$）通过了这些过滤器。

第一轮 (48x):
来源：10k代数过滤：

- ⟨ AlphaFold 和 Rosetta质量过滤器 ⟩

选择：

1. 12个序列与PDB中最近邻的TM-score< 0.64

2. 36个序列没有显著的BLAST命中；是每个结构簇中前2个的子样本。

3. 结构簇由基于成对结构相似性的聚合聚类定义，其TM-score阈值为0.75，适用于所有计算机模拟过滤的世代。

结果：（已评估 / 可溶 / 成功 / + 单分散）

	结果
世代	48 / 46 / 22 / 11

第二轮（手动选择）（24x）：
来源：10k 代
过滤器：⟨ AlphaFold 和 Rosetta质量过滤器 ⟩
选择：手动选择一个在结构上有趣且多样的子集，包含经过计算机模拟验证的蛋白质。
结果：（已评估 / 可溶 / 成功 / +单分散）

	结果
代数	24 / 22 / 14 / 8

第二轮（远距离世代）（57倍）：
来源：25k世代
过滤器：⟨ AlphaFold 和 Rosetta质量过滤器 ⟩选择：

1. 从过滤后的蛋白质集中，我们选择了一小部分设计进行实验评估，这些设计与天然蛋白质相隔较远。在蛋白质的序列新颖性方面，我们没有遵循附录A.5 中所述的方法，而是使用一个单独的工具（BLAST）来评估序列的新颖性，以便在图4D中选择更多样化的蛋白质。

2. 在BLAST中没有显著匹配的序列（最小E值 $>$

1) 针对UniRef90进行选择。

3. 在上述结果中，以Jackhmmer的最佳命中序列的TM-score < 0.5 进行选择。结果：（已评估 / 可溶 / 成功 / +单分散）结果世代

	57 / 56 / 35 / 21

A.7. 实验评估

A.7.1. 质粒构建

用于表达蛋白质的质粒是根据以下程序，由合成DNA构建的，如同在（75）所示：线性DNA片段（综合DNA技术，IDT eblocks）编码设计序列，并包括适合BsaI限制性消化的突出部分，使用金门组装法克隆到定制目标载体中。所有的亚克隆反应都产生了C端带有HIS标签的构建体：MSG-design-GSGSHHWGSTHHHHHH（入口载体LM627），下划线部分是用于切割HIS标签的SNAC标签（68），（此工作中未使用切割），它还包含一个色氨酸残基，以确保蛋白质在280纳米处具有可测量的吸光度。

金门克隆的入口载体是经过修改的pET29b+载体，这些载体在BsaI限制性位点之间包含一个致死的ccdb基因，该基因是由一个组成性启动子控制，并且在T7阅读框内。致死基因通过确保不包含插入序列的质粒（因此仍携带致死基因）杀死转化体，减少了背景。载体在对ccdb耐药的NEB稳定细胞中传播（新英格兰生物实验室C3040H，总是从新鲜转化体中培养）。LM627可通过addgene获取（ID: 191551）

金门反应（每孔1微升）是在使用ECHO声学液体处理器（Labcyte ECHO 525，Beckmann Coulter）的96孔PCR板上设置的：

- 10× T4缓冲液 0.5微升 10× T4缓冲液（新英格兰生物实验室 B0202S）
- 载体 3 fmol 载体（LM627）
- BsaI-HFv2 3U 0.0.06微升 BsaI-HFv2（新英格兰生物实验室

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R3733L)
- T4 Ligase 100U 0.1uL T4 Ligase (New England Biolabs M0202L)
- (6 fmol) linear DNA fragment, at typically of 4 ng/uL stock
- Complete with nuclease-free water to 5 uL total reaction volume.

The reactions were incubated at 37 °C for 20 minutes, followed by 5 min at 60 °C (IKA Dry Block Heater 3).

A.7.2. Small-scale protein solubility screen

For experimental screens, Golden Gate reaction mixtures were transformed into BL21(DE3) (New England Biolabs) as follows: 1 uL of reaction mixture was incubated with 6 uL of competent cells on ice in a 96 well PCR plate. The mixture was incubated on ice for 30 minutes, then heat-shocked for 10 s at 42 °C in a block heater (IKA Dry Block Heater 3), then rested on ice for 2 minutes. Subsequently, 100 uL of room temperature SOC media (New England Biolabs) was added to the cells, followed by incubation at 37 °C with shaking at 1000 rpm on a Heidolph Titramax1000 / Incubator 1000.

The transformations were then grown in a 96 well deep-well plate (2 mL total well volume) in autoclaved LB media supplemented with 50 μg mL-1 Kanamycin at 37 °C and 1000 rpm. In the following protocols all growth plates were covered with breathable film (Breathe Easier, Diversified Biotech) during incubation.

The following day, glycerol stocks were made from the overnight cultures (100 uL of 50% [v/v] Glycerol in water mixed with 100 uL bacterial culture, frozen and kept at -80 °C. Subsequently, two 96 deep well plates were prepared with 900 uL per well of autoclaved Terrific Broth II (MP biomedicals) supplemented with 50 μg mL-1 Kanamycin, and 100 uL of the overnight culture were added and grown for 1.5 h at 37 °C, 1200 rpm (Heidolph Titramax1000 / Incubator 1000). The cultures were then induced with IPTG by adding 10 uL of 100 mM (final concentration approximately 1 mM) per well with an electric repeater pipette (Eppendorf, E4x series), and grown for another 4 h at 37 °C, 1200 rpm. Cultures were combined into a single 96 well plate for a total culture volume of 2 mL and harvested by centrifugation at 4000 x g for 5 min. Growth media was discarded by rapidly inverting the plate, and harvested cell pellets were either processed directly, or frozen at -80 °C.

Proteins were purified by HIS tag-based Immobilized metal affinity chromatography (IMAC). Bacterial pellets were resuspended and lysed in 100 uL per 1 mL of culture volume B-PER chemical lysis buffer (Thermo Fisher Scientific) supplemented with 0.1 mg mL-1 Lysozyme (from a 100 mg mL-1 stock in 50% [v/v] Glycerol, kept at -20 °C, Millipore Sigma), 50 Units of Benzonase per mL (Merck/Millipore Sigma, stored at - 20 °C), and 1 mM PMSF (Roche Diagnostics, from a 100 mM stock kept in Propan-2-ol, stored at room temperature). The plate was sealed with an aluminum foil cover and vortexed for several minutes until the bacterial pellet was completely resuspended (on a Vortex Genie II, Scientific Industries). The lysate was incubated, shaking for 5 minutes, before being spun down at 4000 x g for 15 minutes. In the meantime, 50 uL of Nickel-NTA resin bed volume (Thermo Scientific, resin was regenerated before each run and stored in 20% [v/v] Ethanol) was added to each well of a 96 well fritted plate (25 μm frit, Agilent 200953-100). To increase wash step speed, the resin was equilibrated on a plate vacuum manifold (Supelco, Sigma) by drawing 3 x 500 uL of Wash buffer (20 mM Tris, 300 mM NaCl, 25 mM Imidazole, pH 8.0) over the resin using the vacuum manifold at its lowest pressure setting.

The supernatant of the lysate was extracted after the spin down and applied to the equilibrated resin and allowed to slowly drip through over 5 minutes. Subsequently the resin was washed on the vacuum manifold with 3 x 500 uL per well of Wash buffer. Lastly the fritted plate spouts were blotted on paper towels to drain excess Wash buffer. Then 200 uL of Elution buffer (20 mM Tris, 300 mM NaCl, 500 mM Imidazole, pH 8.0) was applied to each well and incubated for 5 minutes before eluting the protein by centrifugation at 1500 x g for 5 minutes into a 96 well collection plate. Eluate was stored at 4 °C.

A.7.3. Size exclusion chromatography

Designs were subject to a solubility screen and size exclusion chromatography (SEC), in the laboratory, using an S75 5/150 column (Cytiva) at 0.45 mL / min run speed in 20 mM Phosphate, 100 mM NaCl at pH 7.4 on an Akta pure (Cytiva) with an autosampler module. Absorbance was monitored at 280 nm. All designs and buffers were sterile filtered through 0.2 micrometer filters before being run on the instruments.

A.7.4. Classification of experimental outcomes

Designs are labeled as soluble if the total soluble yield (in mg) from the 4x1mL prep is $\leq$ 0.05 mg. Designs are labeled as successful if they are soluble and if rightmost peak returned by `scipy.signal.find_peaks(SEC_trace_y_vals, height=0.1, prominence=0.01)` (where `SEC_trace_y_vals` is normalized to the range [0,1]), is within one standard deviation of a calibration curve relating elution volume to hydrodynamic radius, described above. All ground-truth controls eluted at their expected retention volume or slightly after, thus confirming their monomeric states (except for 1QYS, which is known from the literature to form a homodimer (76). Designs are additionally considered monodisperse if the `find_peaks()` call returns a single peak at the expected elution volume for the given molecular weight as assessed by the calibration curve. The calibration curve was recorded with the Lower Molecular Weight calibration kit (LMW kit, Cytiva) on the S75 5/150 column (Cytiva) in the same running buffer as used for the designs.

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R3733L)
- T4连接酶 100U 0.1uL T4连接酶（新英格兰生物实验室 M0202L ）
- (6 fmol) 线性DNA片段，通常为4 ng/uL的库存
- 用无核酸酶的水补充至5 uL的反应总体积。

反应在37 °C下培养20分钟，然后在60 °C下培养5分钟（IKA干块加热器3）。

A.7.2. 小规模蛋白质溶解性筛选

在实验筛选中，黄金门反应混合物被转化为BL21(DE3)（新英格兰生物实验室），具体如下：1 µL的反应混合物与6 µL的感受态细胞在冰上于96孔PCR板中孵育。混合物在冰上孵育30分钟，然后在42 °C的加热器（IKA干块加热器3）中热激10秒，之后在冰上静置2分钟。随后，向细胞中添加100 µL室温SOC培养基（新英格兰生物实验室），然后在37 °C和1000 rpm的摇床上孵育。

转化产物随后在灭菌的LB培养基中于96孔深孔板（每孔总容积2 mL）中生长，培养基中补充了50 μg mL-1的卡那霉素，培养温度为37 °C，转速为1000 rpm。在接下来的所有培养过程中，生长板都用透气膜（透气膜，多元生物技术）覆盖。

第二天，从过夜培养物制备了甘油库存（50% [v/v] 甘油与100 µL细菌培养物混合，冷冻保存于-80 °C）。随后，准备了两个96孔深孔板，每孔加入900 µL灭菌的特级肉汤II（MP生物医学），并补充50 μg mL-1的卡那霉素，接着添加100 µL的过夜培养物，在37 °C、1200 rpm（海多夫 Titramax1000 / 培养箱1000）下培养1.5小时。然后通过在每孔中添加10 µL的100 mM IPTG（最终浓度约为1 mM）来诱导培养，并继续在37 °C、1200 rpm下培养4小时。将培养物合并到一个96孔板中，最终培养体积为2 mL，通过以4000 x g离心5分钟进行收获。快速颠倒板子丢弃生长培养基，收获的细胞沉淀可直接处理，也可冷冻保存于-80 °C。

蛋白质通过基于HIS标签的固定化金属亲和色谱(IMAC)进行纯化。细菌沉淀在每毫升培养体积100 uL B-PER化学裂解缓冲液（赛默飞世尔科技）中重新悬浮并裂解，该缓冲液中补充了0.1 mg mL-1 溶菌酶（来自50% [v/v] 甘油的100 mg mL-1原液，保存在-20 °C，米利步西门子），每毫升50单位苄腺苷酶（默克/米利步西门子，存放于-20 °C），以及1 mM PMSF（罗氏诊断，来自存放于室温的丙醇的100 mM原液）。该板被密封。

铝箔盖并涡旋几分钟，直到细菌沉淀完全重新悬浮（在Vortex Genie 上）II，科学工业）。裂解液在摇动下孵育5分钟，然后在4000 x g下离心15分钟。与此同时，将50 uL的镍-NTA树脂床体积（赛默飞科学，树脂在每次运行前再生并储存在20%[v/v] 乙醇中）添加到96孔滤板的每个孔中（25 μm滤，安捷伦200953-100）。为了提高洗涤步骤的速度，树脂在真空歧管板（Supelco，Sigma）上平衡，通过真空歧管以最低压力设置抽取3 x 500 uL的洗涤缓冲液（20 mM Tris，300 mM 氯化钠，25 mM 咪唑，pH 8.0）在树脂上流动。

离心后提取裂解液的上清液，施加到已平衡的树脂上，并让其在5分钟内缓慢滴流。随后，树脂在真空歧管上以每孔3 x 500 uL的洗涤缓冲液进行洗涤。最后，将滤板的喷嘴在纸巾上轻轻拍打以排除多余的洗涤缓冲液。然后，向每个孔中加入200 uL的洗脱缓冲液（20 mM Tris，300 mM 氯化钠，500 mM 咪唑，pH 8.0），并孵育5分钟，然后通过在1500 x g的离心将蛋白质洗脱到96孔收集板中。洗脱液在4 °C储存。

A.7.3. 尺寸排除色谱

在实验室中，设计通过可溶性筛选和尺寸排除色谱（SEC）进行测试，使用赛信维的 S75 5/150柱，以 0.45 mL / min 的运行速度在 20 mM 磷酸盐、100 mM 氯化钠，pH 7.4 的条件下，用 Akta pure（赛信维）和自动取样模块进行。 在 280 纳米处监测吸光度。所有设计和缓冲液在运行仪器之前通过 0.2微米滤器进行无菌过滤。

A.7. 4. C实验结果的分类如果从 4x1mL 准备中获得的总可溶产量（以毫克为单位）$\leq$ 0 $\leqslant$ 0.05 mg，则设计被标记为可溶。如果设计可溶，并且由`scipy.signal.findpeaks(SEC trace y vals,height=0`= 0.1, prominence=0= 0.01)(其中`SECtrace y vals` 被归一化到范围 [0,1])，那么其最右侧峰值在与上面描述的与洗脱体积相关的水动力半径的校准曲线的一个标准差以内。所有基准控制都在其预期的保留体积处或稍后洗脱，从而确认了其单体状态（除了 1QYS，它在文献中被认为形成一个同源二聚体（76）。如果`find peaks()`调用在所给分子量的预期洗脱体积处返回一个单峰，则设计被视为单分散。校准曲线是在使用与设计相同的运行缓冲液的 S75 5/150柱（赛信维）上使用低分子量校准工具包（LMW 工具包，赛信维）记录的。

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B. Supplementary Figures

Overview of Supplementary Figures:

- Fig. S1: Overview of the *De Novo* Target Set.
- Fig. S2: The language model understands *de novo* proteins.
- Fig. S3: Language model understanding of experimentally tested *de novo* targets.
- Fig. S4: Fixed backbone designs succeed on all backbones tested experimentally.
- Fig. S5: Analysis of fixed backbone designs across methods.
- Fig. S6: Fixed backbone designs, comparison to natural proteins.
- Fig. S7: Detailed Analysis of Motifs.
- Fig. S8: Free Generation: Experimental Successes
- Fig. S9: Free Generations, comparison to natural proteins
- Fig. S10: Top structure-based matches in PDB for free generations
- Fig. S11: Overview of Experimental Evaluations for all tested designs.

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B. 补充图

补充图表概览：

- 图.S1：新设计目标集合的概述。
- 图.S2：语言模型理解新设计蛋白质。
- 图.S3：语言模型对实验测试的新设计目标的理解。
- 图.S4：固定骨架设计在所有实验测试的骨架上均取得成功。
- 图。S5：针对不同方法的固定骨架设计分析。
- 图。S6：固定骨架设计，与天然蛋白质的比较。
- 图。S7：基序的详细分析。
- 图。S8：自由生成：实验成功案例
- 图。S9：自由生成，与天然蛋白质的比较
- 图。S10：自由生成的最佳结构匹配在PDB中的表现
- 图。S11：所有测试设计的实验评估概述。

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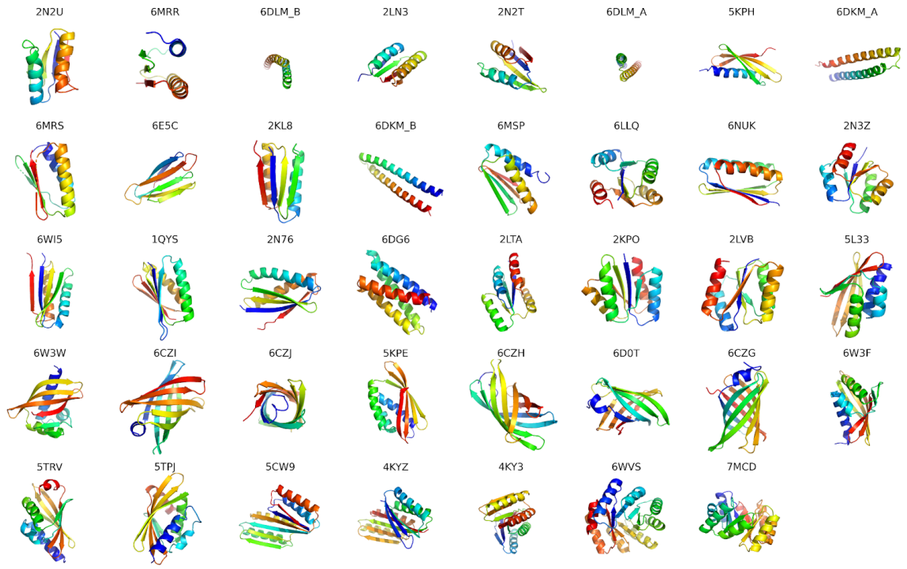

Figure S1. Overview of the De Novo *Target Set.* Crystal/NMR structures for all proteins in the *de novo* target set (N = 39). Targets are sorted by increasing sequence length (range $[67, 184]$). Residues are rainbow-colored from N- to C- terminus. Targets were hand-selected for being *de novo* designed, possessing a high quality experimental structure, and for being structurally diverse: targets possess a wide variety of folds (e.g. alpha-bundle, Rossman, NTF2, Beta-barrel, Ferredoxin, TIM-barrel) and secondary structure content.

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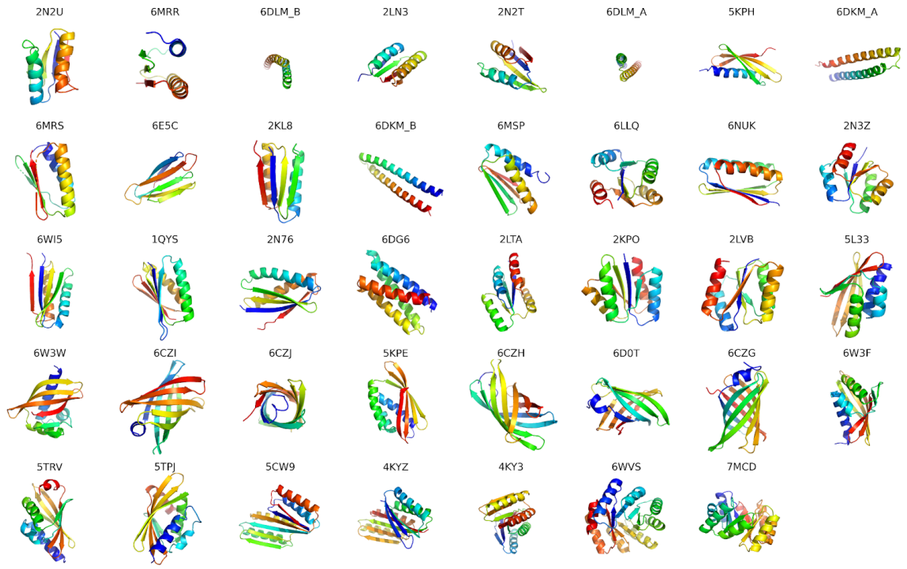

图 *S1*。 目标集概述新设计目标集。所有蛋白质在新设计目标集中（N = 39）的晶体/NMR结构。目标按序列长度递增排序（范围 $[67, 184]$）。残基从N-端到C-端呈彩虹色。目标是手动选择的，需满足条件为新设计，拥有高质量的实验结构，以及在结构上具有多样性：目标具有多种折叠形式（例如，α-束、罗斯曼、NTF2、β-桶、铁氧还蛋白、TIM桶）和二级结构内容。

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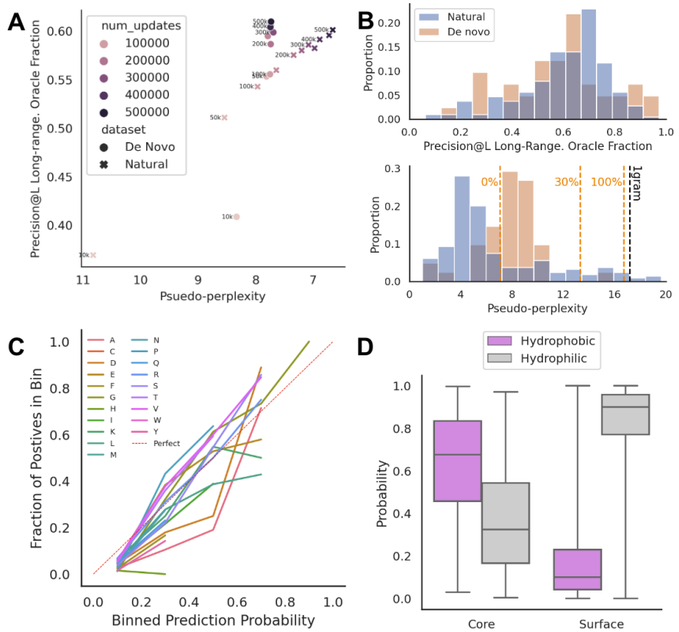

Figure S2. The language model understands de novo *proteins.* (**A**) Contact- and sequence- prediction statistics across ESM2_650k pretrain checkpoints. X-axis shows pseudo-perplexity of sequences under ESM2. Y-axis shows precision of top-L predicted long-range ($\geq$ 24 separation) contacts by the structure projection as a fraction of the maximum achievable value, where L is sequence length. (**B**) Histograms of contact- and sequence- prediction statistics (normalized by dataset size) for natural (blue) and *de novo* (orange) proteins, according to the final ESM2_650k model checkpoint, which is used throughout this study. Despite only being trained on natural sequences, the structure projection from the language model achieves similar structural scores for the considered sets of natural and *de novo* proteins (top). (Bottom) Pseudo-perplexity is better (lower) for natural sequences, but both natural and *de novo* sequences are well understood compared to 30% scrambled *de novo* sequences, 100% scrambled *de novo* sequences, and a unigram model of amino acid frequencies in UniRef50 2018_03, as baselines. (**C**) Calibration plot for predictions of masked amino acids in *de novo* sequences, by the language model. Perfect calibration is a diagonal line from (0, 0) to (1, 1), indicated in dashed red. Due to the low number of sequences in the *de novo* target set (N = 39), true positive counts for binned probabilities with $<$ 5 samples were omitted. (**C**) Masked amino acid prediction correctly places hydrophobic residues in the cores of *de novo* protein structures. Boxplot shows total probability mass for hydrophobic (pink) and hydrophilic (light gray) amino acids during mask-1-out prediction, on *de novo* sequences. Core and surface labels are determined by the number of C-alpha neighbors within 10Åof each C-alpha atom (core: $\geq$ 24 neighbors; surface: $<$ 16 neighbors).

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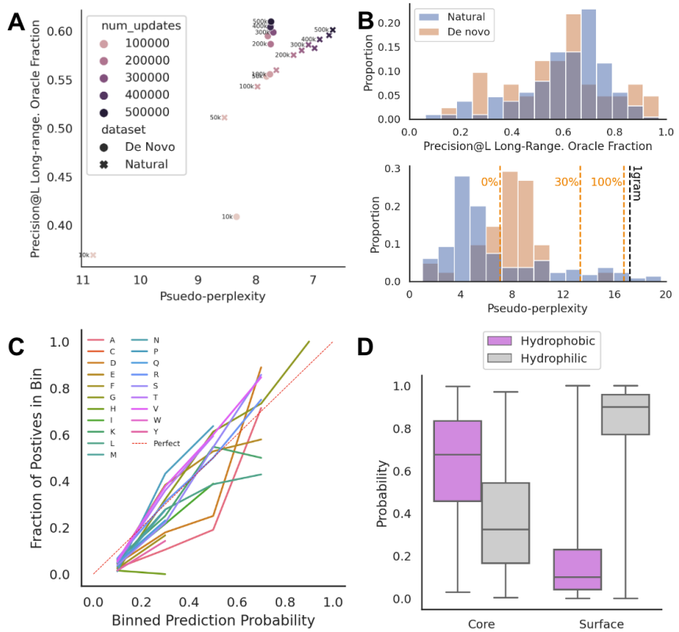

图 S2. 语言模型理解新型蛋白质。 (A) 在 ESM2 650k 预训练检查点下接触和序列预测统计数据。X 轴显示 ESM2 下序列的伪困惑度。Y 轴显示通过结构投影预测的长程接触（$\geq$ 24 分离）中前 L 的精确度，相对于最大可实现值的比例，其中 L 为序列长度。 (B) 根据最终的 ESM2 650k 模型检查点（在本研究中使用），天然（蓝色）和新型（橙色）蛋白质的接触和序列预测统计的直方图（按数据集大小标准化）。尽管只在天然序列上进行训练，但语言模型的结构投影对考虑的天然和新型蛋白质集合（顶部）获得了类似的结构分数。 (底部) 对于天然序列，伪困惑度更好（更低），但与 30% 混合的新型序列、100% 混合的新型序列和 UniRef50 2018 03 中氨基酸频率的单词模型相比，天然序列和新型序列的理解良好，作为基线。 (C) 语言模型对新型序列中被掩蔽氨基酸的预测的校准图。完美的校准是一条从 (0, 0) 到 (1, 1) 的对角线，以虚线红色表示。由于新设计靶点集中序列数量较少（N = 39），省略了有 $<$ 5 样本的分箱概率的真实正例计数。 (C) 被掩蔽氨基酸预测正确地将疏水性残基放置在新型蛋白质结构的核心中。箱线图显示在新型序列的掩蔽-1-输出预测中疏水性（粉色）和亲水性（浅灰色）氨基酸的总概率质量。核心和表面标签由每个 C-alpha 原子内 10 ˚Å 内的 C-alpha 邻居数（核心： $\geq$ 24 邻居；表面： $<$ 16 邻居）决定。

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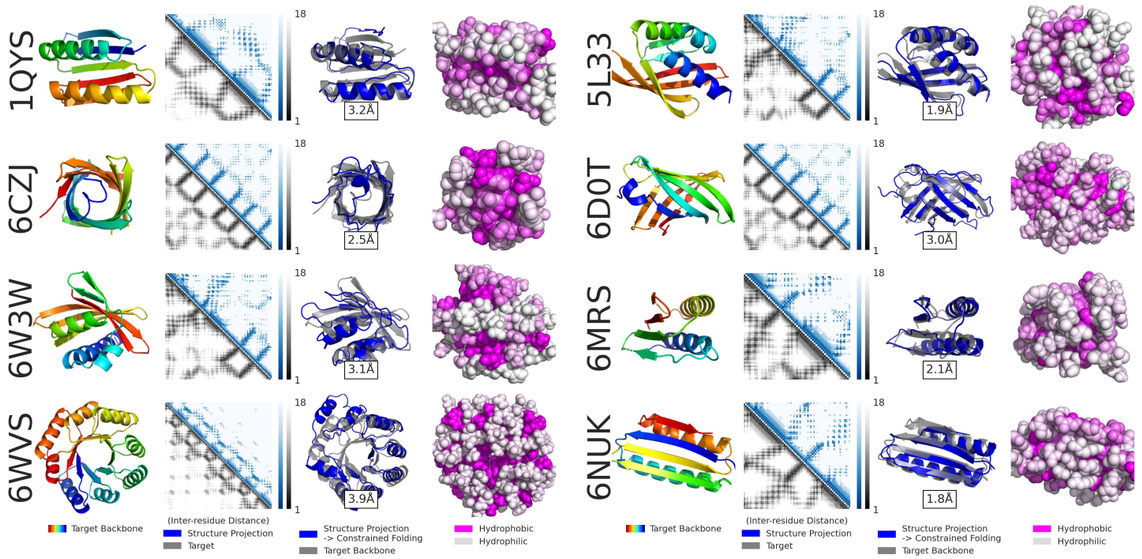

Figure S3. Language model understanding of experimentally tested de novo *targets.* This figure showcases understanding of *targets*, given their (ground-truth) sequences. Fixed backbone designs were produced for all targets in the *de novo* set, but 8 targets in particular had their designs tested experimentally. Each of the rows in the two overall columns above showcases understanding of a single target. The leftmost column in each row shows the target (backbone, x-ray crystal) structure, rainbow-colored from N- to C- terminus. Second and third columns show structural understanding of the target structure by the language model's structure projection, given only the held-out (Appendix A.1.2) *de novo* sequence. The second column compares predicted and true binned inter-residue distances, the structure projection's native output. The third column compares the target backbone (gray) with the backbone derived from constrained folding of the language model's structure projection distogram (blue), folded with trRosetta2's folding script (69). RMSDs in this column range from 1.8Åto 3.9Å. The fourth column shows total probability mass of hydrophobic (magenta) vs. hydrophilic (white) amino acid predictions from the language model, after sequentially masking each position in the ground truth sequence. Side chains on the surface of *de novo* structures are generally predicted to be more hydrophilic than those in the core.

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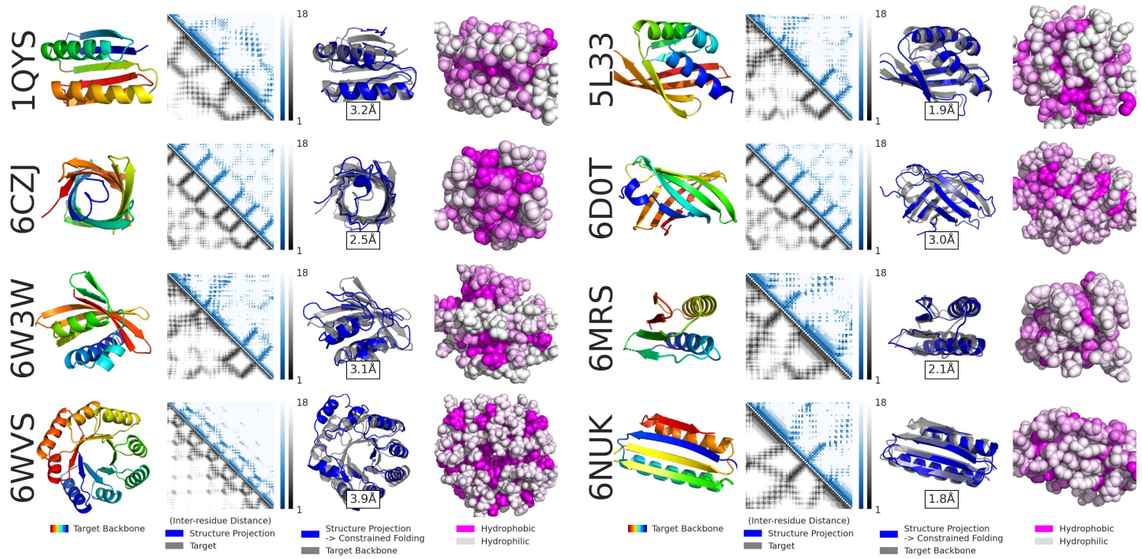

图 *S3*。 语言模型对实验测试的 新设计 目标的理解。 该图展示了对 目标的理解，基于它们（真实值）序列。针对 新设计 集中的所有目标都产生了固定骨架设计，但其中 8 个目标的设计经过了实验测试。上述两个整体列中的每一行展示了单个目标的理解。每一行中最左侧的列显示了目标（主链，X 射线晶体）结构，呈现从 N 端到 C 端的彩虹色。第二和第三列显示了语言模型的结构投影对目标结构的结构理解，仅基于保留的（附录 A.1.2） 新设计 序列。第二列比较了预测的和真实的分箱残基间距离，这是结构投影的本地输出。第三列比较了目标主链（灰色）与来自语言模型的结构投影 distogram（蓝色）受限折叠生成的主链，后者通过 trRosetta2 的折叠脚本（69）进行折叠。该列中的均方根偏差范围为 1。8 ˚A 到 3。9 ˚A。第四列展示了语言模型的疏水性（品红色）与亲水性（白色）氨基酸预测的总概率质量，经过逐个遮罩真实值序列中的每个位置。 新设计 结构表面的侧链通常被预测为比核心中的更亲水。

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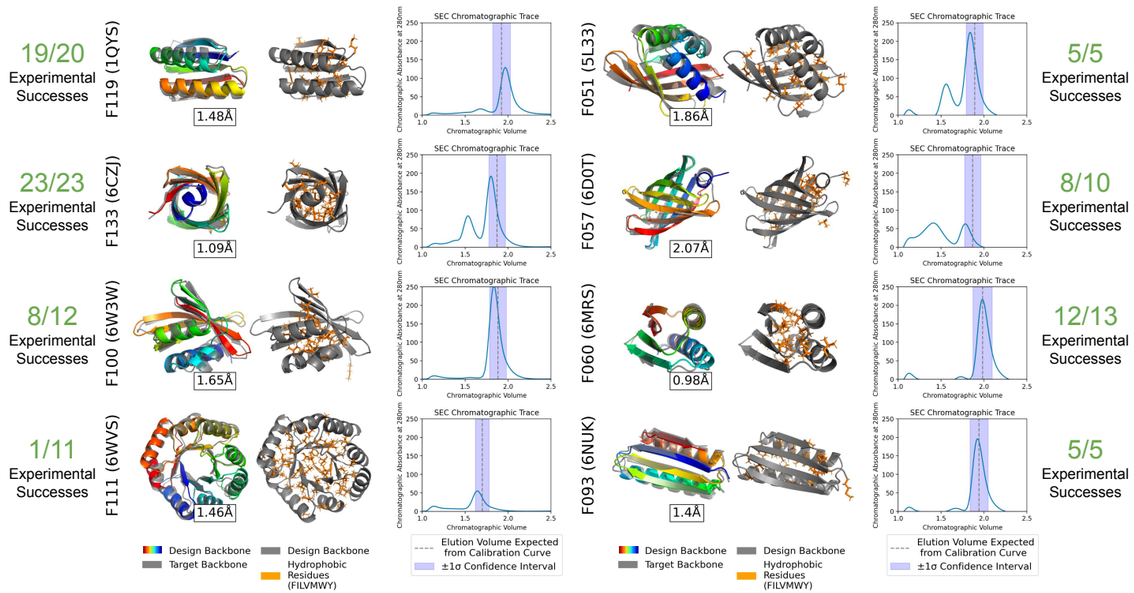

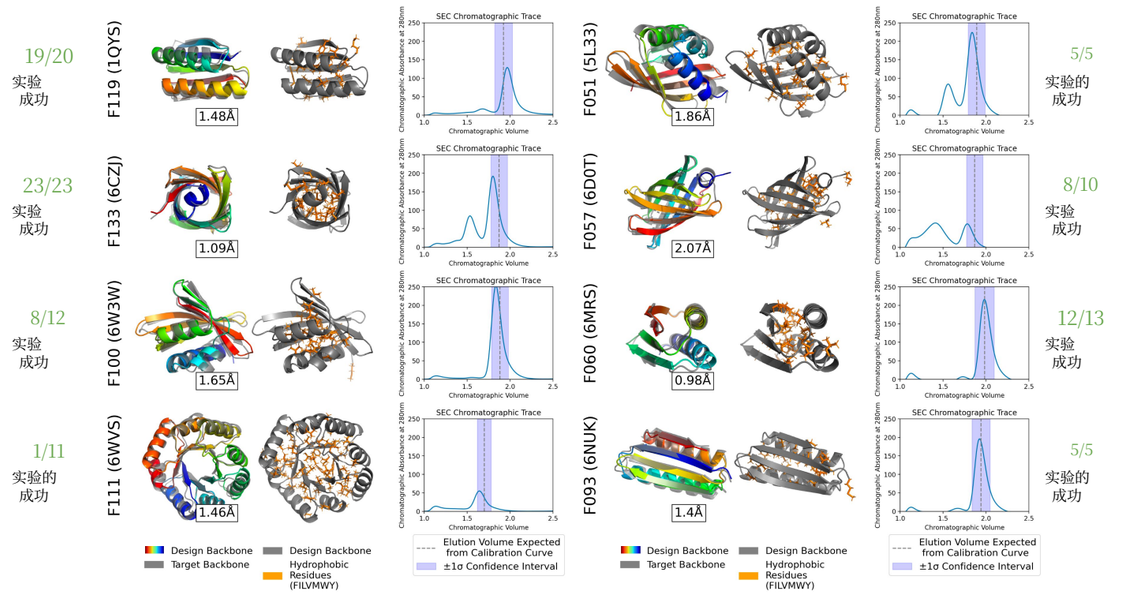

Figure S4. Fixed backbone designs succeed on all backbones tested experimentally. This figure showcases *designs* for the targets in Fig. S3. All target backbones whose designs were tested experimentally have at least one successful design. Targets span a range of lengths (L = [77, 182]) and folds (beta-barrel, NTF2, alpha-beta-mix, TIM-barrel). Each row shows the successful design with minimum RMSD to the target, according to the structure oracle, with overall fixed backbone design experimental outcomes for that target in the margin. The first column in each row shows the oracle prediction of the design's structure, rainbow-colored from N- to C- terminus overlayed on the target crystal structure, in gray. The second column shows placement of hydrophobic residues, with the predicted backbone in gray and hydrophobic side chains colored orange. The third column shows the chromatographic trace from SEC, with the expected elution volume and a one standard deviation confidence interval in dashed gray and light blue, respectively. All designs have a peak within the expected range of elution volume under SEC, indicative of a properly folded monomeric species. Two designs, F093 (6NUK) and F100 (6W3W) are monodisperse - the only peak detected is the one at the expected elution volume.

图 S4. 展完备架设计在所有实验测试的骨架上取得成功。该图展示了针对 图中目标的设计 S3。所有经过实验测试的目标骨架至少有一个成功设计。目标跨越不同的长度 (L = [77, 182]) 和构型 (β-桶, NTF2, α-β-混合, TIM桶)。每一行展示了与目标的最小均方根偏差 (RMSD) 相关的成功设计，根据结构甲骨文。边缘列出了该目标的整体固定骨架设计实验结果。每行的第一列显示了设计结构的甲骨文预测，彩虹色从N端到C端覆盖在目标晶体结构上，呈灰色。第二列显示了疏水残基的排布，预测的主链呈灰色，疏水侧链呈橙色。第三列展示了来自SEC的色谱追踪，预期的洗脱体积和一个标准差置信区间分别以虚线灰色和浅蓝色表示。所有设计的峰值均位于SEC下预期洗脱体积的范围内，表明适当折叠的单体物种。两个设计，F093 (6NUK) 和 F100 (6W3W) 是单分散的——唯一检测到的峰是位于预期洗脱体积的峰。

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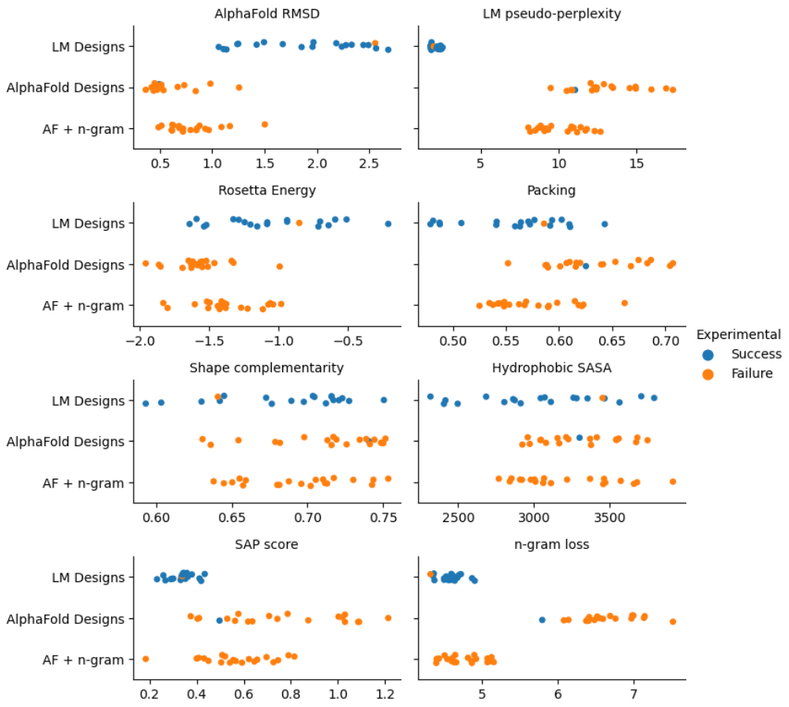

Figure S5. Analysis of fixed backbone designs across methods. Evaluation metrics of the sequences designed by Language Model vs designed by AlphaFold without LM, vs designed by AlphaFold with n-gram term. We present results for the best 5 designs for each of the four targets selected for direct comparison (PDB target IDs: 5L33, 6D0T, 6MRS, 6NUK). AlphaFold RMSD is lower (better) for the designs by AlphaFold. The Rosetta Energies are negative (good) for both sets and are not able to discriminate experimental outcomes, even though the Rosetta Energy function was developed for protein modeling and design (we use the beta_nov16 Rosetta energy function, length-normalized). LM pseudo-perplexity identifies the sequences designed without strong LM as improbable (low pseudo-perplexity), and is predictive of experimental success in this comparison. The *in silico* quality metrics (Appendix A.4) indicate that the AlphaFold designs without LM are not easily distinguished based on packing or shape complementarity, but tend to have more surface hydrophobics and higher (worse) SAP score. Adding the n-gram LM term to the AlphaFold fixed backbone optimization objective (Appendix A.3) improves the n-gram (or k-mer) statistics as intended, and slightly improves the SAP score, but has a 100% failure rate (vs 95% failure without n-gram). The aggregate statistics of this comparison are also reported in Table S2.

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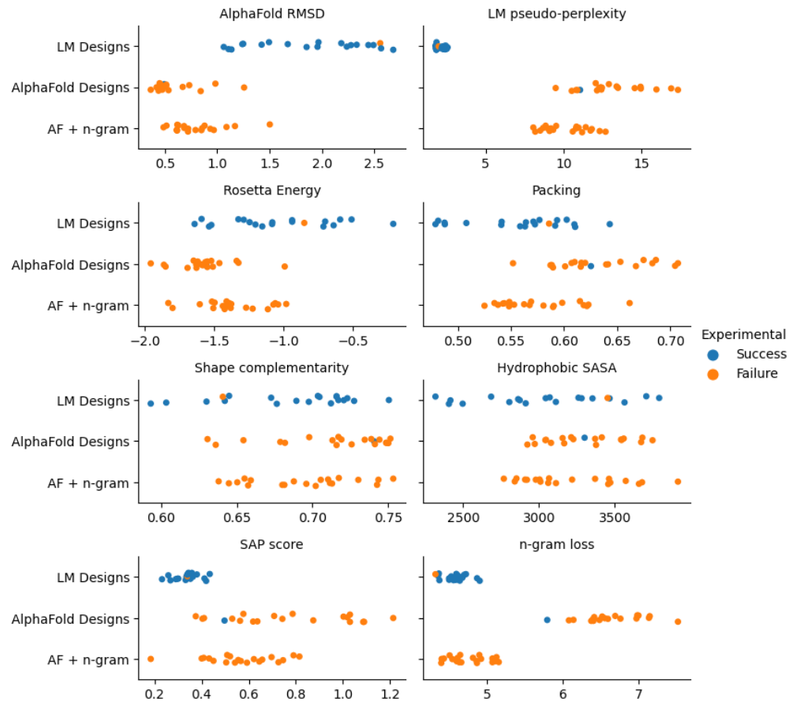

图 *S5.* 不同方法中固定骨架设计的分析。 由语言模型设计的序列与无 LM 设计、使用 n-gram 项的 AlphaFold 设计的序列的评估指标。我们展示了对每个四个直接比较的目标所选的最佳 5 个设计的结果 (PDB 目标 ID: 5L33、6D0T、6MRS、6NUK)。 AlphaFold 的 RMSD 对于 AlphaFold 设计的结果较低（更好）。罗塞塔能量对两种设计的结果都为负值（好），但无法区分实验结果，尽管罗塞塔能量函数是为蛋白质建模和设计而开发的 (我们使用 beta nov16 罗塞塔能量函数，长度标准化)。 LM 伪困惑度将没有强 LM 设计的序列识别为不太可能（低伪困惑度），并且可以预测该比较中的实验成功。 计算机模拟 质量指标 (附录 A.4) 表明，没有 LM 的 AlphaFold 设计在包装或形状互补性方面不容易区分，但往往具有更多的表面憎水性和更高（更差）的 SAP 得分。将 n-gram LM 项添加到 AlphaFold 固定骨架优化目标 (附录 A.3) 按预期改善 n-gram（或 k-mer）统计，并轻微提高 SAP 得分，但失败率为 100%（相比之下无 n-gram 失败率为 95%）。该比较的聚合统计数据也报告在表 S2。

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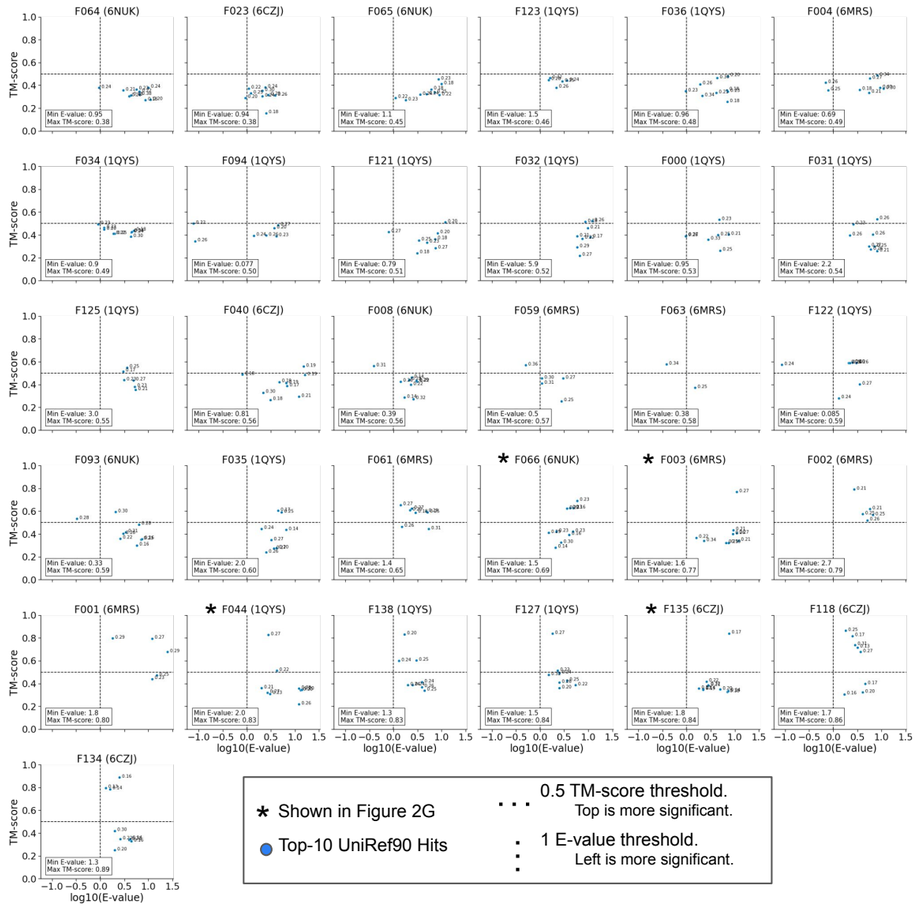

Figure S6. Fixed backbone designs, comparison to natural proteins. Details are shown for the comparison of select successful fixed backbone designs to natural proteins. Each plot shows sequence- and structural- match statistics of the top-10 most significant Jackhmmer hits (blue dots), when querying with the designed sequence against UniRef90 (Appendix A.5.3). We showcase a subset of 31 successful designs from the union of two sets: the 17 designs with no significant sequence hits, and the 19 designs with maximum TM-score < 0.6 to the neighbors' predicted structures. X-axes show the (sequence-based) significance of matches, according to log10(E-value) of the best domain. Hits to the left of the dashed vertical line at E-value $= 1$ are considered significant. Across all hits shown in the figure, only 18 are significant (E-value < 1) and only 3, for design {F094,F122} have E-value < 0.1. Hits are also labeled with their sequence-identity to the designed sequence. Significant hits have a median sequence-identity of 26%, and 14/17 are ¡ 30%. Y-axes compare the design and its top hits structurally, via TM-score between AlphaFold-predicted structures (Methods; Comparison). Plots are sorted in order of increasing maximum TM-score. Designs at the bottom of the figure may be using homology beyond our significance threshold, but many of the designs have no strong structural matches to their top hits. Structures for designs {F044,F135,F003,F066} and their top-significance hit are featured in Fig. 2G.

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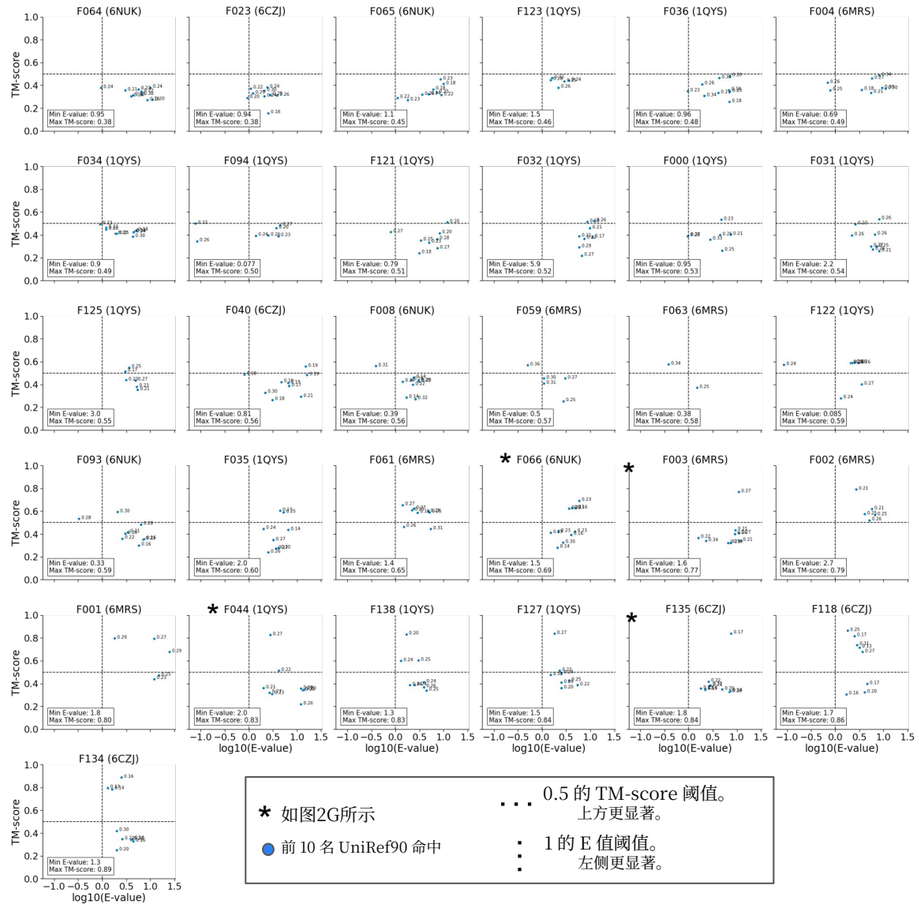

图*S6*。固定骨架设计，与天然蛋白质的比较。显示了选择的成功固定骨架设计与天然蛋白质的比较的详细信息。每个图展示了前10个最显著的Jackhmmer命中（蓝点）的序列和结构匹配统计数据，当用设计的序列查询UniRef90时（附录A.5.3）。我们展示了两个集合的联合中31个成功设计的子集：17个没有显著序列命中的设计和19个与邻近预测结构的最大TM-score $6 < 0$。X轴显示了根据最佳域的log10(E值)的（基于序列的）匹配显著性。位于E值的虚线垂直线左侧的命中被认为是显著的。在图中展示的所有命中中，仅有18个是显著的（E值）而仅有3个，设计F094,F {的E值为0.1。命中还标记了它们与设计序列的序列相似度。显著命中的序列相似度中位数为26%，且14/17 ⩾ 30%。Y轴通过AlphaFold预测结构之间的TM-score来比较设计及其顶级命中（方法；比较）。图形按照最大TM-score递增的顺序排序。位于图底部的设计可能使用了超出我们显著性阈值的同源性，但许多设计与它们的顶级命中没有强结构匹配。设计F044,F135,F003,F {及其最具显著性的命中在图2G中展示。

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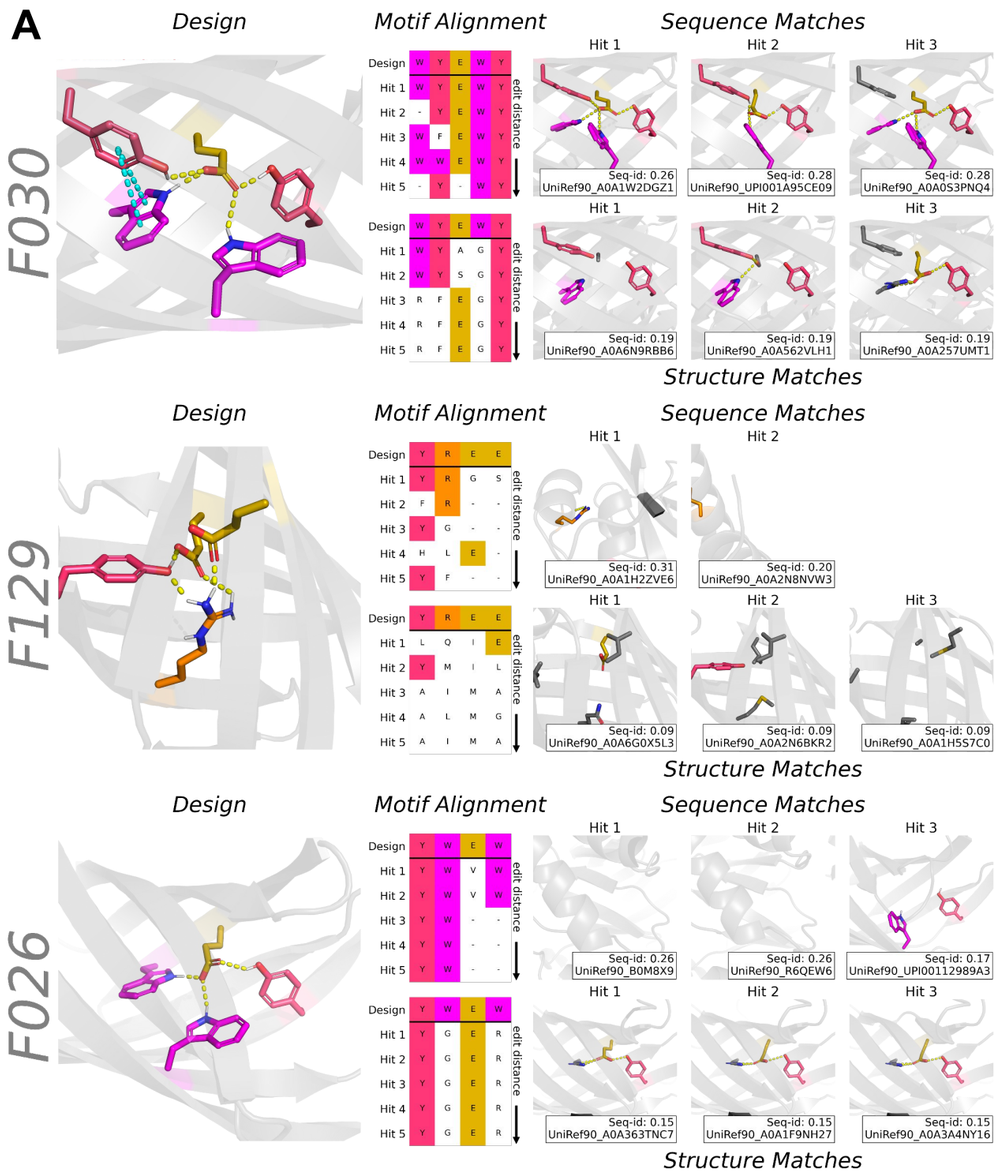

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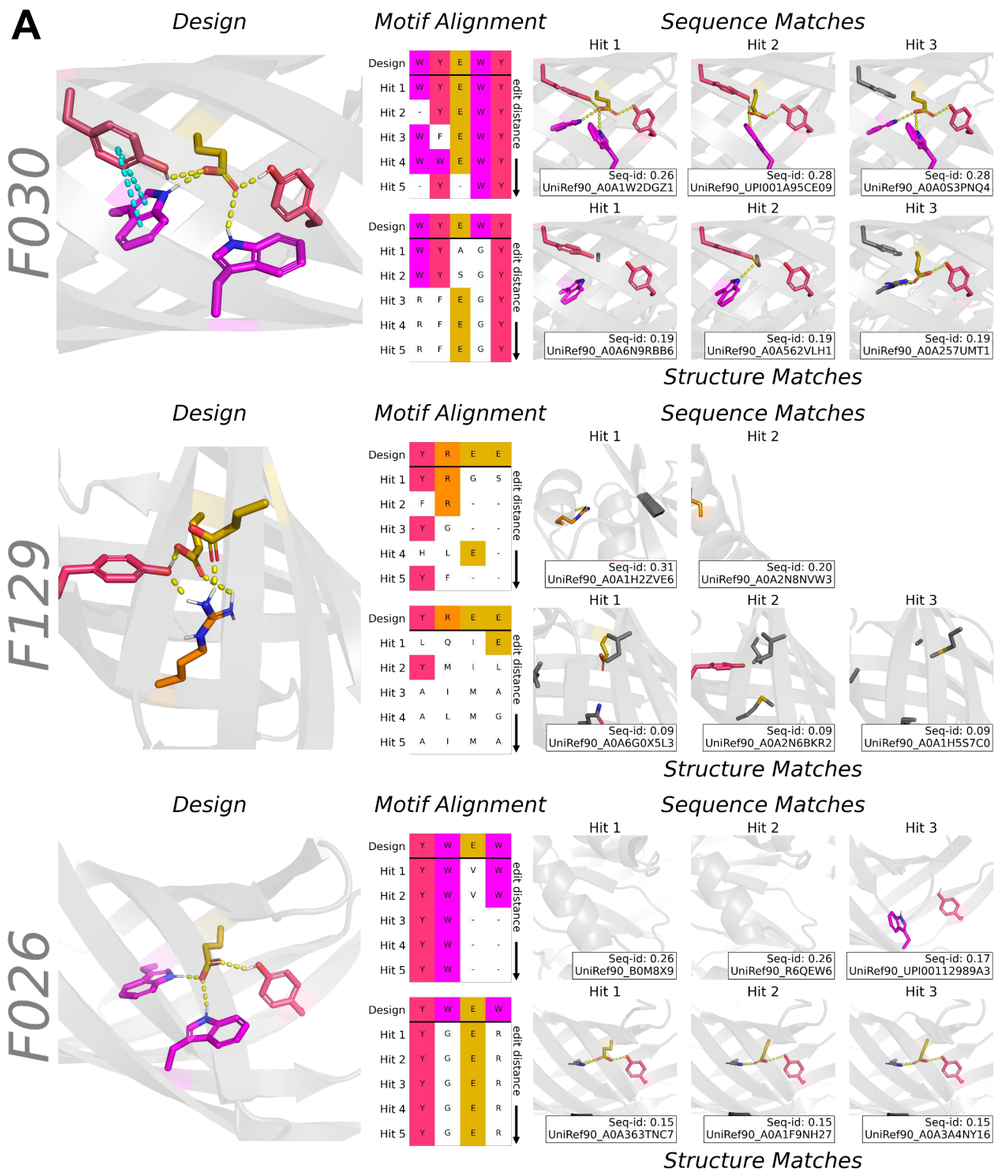

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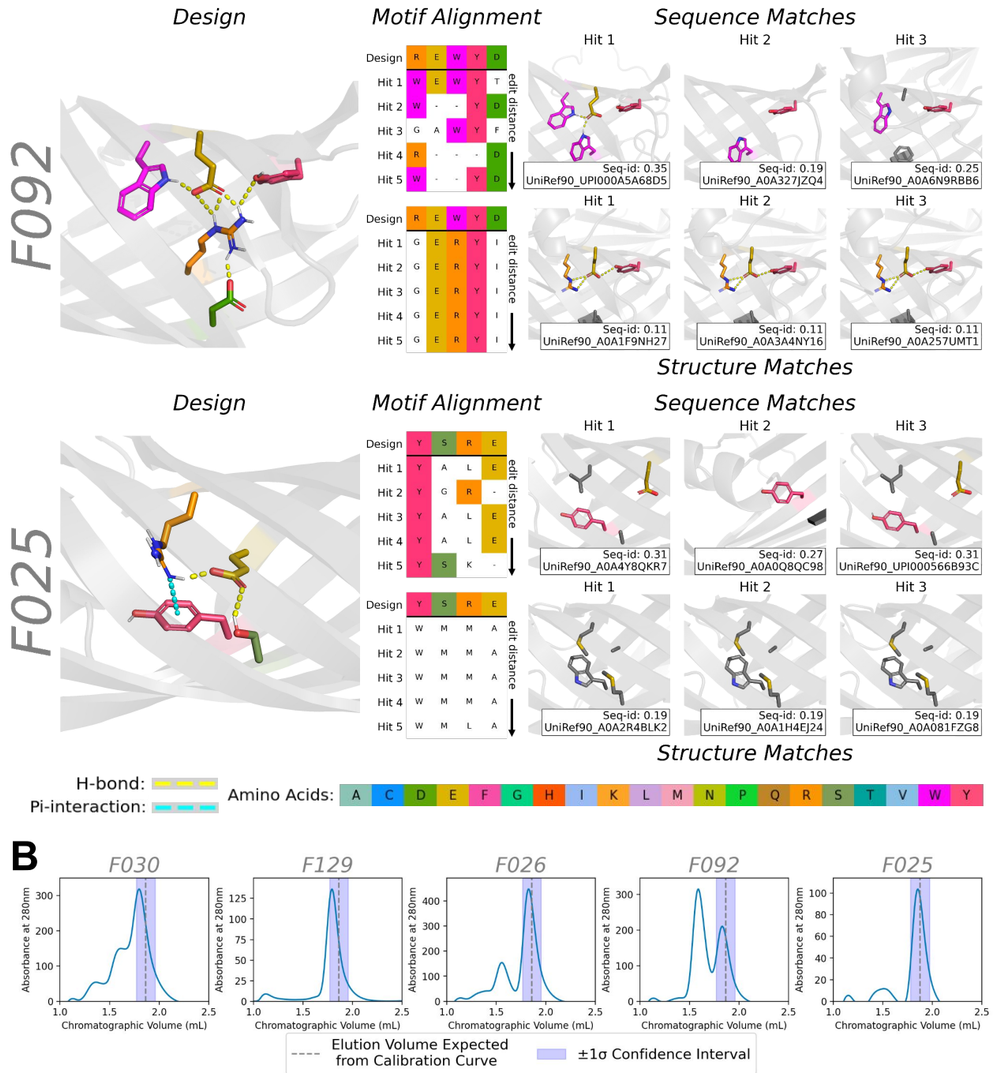

Figure S7. Detailed Analysis of Motifs. (**A**) Comparisons of hydrogen-bond network motifs in designs to aligned positions in natural neighbors. Compared with the views in Fig. 3D,3E, 2 additional designs are shown (F026, F025) and the top-3, rather than top-1, aligned sequence and structure search neighbors are shown. Otherwise, views are the same as in Fig. 3D,3E. The design is shown with side chains enabled for the motif, and bond networks drawn as dashed lines. Neighbors from Jackhmmer search of natural sequences in Uniref90 and Foldseek search of natural structures in AlphaFold DB are performed. The full, MSAs from both of these searches are sorted by edit distance at the positions aligned to that of the motif in the design. Minimum edit distance neighbors are shown with side chains shown at aligned positions. Sidechains are colored gray where matched amino acids in neighbors are not in the designed motif. (**B**) Size exclusion chromatography (SEC) traces are shown at the bottom of the figure. In all cases, there is a peak detected near the expected elution volume indicative of a properly folded monomeric species, according to a calibration curve (Appendix A.7). In 4/5 cases, the peak at expected elution volume is dominant, higher than any other peak. F129 is monodisperse - the only peak detected is the one at the expected elution volume.

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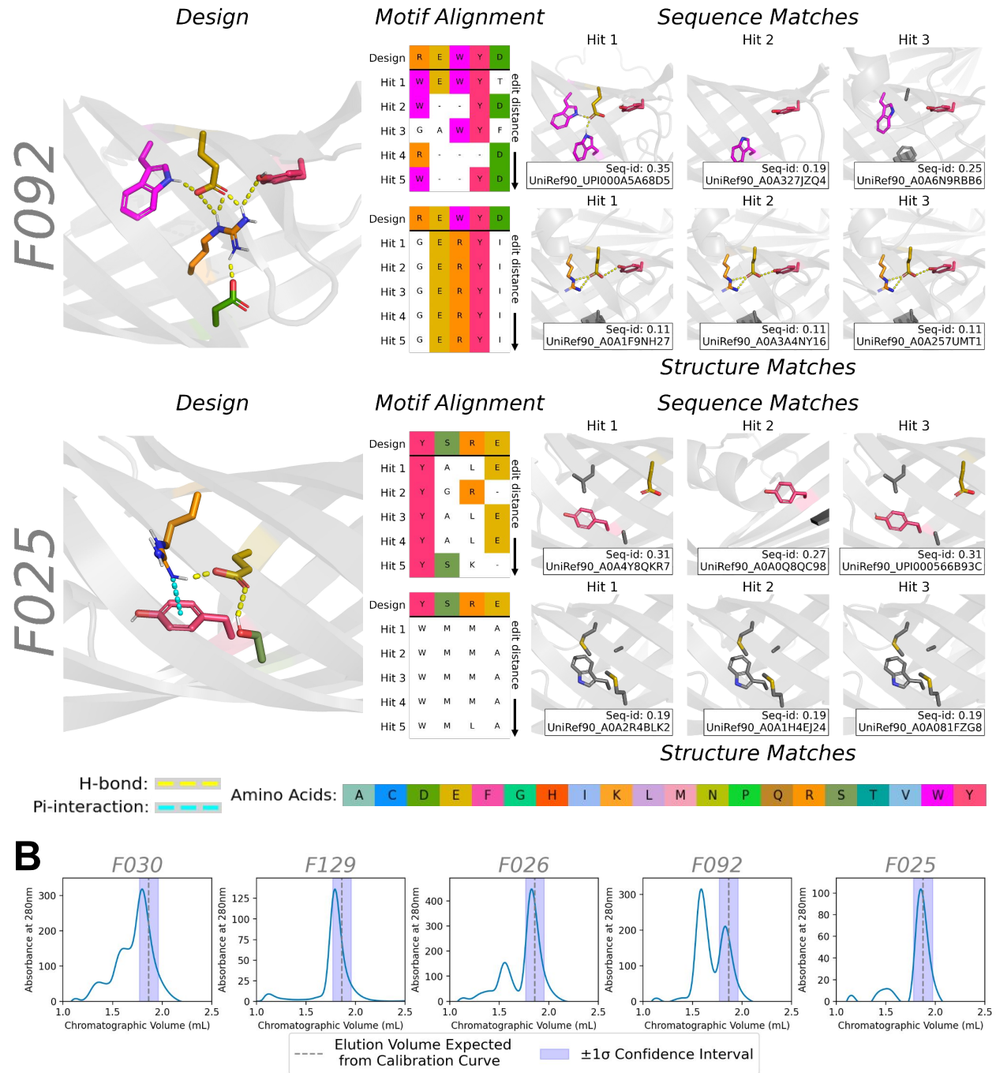

图 *S7*。基序的详细分析。(**A**) 设计中的氢键网络基序与天然邻居对齐位置的比较。与图中的视图相比，3D，3E，展示了另外 2 个设计（F026，F025）以及前 3 个而非前 1 个对齐的序列和结构搜索邻居。否则，视图与图中的相同。3D，3E。该设计显示了基序的侧链，键网络以虚线形式绘制。从 Uniref90 中天然序列的 Jackhmmer 搜索和 AlphaFold 数据库中天然结构的 Foldseek 搜索中获得邻居。这两个搜索的完整 MSA 按照与设计中基序对齐位置的编辑距离进行排序。显示了最小编辑距离邻居，侧链在对齐位置上显示。当邻居中的匹配氨基酸不在设计基序中时，侧链呈灰色。(**B**) 尺寸排除色谱 (SEC) 档案显示在图的底部。在所有情况下，在预期洗脱体积附近检测到一个峰，表明根据校准曲线（附录A.7）存在正确折叠的单体种类。在 4/5 的情况下，预期洗脱体积处的峰是主导的，高于任何其他峰。F129 是单分散的 - 唯一检测到的峰是预期洗脱体积处的峰。

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Language models generalize beyond natural proteins

语言模型超越天然蛋白质的泛化能力

35

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Figure S8. Free Generation: Experimental Successes Overview of predicted structure for all 71 free generations (except for G230, ommitted randomly due to space constraints) that were experimentally successful. Designed structures from the *in silico* structure oracle (AlphaFold) are shown, colored by pLDDT, a measure of local prediction confidence. Statistics (sequence identity, TM-score, and significance) of each design's most significant sequence-search hit in AlphaFold DB shown. The first 31 designs shown are those from the bottom-left, *de novo* quadrant of Fig. 4D, meaning they were found distant from natural sequences, after searching them against AlphaFold DB (Appendix A.5.2).

图 ***S8***。自由生成：实验成功 所有 71 个实验成功的自由生成的预测结构概述（G230 除外，由于空间限制随机省略）。所设计的结构来自计算机模拟 结构甲骨文（AlphaFold），以 pLDDT 着色，pLDDT 是局部预测置信度的测量。每个设计在 AlphaFold 数据库中最显著的序列搜索命中的统计数据（序列相似性、TM-score 和显著性）显示。前 31 个设计来自图 ***4D*** 的左下方 新设计 象限，这意味着在与 AlphaFold 数据库搜索后，它们从天然序列中找到的比较遥远（附录 A.5.2）。

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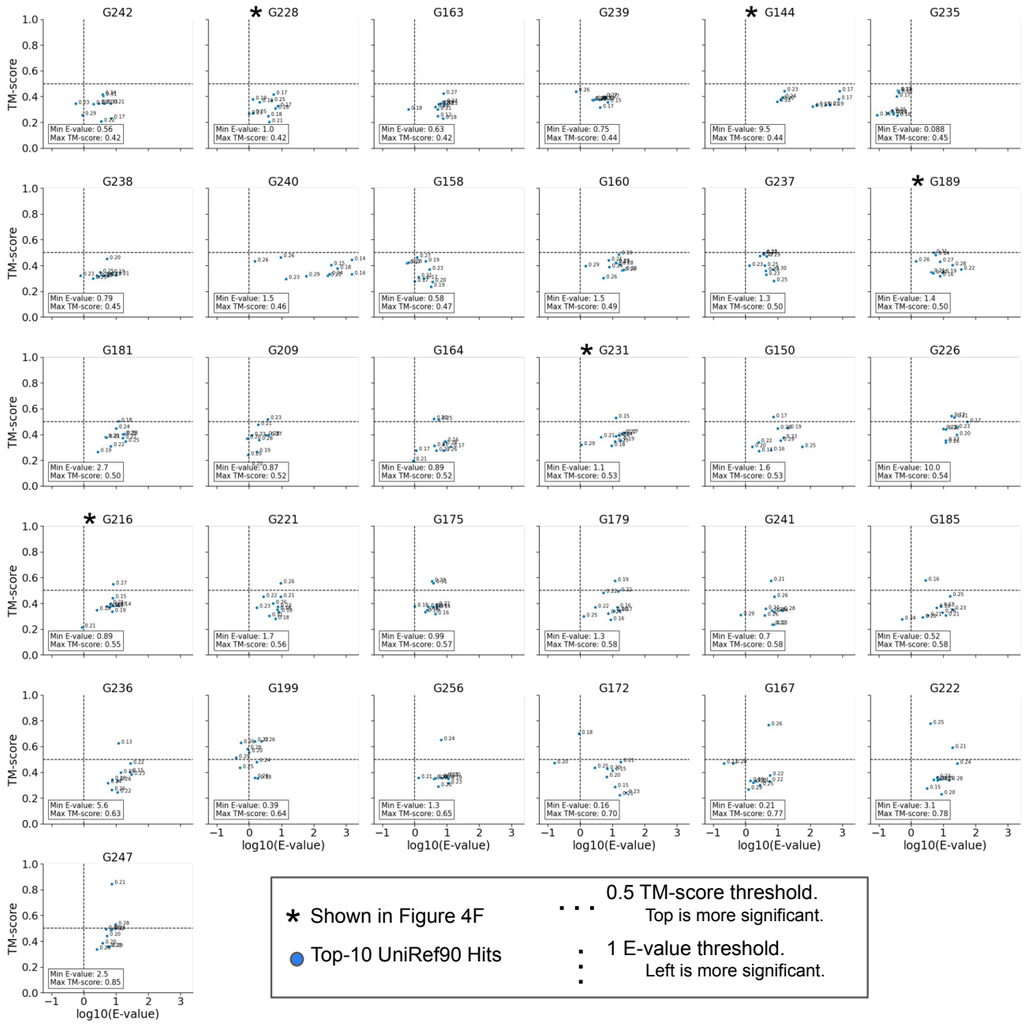

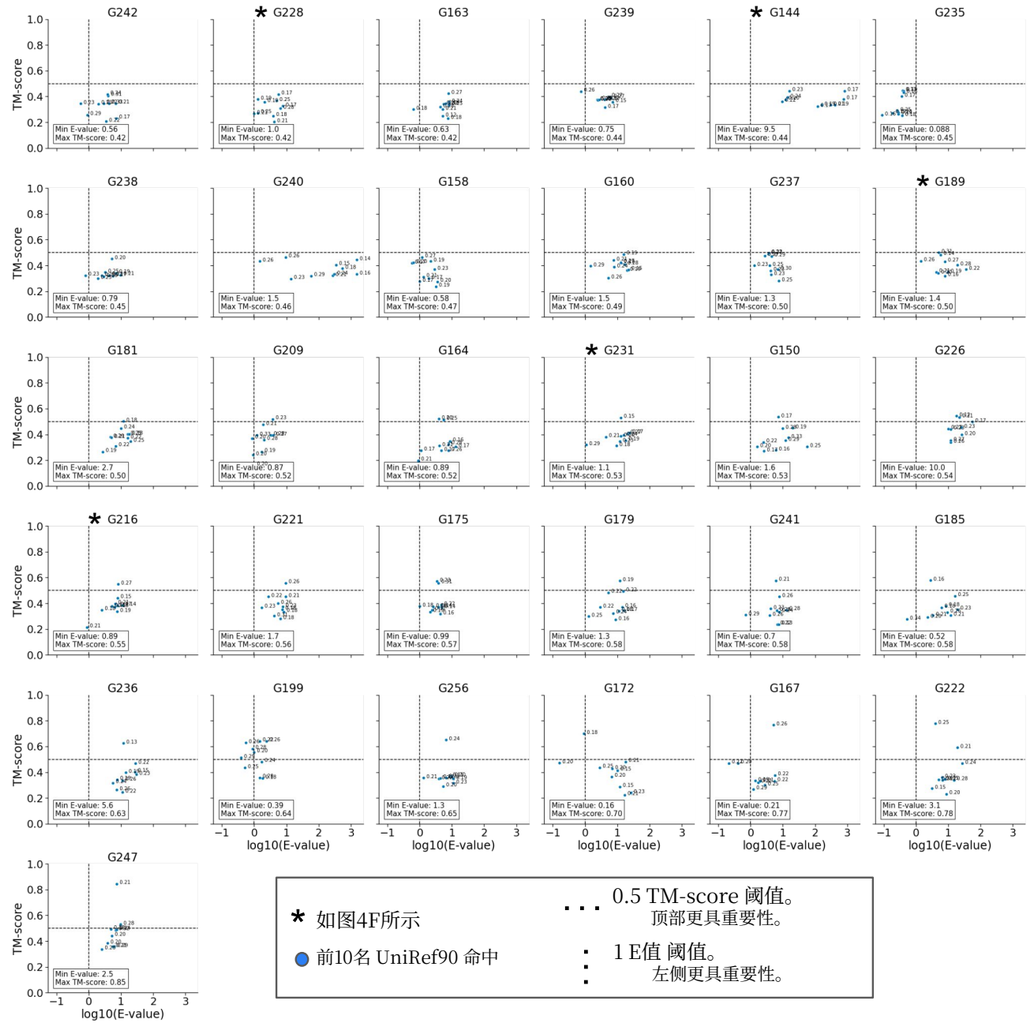

Figure S9. Free Generations, comparison to natural proteins As in Fig. S6, we show verbose sequence search statistics from the comparison of successful, distant free generations to natural proteins. Each panel represents one of the 31/49 experimentally successful proteins in the lower left quadrant of Fig. 4D, which was distant from its top sequence hit in UniProt 2021_04 / AlphaFold DB. For these 31 successful free generations, we did a more thorough analysis: comparing to UniRef90, which fully contains the language model's training set, and considering more than just the top hit (Appendix A.5.3). Plots are formatted identically to those in fig. S6: each plot is for one free generation, the top-10 Jackhmmer hits from searching UniRef90 are shown as blue dots, x-axes shows sequence match strength, y-axes shows TM-score comparison of predicted structure, and sequence identity is annotated for each dot. Plots are sorted in order of ascending maximum TM-score. In general, there is strong agreement between the results of this UniRef90 search, and their classification as distant from searching AlphaFold DB. 16/31 successes have no significant (E-value < 1) hits, and no hits with E-value < 0.1 are detected among all 31. Comparison of predicted structures further confirms the dissimilarity of each generation from its top natural sequence hits. 12/31 designs have all top-10 sequence hits likely to possess a different fold (max TM-score < 0.5). Those few hits with high TM-score (> 0.7) generally possess E-values in the 3 to 10 range. Structures for designs {G216,G228,G231,G189,G144} and their top-significance hit are featured in Fig. 4F.

图*S9*。自由生成，与天然蛋白质的比较如图。S6，我们展示了成功的、遥远的自由生成与天然蛋白质的比较的详细序列搜索统计数据。每个面板代表了图4D的左下象限中31/49个实验成功的蛋白质之一，该蛋白质在UniProt 2021 04 / AlphaFold数据库中与其最高序列命中相距较远。对于这31个成功的自由生成，我们进行了更深入的分析：与完全包含语言模型训练集的UniRef90进行比较，并考虑的不仅仅是最高命中 (附录A.5.3)。图表格式与图S6中的图表相同：每个图表对应一个自由生成，从搜索UniRef90中得到的前10个Jackhmmer命中显示为蓝色点，x轴表示序列匹配强度，y轴显示预测结构的TM-score比较，并为每个点标注序列相似性。图表按升序最大TM-score排序。总体而言，UniRef90搜索的结果与其在搜索AlphaFold数据库时的分类为遥远的结果之间存在强的吻合。16/31的成功案例没有显著的（E值 <

1) 命中，并且在所有中未检测到 E 值为 < 0.1 的命中
31. 预测结构的比较进一步证实了每一代与其顶级自然序列命中的不相似性。12/31 的设计中，所有前 10 名序列命中可能具有不同的折叠（最大 TM-score < 0.5）。那几个具有高 TM-score 的命中（> 0.7）一般具有 3 到 10 范围内的 E 值。设计的结构 {G216,G228,G231,G189,G144} 及其顶级显著命中在图中展示。4F.

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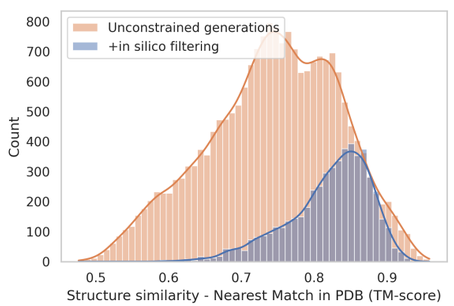

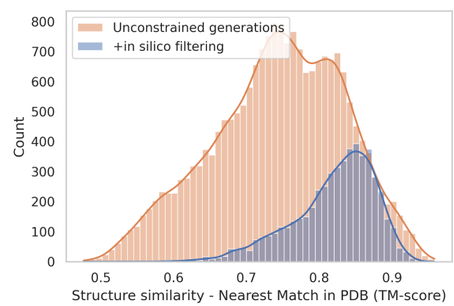

Figure S10. Top structure-based matches in PDB for free generations We show the distribution of the similarity to the nearest match amongst all known protein structures in the Protein Data Bank (PDB), for each free (unconstrained) generation. The nearest neighbor is defined by a structure-based search using foldseek, and similarity is TM-score from TMalign (between 0 and 1, where 0.5 is typically seen as a threshold for belonging to another fold). We believe that the designs' structural matches may be explained by the relatively short length (L=100) of free generations, which makes them likely to partially match a larger structure. Even though structural matches were found for the generated proteins, for many of the designs none of the matches could be found based on homology sequence search (Fig. 4D).

图 *S10.* 自由生成的蛋白质数据库中的顶部基于结构的匹配 我们展示了在蛋白质数据银行（PDB）中，对于每个自由（无约束）生成，已知蛋白质结构中与最近匹配的相似度的分布。最近邻是通过使用 Foldseek 进行基于结构的搜索来定义的，相似度是来自 TMalign 的 TM-score（范围在 0 到 1 之间，其中 0.5 通常被视为属于另一个折叠的阈值）。我们认为设计的结构匹配可能可以通过自由生成的相对较短长度（L=100）来解释，这使得它们可能部分匹配更大的结构。尽管为生成的蛋白质找到了结构匹配，但对于许多设计，基于同源序列搜索并未找到任何匹配（图 4D）。

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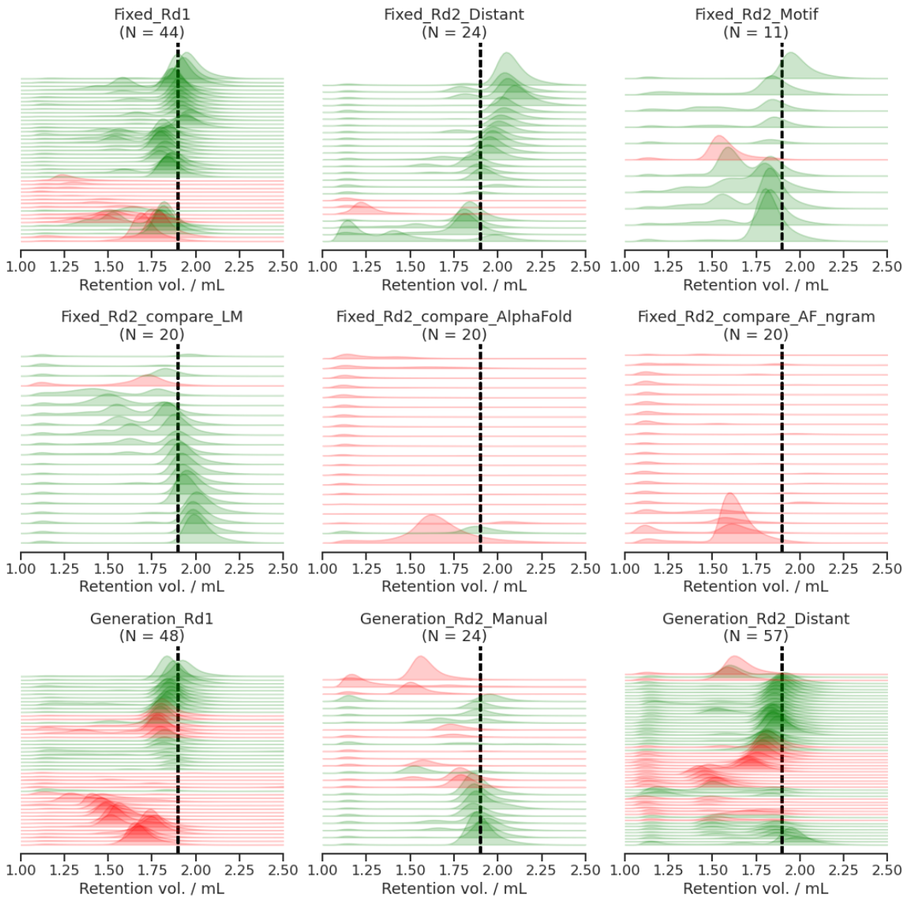

Figure S11. Overview of Experimental Evaluations for all tested designs. In total, 268 designed proteins were tested experimentally for their solubility and for having expected hydrodynamic radius via size-exclusion chromatography (SEC). Shown here are all SEC traces for those 268 evaluated proteins, grouped according to the categories described in (Appendix A.6). Designs for the comparison of LM vs. "no-LM" are split on the middle row, according to the model used for designs. Plots show chromatographic absorbance at 280 nm (y-axis) vs. retention volume (x-axis). Particles with larger radius flow faster through a porous column, and elute at lower volumes (to the left). Monomeric species are the smallest particles and give a peak most to the right. Expected elution volume is different for each sequence, but as a visual guide, we annotate the average expected elution volume (1.9 mL) for a length 100 sequence, in dashed black. Traces are colored according to the definition of experimental success: green for success, red for failure (Appendix A.7.4).

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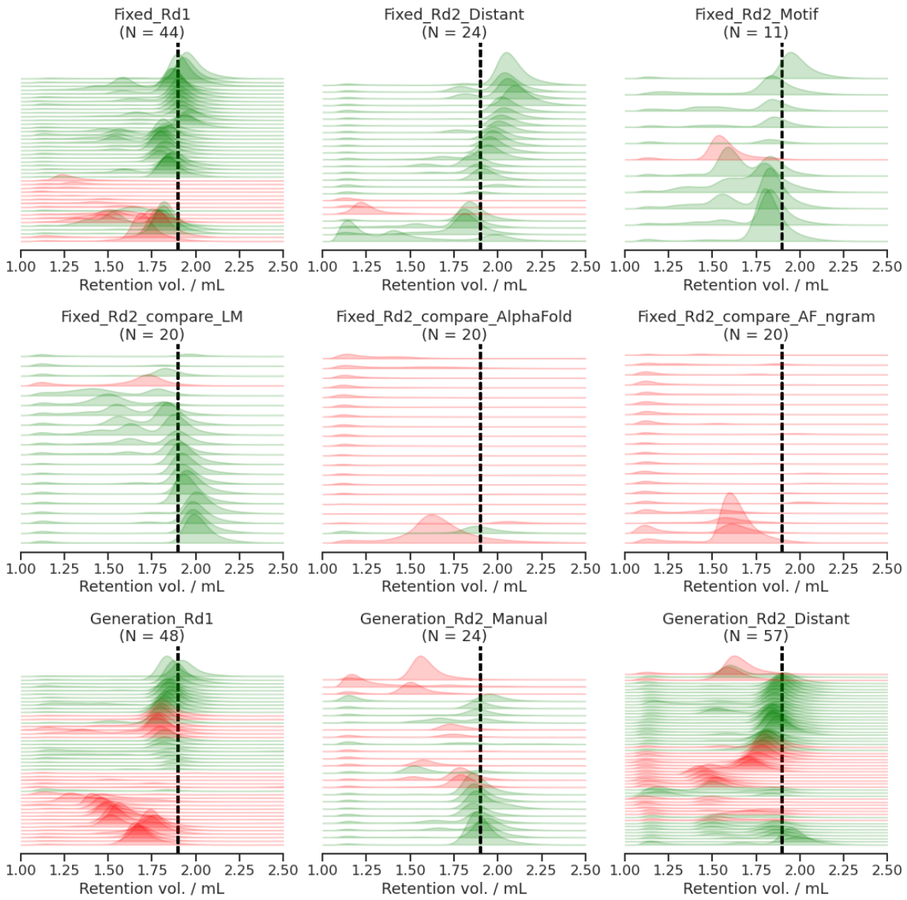

图 *S11.* 所有测试设计的实验评估概述。共测试了268种设计蛋白质的溶解性和通过大小排阻色谱法（SEC）获得预期的水动力半径。此处显示的是所有268种已评估蛋白质的SEC痕迹，按照在（附录A.6）中描述的类别进行分组。LM与“无LM”的比较设计在中间行分开，按照设计所用的模型进行分类。图表显示在280纳米（y轴）处的色谱吸光度与保留体积（x轴）的关系。半径较大的颗粒通过多孔柱的流速更快，并且在较低的体积处洗脱（向左）。单体物质是最小的颗粒，且其峰值位于最右侧。每种序列的预期洗脱体积不同，但作为视觉指南，我们用虚线黑色标注长度为100的序列的平均预期洗脱体积（1.9 mL）。痕迹根据实验成功的定义着色：绿色表示成功，红色表示失败（附录A.7.4）。

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C. Supplementary Tables

Overview of Supplementary Tables:

- Table S1: Comparison of sequence and structure understanding of ESM2 and baselines.
- Table S2: Comparisons for fixed backbone designs.
- Table S3: Line sweep of n-gram LM loss coefficient for AlphaFold + n-gram LM.
- Table S4: Analysis of fixed backbone designs across methods.
- Table S5: Comparison of different approaches of free generation using the Language Model.

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C. 补充表

补充表概述：

- 表S1：ESM2与基线的序列和结构理解比较。
- 表S2：固定骨架设计的比较。
- 表S3：AlphaFold的n-gram语言模型损失系数的线性扫描 + n-gram语言模型。
- 表 S4：不同方法下固定骨架设计的分析。
- 表 S5：使用语言模型的不同自由生成方法的比较。

40

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	Precision@L Long-Range		Pseudo-Perplexity	
	De Novo	Natural	De Novo	Natural
Best Achievable	0.82	0.94	1.00	1.00
Prior	0.05	0.04	17.17	18.72
ESM2 untrained	0.05	0.04	17.59	19.42
ESM2 trained (≈500k updates)	0.49	0.57	7.73	6.66

Table S1. Comparison of sequence and structure understanding of ESM2 and baselines. Comparison of structural and sequence understanding of ESM2 and baselines. Predictors on rows, metrics and datasets on columns. (Columns) The first major column characterizes structural understanding of the language model with minimal structure projection introduced in Appendix A.2.2. The metric shown is precision of the top-L predicted long-range ($separation \geq 24$ backbone positions) contacts, where L is sequence length. The second major column characterizes sequence understanding. The metric shown is sequence perplexity, or pseudo-perplexity in the case of ESM2. (Rows) The "Best Achievable" row oracle shows the best achievable score for each metric. The "Prior" row for structure shows the score of a per-sequence-length background model of structure, defined as the averaged predicted distograms of 500 randomly selected natural sequences of length L, as predicted by the trained structure projection used in this paper. The "Prior" row for sequence shows the perplexity of the unigram model trained on amino acid frequencies in UniRef50 (2018_03) Appendix A.2.3. The final two rows of the table show the performance of untrained and fully-trained ESM2, in that order.

Score	RMSD	AlphaFold (pLDDT)	ESM (pseudo-perplexity)
Ground Truth	0.00	91.11	7.27
AlphaFold	0.58	95.18	13.05
AlphaFold + n-gram	0.80	92.62	10.14
ESM Inverse Folding	0.99	90.76	5.15
ProteinMPNN	1.03	91.13	4.96
ESM2 (ours)	1.90	87.94	2.1

Table S2. Comparisons for fixed backbone designs. We present *in silico* metrics for additional baseline fixed backbone design methods: using ESM Inverse Folding (14) and ProteinMPNN (15). The results for each method (all rows but "Ground Truth") are for 20 sequence designs over four different *de novo* backbone targets used for the main comparison results (Figs. 2, S3 and S5). The sequences designed using the two inverse-folding models are sampled with the default temperature of 0.1. The oracle structure predictions for the inverse folding designs are close to the target backbone, with RMSD close to 1Å. The sequences were also seen as plausible by the AlphaFold Oracle (confident structure predictions with pLDDT > 90). ESM2 pseudo-perplexity of inverse folding designs is low compared to AlphaFold designs and even ground truth sequences, meaning the sequences are plausible under ESM. In light of the high experimental success rates demonstrated with ProteinMPNN on other targets, the results support the hypothesis that ESM2 can understand design patterns to the level where it is indicative of experimental success.

	λ_{ngram}	Oracle RMSD	Oracle pLDDT	E_{ngram}
LM Designs (reference)	1	2.1	86	4.5
AlphaFold + n-gram Designs	1	0.82	92	5.23
	2 (selected)	**0.82**	**92**	**5.09**
	5	1.07	90	4.89
	7	1.27	87	4.76
	10	1.97	83	4.67
	15	2.19	77	4.57
	20	3.41	71	4.42
	30	6.85	61	4.32
	50	8.87	54	4.21

Table S3. Line sweep of n-gram LM loss coefficient for AlphaFold + n-gram LM. A line sweep was performed to determine λ_{ngram}, the coefficient for E_{ngram} for Alphafold + n-gram designs. Each row below shows the average statistics of 40 total designs, 10 designs for each of 4 target backbones (5L33, 6D0T, 6MRS, 6NUK). For the top row, statistics are shown randomly selecting from the 200 fixed backbone design produced for each backbone. For each other row, (4*10 = 40) fresh designs were produced by AlphaFold-based design with a specific n-gram energy function coefficient. The same oracle (AlphaFold) structure prediction pipeline was applied to all designed sequences. A coefficient 2 was chosen from the line sweep, as it is the highest value that does not degrade oracle structure accuracy (RMSD) and confidence (pLDDT) metrics. After following our full generation, filtering, and selection protocol (Appendix A.6.2), final E_{ngram} values were roughly matched (4.59 vs. 4.77) for LM and AlphaFold+n-gram designs (Table S4).

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	精度@L 长期		伪困惑度	
	新设计	自然	新设计	自然
最佳可达	0.82	0.94	1.00	1.00
先验	0.05	0.04	17.17	18.72
未训练的 ESM2	0.05	0.04	17.59	19.42
训练后的 ESM2 (≈500k 更新)	0.49	0.57	7.73	6.66

表 *S1.ESM2*与基线的序列和结构理解比较。ESM2与基线的结构和序列理解比较。行表示预测器，列表示指标和数据集。（列）第一大列描述了语言模型的结构理解，附录中介绍的最小结构投影A.2.2 的特点。所示指标是预测的最长距离（$separation \geq 24$ 主链位点）接触的精度，其中 L 为序列长度。第二大列描述了序列理解。所示指标是序列的困惑度，或者在ESM2的情况下是伪困惑度。（行）“最佳可达”行甲骨文显示每个指标的最佳可达分数。“先验”行对于结构显示每个序列长度的背景模型分数，该模型被定义为 500 个随机选择的自然序列的平均预测 Distograms，其长度为 L，由本文使用的训练结构投影预测。“先验”行对于序列显示基于 UniRef50 中氨基酸频率训练的 unigram 模型的困惑度（2018年03）附录A.2.3。表中的最后两行按顺序显示未训练和完全训练的 ESM2 的性能。

分数	均方根偏差	AlphaFold (pLDDT)	ESM (伪困惑度)
真实值	0.00	91.11	7.27
AlphaFold	0.58	95.18	13.05
AlphaFold + n-gram	0.80	92.62	10.14
ESM 逆折叠	0.99	90.76	5.15
ProteinMPNN	1.03	91.13	4.96
ESM2（我们的）	1.90	87.94	2.1

表 *S2*。固定骨架设计的比较。我们提供了计算机模拟的额外基准固定骨架设计方法的指标：使用 ESM 逆折叠 (14) 和 ProteinMPNN (15)。每种方法的结果（除“真实值”外的所有行）是针对四种不同的主要比较结果所使用的 20 个序列设计所得到的新设计骨架目标（图。2、S3 和 S5）。使用这两个逆折叠模型设计的序列是在默认温度为 0.1 的情况下进行采样的。逆折叠设计的甲骨文结构预测接近目标骨架，均方根偏差接近 1 ˚A。这些序列也被 AlphaFold 甲骨文视为可信的（具有 pLDDT $>$ 90的自信结构预测）。ESM2

与 AlphaFold 设计和真实序列相比，逆折叠设计的伪困惑度较低，这意味着这些序列在 ESM 下是合理的。考虑到 ProteinMPNN 在其他目标上展示的高实验成功率，这些结果支持了 ESM2 能够理解设计模式到可指示实验成功级别的假设。

	λ_{ngram}	甲骨文 RMSD	甲骨文 pLDDT	E_{ngram}
语言模型设计（参考）	1	2.1	86	4.5
AlphaFold + n-gram 设计	1	0.82	92	5.23
	2（已选择）	**0.82**	**92**	**5.09**
	5	1.07	90	4.89
	7	1.27	87	4.76
	10	1.97	83	4.67
	15	2.19	77	4.57
	20	3.41	71	4.42
	30	6.85	61	4.32
	50	8.87	54	4.21

表 *S3. AlphaFold* 的 *n-gram LM* 损失系数线性扫描 + *n-gram LM*. 进行了一次线性扫描，以确定 λ_{ngram}，用于E_{ngram} AlphaFold + n-gram 设计的系数。下面每一行显示了 40 个总设计的平均统计数据，每个目标主链（5L33、6D0T、6MRS、6NUK）有 10 个设计。对于顶行，统计数据是从为每个主链生成的 200 个固定主链设计中随机选择的。对于其他每一行，(4*10 = 40) 通过基于 AlphaFold 的设计用特定 n-gram 能量函数系数生成了新设计。相同的甲骨文 (AlphaFold) 结构预测流程被应用于所有设计序列。从线性扫描中选择了系数 2，因为它是最高的值，且不会降低甲骨文结构的准确性 (均方根偏差) 和置信度 (pLDDT) 指标。在遵循我们的完整生成、过滤和选择协议后 (附录A.6.2)，最终 E_{ngram} 值大致匹配 (4.59 vs. 4.77)，用于 LM 和 AlphaFold+n-gram 设计 (表S4).

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	LM Designs	AlphaFold Designs	AF + n-gram
Experimental Success	0.95	0.05	0.00
Oracle (AlphaFold) RMSD	1.90	0.58	0.80
LM pseudo-perplexity	2.10	13.05	10.14
Rosetta Energy	-1.04	-1.57	-1.37
Packing	0.56	0.63	0.58
Shape complementarity	0.68	0.71	0.69
Hydrophobic SASA	3043.98	3299.73	3216.39
SAP score	0.34	0.76	0.57
n-gram loss	4.59	6.61	4.77

Table S4. Analysis of fixed backbone designs across methods. This table shows the aggregate statistics corresponding to the plots in Fig. S5, presenting a comparison between fixed backbone designs from Language Model vs. AlphaFold (No LM) vs. AlphaFold + n-gram LM on 20 sequences designed over four different *de novo* backbone targets. Please refer to the figure caption for more details.

Method	N (number of samples)	pLDDT	pTM	Clusters by TM-score
p(x,y) sampling (subset)	400	81.49	0.67	242/400 clusters
p(x) sampling	400	76.22	0.53	372/400 clusters
Distogram KL (11)	400	80.22	0.56	282/400 clusters
p(x,y) sampling (full)	25,000	81.09	0.66	7663/25000 clusters

Method	Pairwise sequence identity	Rosetta Energy	SAP Score	helix/sheet/loop ratio
p(x,y) sampling (subset)	0.13	-0.19	0.45	0.51/0.23/0.26
p(x) sampling	0.15	1.93	0.65	0.15/0.36/0.50
Distogram KL (11)	0.13	0.65	0.73	0.40/0.29/0.31
p(x,y) sampling (full)	0.12	-0.21	0.46	0.52/0.21/0.26

Table S5. Comparison of different approaches of free generation using the Language Model. In addition to the Blocked Gibbs sampling method described in this paper for free generation of proteins, we tested two other procedures to sample protein sequences: p(x) sampling - in which only ESM2 and an ngram term were used to sample probable amino acid sequences using a Markov Chain, but no structure sampling was used. In addition, we followed the distogram KL maximization procedure (11) where the sampling objective for the structure step is a KL divergence from the distogram to a background distribution. Maximizing this KL can be thought of as minimizing the entropy of the distogram, meaning the objective steers towards confident structure prediction. The table above compares key statistics between the approaches. Notably, we observed that p(x) sampling often produces repeat sequence patterns. The distogram KL approach, applied with the low-capacity structure prediction head, has disadvantages to the proposed Blocked Gibbs approach. The distogram KL approach tends to generate structures with almost no mixture of alpha-helix and beta sheets in the same design, and worse pTM, Rosetta Energy and SAP scores.

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	语言模型设计	AlphaFold 设计	AF + n-gram
实验成功	0.95	0.05	0.00
甲骨文 (AlphaFold) 均方根偏差	1.90	0.58	0.80
LM 伪困惑度	2.10	13.05	10.14
罗塞塔能量	-1.04	-1.57	-1.37
打包	0.56	0.63	0.58
形状互补性	0.68	0.71	0.69
疏水性SASA	3043.98	3299.73	3216.39
SAP得分	0.34	0.76	0.57
n-gram损失	4.59	6.61	4.77

表 *S4.* 不同方法下固定骨架设计的分析。本表显示了与图 S5 中的图表对应的汇总统计数据，展示了语言模型与AlphaFold（无LM）与 AlphaFold + n-gram LM 在四种不同的 新设计 骨架目标上设计的20个序列之间的比较。有关更多细节，请参考图例说明。

方法	N（样本数量）	pLDDT	pTM	根据 TM-score 的簇
p(x,y) 采样（子集）	400	81.49	0.67	242/400 簇
p(x) 采样	400	76.22	0.53	372/400 簇
距离图 KL（11）	400	80.22	0.56	282/400 集群
p(x,y) 采样（完整）	25,000	81.09	0.66	7663/25000 集群

方法	成对序列相似性	罗塞塔能量	SAP得分	螺旋/片段/环比
p(x,y) 采样 (子集)	0.13	-0.19	0.45	0.51/0.23/0.26
p(x) 取样	0.15	1.93	0.65	0.15/0.36/0.50
距离图 KL (11)	0.13	0.65	0.73	0.40/0.29/0.31
p(x,y) 采样 (完整)	0.12	-0.21	0.46	0.52/0.21/0.26

表 *S5.* 使用语言模型自由生成的不同方法的比较。除了描述的阻塞吉布斯采样方法
在本文中，我们测试了两种其他程序，以进行蛋白质的自由生成：p(x) 取样 - 其中仅使用 ESM2 和一个 ngram 项，通过马尔可夫链取样可能的氨基酸序列，但没有使用结构采样。此外，我们遵循了距离图 KL 最大化过程（11），其中结构步骤的采样目标是距离图与背景分布之间的 KL 散度。最大化这个 KL 可以被视为最小化距离图的熵，这意味着目标引导到自信的结构预测。上表比较了这些方法之间的关键统计数据。值得注意的是，我们观察到 p(x) 取样常常产生重复的序列模式。距离图 KL 方法，结合低能力结构预测头，其缺点在于提出的阻塞吉布斯方法。距离图 KL 方法倾向于生成几乎没有 α-螺旋和 β-片混合的结构，并且 pTM、罗塞塔能量和 SAP 得分更差。